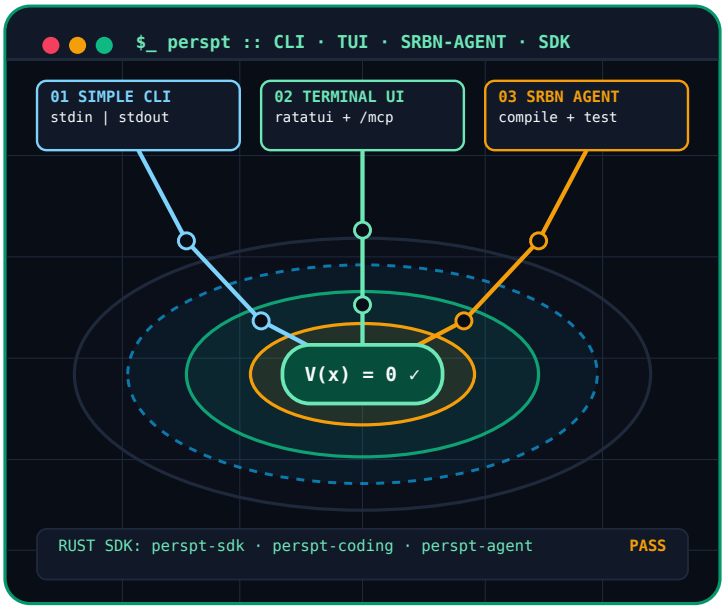
PERSPT :: DEVELOPER HANDBOOK

v0.6.6

Build with AI in your terminal.
Ship code verified by your compiler.

The complete guide to Perspt's command-line pipes, interactive TUI, compiler-verified SRBN coding agent, and embeddable Rust SDK.



01 · SIMPLE CLI & TUI
Pipe-ready `simple-chat` and full-screen `perspt chat` across 9+ providers.

02 · SRBN AGENT & RUST SDK
Compiler-verified `perspt agent` and reusable `perspt-sdk` crates.

Vikrant Rathore

EDITION 0.6.6 · HÓZHÓ

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The keyboard hums, the screen aglow,
AI's wisdom, a steady flow.

Will robots take over, it's quite the fright,
Or just provide insights, day and night?

We ponder and chat, with code as our guide,
Is AI our helper or our human pride?

Build with AI in Your Terminal — Ship Code Verified by Your Compiler

Perspt (*Personal Spectrum Pertaining Thoughts*) is a fast, single-binary Rust toolkit and embeddable SDK that brings 9+ cloud and local Large Language Models directly into your command-line workflow.

Instead of forcing you into a single interaction style, Perspt gives you **three ways to work in your terminal** plus a **modular Rust SDK**:

- **Simple CLI** (`perspt simple-chat`) — Stream plain text straight to `stdout` with zero UI chrome. Pipe files, compiler logs, or `git diff` from `stdin` into shell scripts and Unix pipelines.
- **Interactive Terminal UI** (`perspt` / `perspt chat`) — Chat in a full-screen `ratatui` workspace with streaming Markdown, syntax highlighting, tables, LaTeX math, conversation history, read-only MCP tools (`/mcp`), and live model switching.
- **SRBN Coding Agent** (`perspt agent`) — Delegate multi-file tasks to an autonomous coding agent that edits an isolated sandbox copy of your project and runs your real compiler, LSP, and test suite to fix its own mistakes before committing.
- **Embeddable Rust SDK** (`perspt-sdk`, `perspt-coding`, `perspt-research`) — Integrate Perspt's verification gate, capability sandbox, multi-provider router, and DuckDB audit ledger directly into your own Rust applications.

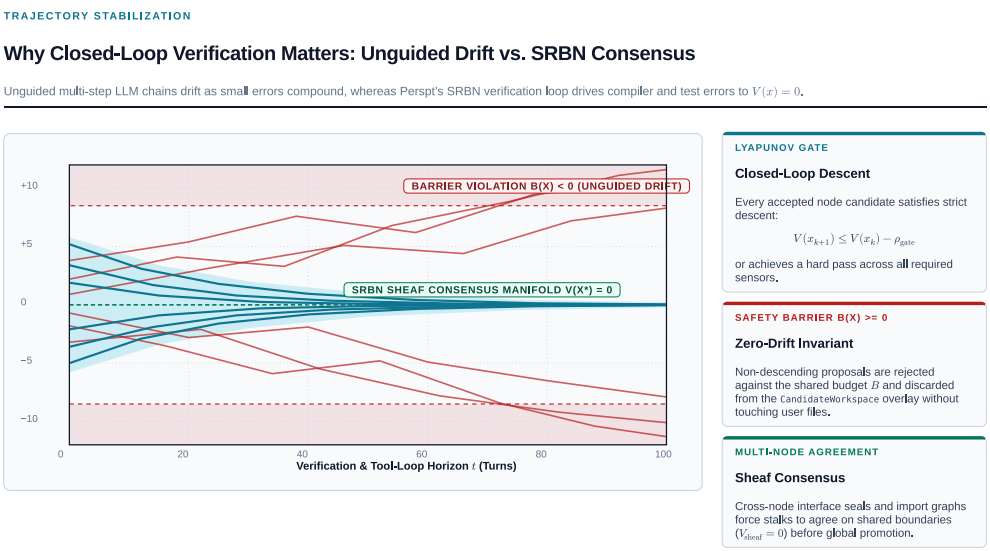


Fig. 1 Why `perspt agent` stays reliable across multi-step tasks: unguided LLM loops drift as small mistakes compound, whereas Perspt's internal SRBN verification loop checks every step against your compiler and test suite until errors reach zero ($V(x) = 0$).

> **Note**
>
> What is SRBN?
>
> **SRBN (Stabilized Recursive Barrier Networks)** is the internal reliability engine inside `perspt agent`, based on the *Stability is All You Need* research paper series. You never need to configure or know the math to use `perspt agent`—SRBN works automatically behind the scenes by running your compiler, language server, and tests after each edit, scoring remaining errors as $V(x)$, and discarding bad guesses so the code Perspt generates actually compiles and matches your task.

Introduction to Perspt

Perspt brings Large Language Models (LLMs) directly into your terminal and your Rust codebase. Built for developers who value speed, simplicity, and code that actually compiles, Perspt provides **three terminal tools in one binary** plus a **reusable Rust SDK**.

The Three Ways to Use Perspt

Whether you want a quick one-liner in a bash pipeline, an interactive chat window in your terminal, or an autonomous coding agent that writes and tests code across multiple files, Perspt gives you a dedicated mode for the job:

1. **Simple CLI** (`perspt simple-chat`) — **Plain Terminal Output** A fast, zero-chrome command-line interface that prints responses directly to `stdout`. Because it adds no TUI borders or control sequences, you can pipe logs, diffs, or source files straight into `perspt simple-chat` and redirect its output to files or other Unix tools:

   ```
   git diff | perspt simple-chat "Write a concise commit message for these changes"
   ```

2. **Terminal UI** (`perspt` or `perspt chat`) — **Interactive Full-Screen Chat** A rich Terminal User Interface built on `ratatui`. It gives you real-time token streaming, syntax-highlighted code blocks, Markdown tables, inline LaTeX math rendering, multi-line paste, conversation history, read-only MCP tool integration (`/mcp`), and instant model switching (`/model`) across 9+ cloud and local providers.
3. **SRBN Coding Agent** (`perspt agent`) — **Verified Autonomous Coding** Give Perspt a coding task and a target directory:

   ```
   perspt agent -w ./my-project "Add CSV export support with unit tests"
   ```

   Instead of blindly writing unverified code to your files, `perspt agent` edits an isolated sandbox copy of your project, runs your real compiler, language server (LSP), and test suite after each change, and automatically fixes compiler or test errors before promoting the finished changes.
4. **Rust SDK** (`perspt-sdk`, `perspt-coding`, `perspt-research`) — **Embed in Your Own Apps** Need to integrate verified AI workflows into your own Rust application or build a custom domain agent? Perspt's engine is published as modular Rust crates so you can reuse its verification gate, capability sandbox, provider portfolio, and DuckDB audit ledger as a library.

Why SRBN? How Perspt Keeps Generated Code Reliable

Every developer who has used an AI coding assistant knows the biggest frustration: **LLMs guess**. On a single function, a guess often works. Across a 10-step, multi-file task, small mistakes

compound rapidly—an incorrect import in step 2 causes a type mismatch in step 4 and a broken test in step 7. Even if a model succeeds 95% of the time on a single step, an unverified 50-step chain succeeds less than 8% of the time ($0.95^{50} \approx 7.7\%$).

Internally, `perspt agent` solves this using **SRBN (Stabilized Recursive Barrier Networks)**, an engineering architecture based on the *Stability is All You Need* research paper series.

> **Tip**
> You don't need to know the math to use `perspt agent`
> SRBN is an internal implementation detail. When you run `perspt agent`, it automatically detects your project's language toolchain and runs the verification loop for you.

Here is what SRBN does behind the scenes on every coding task:

- **Sandboxed Proposals**: The LLM never edits your working tree directly. Every file edit or tool call is applied to a disposable candidate workspace first.
- **Real Toolchain Sensors**: After the model proposes a change, Perspt runs your actual language tools—such as `cargo check`, `rust-analyzer`, `pyright`, `pytest`, or `tsc`—and checks that all required symbols and contracts exist.
- **Single Error Score** ($V(x)$): Perspt turns compiler errors, type mismatches, and failing tests into a single non-negative error score called **Lyapunov energy**:

  $$V(x) = \sum_{e \in E} w_e \parallel r_e(x) \parallel^2$$

  When $V(x) = 0$ and all required build/test stages pass, your code compiles cleanly and passes its tests.
- **Accept Only Real Progress**: A candidate change is accepted only if it passes all checks (*HardPass*) or measurably reduces the error score by at least ρ_{gate}:

  $$V(y) \leq V(x_{\text{best}}) - \rho_{\text{gate}}$$

  If an edit introduces new compiler errors or makes things worse, Perspt automatically discards that attempt, feeds the exact compiler/test diagnostics back to the model, and tries a targeted fix.

The result is simple: **code generated by Perspt's agent is tested against your real toolchain before you ever see it, so it runs and stays faithful to the task you gave it.**

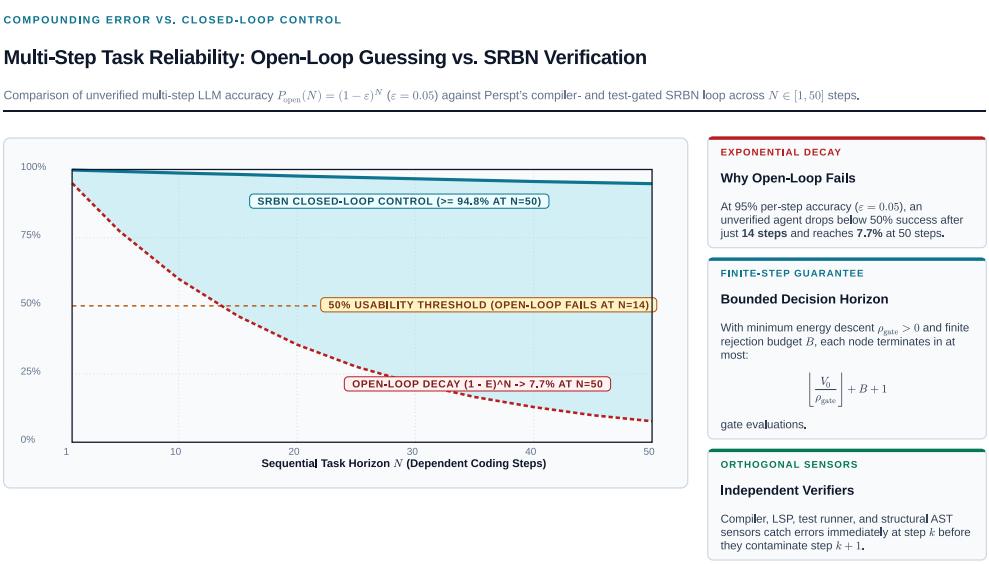


Fig. 2 Why closed-loop verification matters: unguided multi-step LLM chains decay exponentially as errors accumulate, while Perspt's SRBN verification loop catches and repairs errors at each step.

How `perspt agent` Protects Your Repository

When you run `perspt agent`, you might wonder: *What happens if the LLM writes broken code mid-task?*

The answer is that **Perspt never edits your working tree directly**. Instead, every coding task runs through a four-stage sandbox workflow (illustrated in the figure below):

1. **Give a Task**: You pass a plain-English task to `perspt agent`. Perspt detects your project's language toolchain and plans which files need to change.
2. **Edit in Sandbox**: The model applies its proposed edits inside an isolated **Candidate Overlay**—a disposable copy of your project—leaving your real files untouched.
3. **Compile & Test**: Perspt runs your real compiler, language server (LSP), and test suite inside the sandbox and measures any remaining errors as $V(x)$. If errors appear, those exact compiler or test diagnostics feed back to Step 2 so the model can repair them automatically.
4. **Promote Clean Code**: Only after all checks pass ($V(x) = 0$) does Perspt write the verified changes to your repository and record a rollback checkpoint in the local DuckDB ledger.

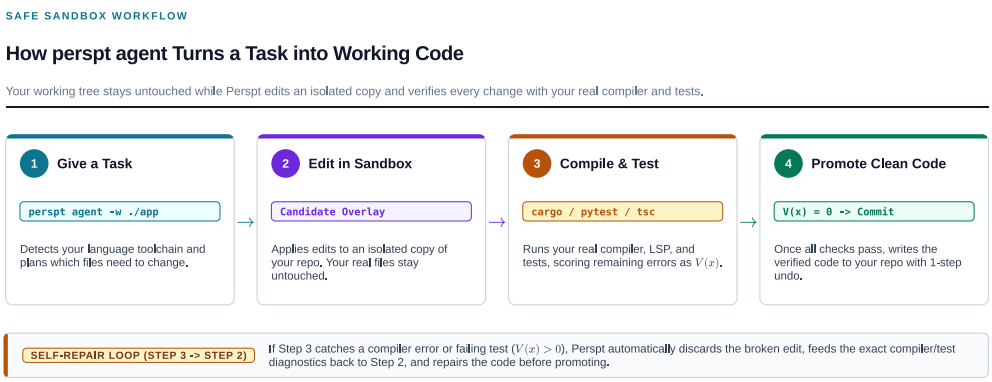


Fig. 3 How `perspt agent` turns a task prompt into working code: edits happen in an isolated sandbox and are checked by your real compiler and test suite before any change touches your repository.

Integrating Perspt via the Rust SDK

Perspt is structured as a fourteen-crate Cargo workspace so you can use its building blocks in your own Rust projects:

- **Embed the Verification Core** (`perspt-sdk`): Use `ResidualEvent`, `EnergyModel`, `score_candidate`, `evaluate_gate`, the five-clause admissibility kernel, and the Merkle-chained replay ledger in any domain—not just software engineering.
- **Reuse Domain Packages** (`perspt-coding`, `perspt-research`): Plug in ready-made language adapters (Rust, Python, TypeScript) or build your own domain package by implementing the `AgentDomainPackage` trait.
- **Embed the Full Agent Runtime** (`perspt-agent`, `perspt-core`, `perspt-store`): Run governed agent sessions programmatically from Rust with custom tool families, Starlark security policies (`perspt-policy`), and OS sandboxing (`perspt-sandbox`).

See API & SDK Reference for crate-by-crate documentation and code examples.

Core Features at a Glance

Feature	What It Does for You
3 Terminal Modes	`perspt simple-chat` for Unix pipes/scripts, `perspt chat` for interactive TUI chat, and `perspt agent` for autonomous verified coding.
9+ LLM Providers	Connect to OpenAI, Anthropic, Google Gemini, Vertex AI, Groq, Cohere, xAI, DeepSeek, local Ollama, or any OpenAI-compatible endpoint.
Compiler & LSP Checks	Automatically verifies candidate code with `cargo check`, `rust-analyzer`, `pyright`, `pytest`, `typescript-language-server`, and `gopls`.

Feature	What It Does for You
Per-Role Model Routing	Mix and match models: use a fast model for read-only repo exploration (`--explorer-model`), a strong coding model for edits (`--actuator-model`), and a reviewer model for diff checks (`--adjudicator-model`).
OS Sandbox & Policy	Protects your machine with Starlark command policies, workspace path confinement, and OS process isolation (Bubblewrap on Linux, `sandbox-exec` on macOS).
Hard Cost & Turn Limits	Strict per-session turn, tool-call, and rejection budgets ensure an agent never loops forever or burns unbounded API credits.
Live Web Dashboard	Pass `--dashboard` to watch the task graph, error energy $V(x)$, live file diffs, and LLM calls in your browser at `http://127.0.0.1:3000`.
Full Audit & Rollback	Every session is stored in a local DuckDB ledger. Replay sessions offline with `perspt replay` or undo any promotion with `perspt ledger --rollback`.

System Command Index

Run `perspt <COMMAND> --help` at any time to see all flags for a command:

Command	Purpose	Example
`chat`	Interactive full-screen Terminal UI chat (default when running `perspt`).	`perspt chat --model gemini-3.1-pro`
`simple-chat`	Plain CLI output with no TUI chrome; built for Unix pipes and scripts.	`perspt simple-chat "Explain this error"`
`agent`	Autonomous SRBN coding agent with compiler and test verification.	`perspt agent -w ./app "Add unit tests"`
`dashboard`	Launch the real-time web dashboard in your browser.	`perspt dashboard`
`status`	Show current session status, node progress, and error energy $V(x)$.	`perspt status`
`ledger`	Inspect session history or roll back the most recent promotion.	`perspt ledger --recent`
`resume`	Resume an interrupted or paused agent session from the DuckDB ledger.	`perspt resume <session-id>`
`replay`	Deterministically replay and verify a session offline without API keys.	`perspt replay <session-id>`
`providers`	List configured LLM providers and optionally probe live connectivity.	`perspt providers --probe`
`config`	View or edit your `config.toml` settings.	`perspt config --show`
`init`	Initialize project memory files and Starlark security policy rules.	`perspt init --memory --rules`

Command	Purpose	Example
`abort`	Safely stop a running agent session by revoking its authority epoch.	`perspt abort`
`prompts`	Inspect and lint compiled prompt section libraries.	`perspt prompts list`
`context`	Explain recorded context paging and recall events for a session.	`perspt context explain-turn --db-path perspt.db <id>`
`audit`	Record delayed safety audit labels for conformal risk calibration.	`perspt audit <sample> --safe`
`db repair`	Back up and repair a local DuckDB database after an unclean shutdown.	`perspt db repair --db-path perspt.db --discard-wal`

Supported LLM Providers

Perspt connects to nine provider families out of the box (plus any OpenAI-compatible local or cloud server):

Provider	Environment Variable	Example Models
Google Gemini (`gemini`)	`GEMINI_API_KEY`	`gemini-3.5-flash`, `gemini-3.1-pro`
OpenAI / Compatible (`openai`)	`OPENAI_API_KEY` (+ `OPENAI_BASE_URL`)	`gpt-4o`, `gpt-4o-mini`, local vLLM / llama.cpp
Anthropic (`anthropic`)	`ANTHROPIC_API_KEY`	`claude-3-5-sonnet-20241022`
Google Vertex AI (`vertex`)	Google Cloud ADC (`gcloud auth`)	`vertex::gemini-3.5-flash`, `vertex::gemini-2.5-pro`
Ollama (Local) (`ollama`)	None (runs locally on port 11434)	`llama3.2`, `qwen2.5-coder`
Groq (`groq`)	`GROQ_API_KEY`	`llama-3.1-8b-instant`
DeepSeek (`deepseek`)	`DEEPSEEK_API_KEY`	`deepseek-chat`, `deepseek-reasoner`
xAI (`xai`)	`XAI_API_KEY`	`grok-beta`
Cohere (`cohere`)	`COHERE_API_KEY`	`command-r-plus`

See Providers for multi-provider portfolio configuration in `config.toml`.

Quick Start

Get Perspt installed and try all three developer modes (**Simple CLI**, **Terminal UI**, and **SRBN Coding Agent**) in under five minutes.

Prerequisites

- **Rust Toolchain**: Rust 1.97.1+ (if building from source).
- **An LLM Provider**: An API key for Gemini, OpenAI, Anthropic, Groq, DeepSeek, xAI, or Cohere—or a local **Ollama** instance (no API key needed).

Installation

From Source (Recommended)
To compile the release binary directly from the source repository:

```
git clone https://github.com/eonseed/perspt.git
cd perspt
cargo build --release
```

The compiled binary is placed at `target/release/perspt`.

Cargo Install
To compile and install the package from the local directory:

```
cargo install --path .
```

With Ollama (No API Key)
To run local models using Ollama:

```
# Start the local Ollama service
ollama serve

# Pull the target model
ollama pull llama3.2
```

Set Environment API Keys

Export the key for your selected provider. The system automatically detects these variables at startup:

```
# Choose one
export OPENAI_API_KEY="sk-..."        # For OpenAI
export ANTHROPIC_API_KEY="sk-ant-..." # For Anthropic
export GEMINI_API_KEY="..."           # For Google Gemini
```

1. Simple CLI Mode (`perspt simple-chat`)

Use `perspt simple-chat` when you want plain text output in your terminal with no UI borders—ideal for Unix pipes, shell scripts, and quick one-liners:

```
# Ask a direct question in the terminal
perspt simple-chat

# Pipe a file or command output into Perspt
cat Cargo.toml | perspt simple-chat "List the crates in this workspace"

# Use a specific model
perspt simple-chat --model gemini-3.5-flash
```

2. Interactive Terminal UI (`perspt` / `perspt chat`)

Run `perspt` (or `perspt chat`) to open the full-screen Terminal UI with streaming Markdown, syntax highlighting, tables, and LaTeX math:

```
# Auto-detects your configured provider from environment variables
perspt

# Or choose a model explicitly
perspt chat --model gemini-3.1-pro
```

Type your dialogue prompt and press **Enter** to submit. Press **Esc** to exit the application. If the selected configuration contains chat-enabled `[[external_tools]]`, type `/mcp` to inspect MCP discovery and the admitted read-only tools. See Model Context Protocol (MCP) for a working server and policy configuration. Normal terminal paste shortcuts insert multiline clipboard content into the chat input because bracketed paste is enabled for the TUI lifecycle.

TUI Key Bindings

Key	Action
Enter	Transmit dialogue input buffer.
Esc	Terminate the TUI process.
Up / Down	Navigate through dialogue command history.
Page Up / Down	Scroll up/down in the chat conversation panel.
/save	Save dialogue log to a local file.

3. SRBN Coding Agent (`perspt agent`)

Give `perspt agent` a coding task and a project folder (`-w`). Behind the scenes, Perspt's **SRBN** engine edits files in a safe sandbox, runs your compiler, LSP, and test suite, and automatically fixes errors before committing the result:

```
# Create a Python package inside a new directory
perspt agent -w ./my-calculator "Create a Python calculator package with add, subtract,
multiply, divide. Include pytest tests."

# Auto-approve all modifications (headless mode)
perspt agent -y -w ./my-api "Build a REST API in Rust with Axum"

# Run with specific models for Actuator and Explorer roles
perspt agent \
  --actuator-model gemini-3.5-flash \
  --explorer-model gemini-3.1-flash-lite \
  -w ./project "Create an ETL pipeline in Python"
```

A headless execution run narrates the governed tool loop: admitted effects, measured energies, and gate decisions. Below is a clinical trace of a typical autonomous run:

```
Domain: coding
PSP-9 agent starting
Task: Create a Python calculator package with add, subtract, multiply, divide. Include pytest
tests.
Workspace: ./my-calculator
  Exploration mapped 1 language groups and 1 package roots
  PSP-9 session 01997a2f using gemini::gemini-3.5-flash
  [implement-1] Coding
  Effect call-1 applied to candidate (mutated=true)
  Effect call-2 applied to candidate (mutated=true)
  Measured implement-1 generation 1: V=2.000, hard_pass=false, residuals=1
  Gate implement-1 generation 1: RejectedNonDescending { delta_v: 0.0 }
  Effect call-3 applied to candidate (mutated=true)
  Measured implement-1 generation 2: V=0.000, hard_pass=true, residuals=0
  Gate implement-1 generation 2: HardPass

Outcome: HardPass
Session: 01997a2f-9c1e-4c30-b7ac-2f5d8f3e6a41
Turns: 5
Ledger head: 9f8a7e...
Promoted paths: pyproject.toml, src/calc/__init__.py, tests/test_calc.py
```

Every effect the model proposes passes the deterministic admissibility kernel before it touches the candidate workspace, and the acceptance gate reads the re-measured candidate — never the model's account of it.

Operational Modes

Choose the appropriate command mode depending on your task requirement:

Mode	Command	Target Use Case
Chat TUI	`perspt` or `perspt chat`	Interactive conversation with formatted terminal rendering.
Agent	`perspt agent "<task>"`	Autonomous multi-file code generation (experimental).
Simple Chat	`perspt simple-chat`	CLI chat without terminal interface, ideal for shell piping.
Exploration	`perspt agent --exploration-only "<question>"`	Read-only repository survey; nothing is mutated or promoted.
Status	`perspt status`	Query metrics of the active agent session.
Providers	`perspt providers --probe`	Print the provider capability matrix with live behavioral probes.
Replay	`perspt replay <session-id>`	Deterministic, credential-free audit replay of a session.
Audit	`perspt audit <sample> --safe`	Ingest delayed audit labels for conformal calibration.
Prompts	`perspt prompts list`	Inspect the compiled prompt section libraries.
Context	`perspt context explain-turn --db-path <DB> <session-id>`	Explain a session's recorded resident-context events.

Essential System Commands

Command	Description
`perspt config --show`	Prints active configuration parameters.
`perspt config --edit`	Opens the TOML configuration file in your editor.
`perspt init --memory --rules`	Instantiates memory files and policy rules in the project workspace.
`perspt status`	Displays per-node states, energy components, and retries.
`perspt abort`	Signals the active agent process to terminate.
`perspt resume --last`	Resumes the most recently interrupted agent session.
`perspt ledger --recent`	Displays recent commits recorded in the Merkle ledger.
`perspt ledger --rollback <session>`	Undoes the named session's newest completed promotion (session id prefix).

Next Steps

Guide	Summary
Tutorials	Step-by-step learning guides.
Configuration Guide	Customize providers and models.
Agent Mode Tutorial	Master autonomous coding.
Architecture	Understand the fourteen-crate design.

Installation

To deploy and execute Perspt, the environment must satisfy the specified system requirements. This document outlines the procedures for building the application, configuring provider keys, and setting up domain-specific tools.

System Requirements

- **Operating System**: Linux, macOS, or Windows. Native Windows supports all interface modes and explicit reduced-isolation coding; use Windows Subsystem for Linux when the coding agent must have an OS sandbox.
- **Rust Toolchain**: Version 1.97.1 or later (required for compilation from source).
- **Terminal Emulator**: Must support 256-color escape codes and UTF-8 encoding.

Building from Source

To compile the executable from the source repository:

```
# Clone the source repository
git clone https://github.com/eonseed/perspt.git
cd perspt

# Compile the project in release mode
cargo build --release
```

The compiled binary will be placed at `target/release/perspt`. To make the command globally accessible, copy the binary to a directory in your system `PATH`, for example:

```
cp target/release/perspt ~/.local/bin/
```

Installing via Cargo

If you possess the Rust toolchain and wish to install directly from the local directory:

```
cargo install --path .
```

Verifying the Installation

To verify that the executable is operational, query the system version and help manual:

```
perspt --version
# Expected output: perspt 0.6.6

perspt --help
```

Configuring API Keys

The application requires access to an LLM provider. Set the appropriate API key as an environment variable. The program will automatically detect the provider from the environment:

```
# For Google Gemini
export GEMINI_API_KEY="your-api-key"

# For OpenAI
export OPENAI_API_KEY="sk-..."

# For Anthropic
export ANTHROPIC_API_KEY="sk-ant-..."
```

If you are using Ollama for local execution, start the local service. No API key is required:

```
ollama serve
ollama pull llama3.2
```

Agent Mode Prerequisites

Autonomous agent execution requires the presence of domain-specific compilers, type checkers, and test runners on the system path.

Native Windows has no Perspt process-sandbox backend. Strict agent mode fails closed before starting compiler, test, lint, inspection, or LSP processes. To accept native host-user execution explicitly, use:

```
perspt agent --allow-unisolated -w . "Implement the requested change"
```

The same durable setting is useful for resume:

```
[verification]
allow_unisolated = true
```

This mode retains candidate overlays, capability checks, verification gates, journaling, and promotion approval, but it does not isolate filesystem or network access by child processes. Prefer WSL when that boundary matters.

Python Domain

The Python adapter requires:

1. **uv**: For dependency management and environment isolation.
2. **ty**: For static type verification.

3. **pytest**: For logical assertion testing.

Install these dependencies as follows:

```
# Install the uv package manager
curl -LsSf https://astral.sh/uv/install.sh | sh

# Install verification tools
pip install ty pytest
```

Rust Domain

The Rust adapter requires:

1. **rustup**: For compiler toolchain management.
2. **rust-analyzer**: For language server diagnostics.

Install these components as follows:

```
# Install the Rust compiler and manager
curl --proto '=https' --tlsv1.2 -sSf https://sh.rustup.rs | sh

# Install the LSP component
rustup component add rust-analyzer
```

JavaScript/TypeScript Domain

The TypeScript adapter requires Node.js and the TypeScript compiler:

```
# Install Node.js via your system package manager, then install the compiler
npm install -g typescript
```

Building the Documentation

To compile the Guidedog and Typst documentation books locally:

```
# Build all Typst diagrams, PSP catalog, book HTML/PDF, and unified site
cargo run -p xtask -- docs

# Or build individual targets directly with guidedog
guidedog dsc build --dir=docs/psps
guidedog build html docs/psps -W -n
guidedog build html docs/perspt_book -W -n
guidedog ssg build docs --out docs/_build/site
```

The generated unified site is located at `docs/_build/site/index.html`, the book PDF is placed at `docs/_build/site/books/perspt-book-en.pdf` (and `docs/perspt.pdf`), and individual PSP PDFs are placed in `docs/psps/_build/pdf/`.

Getting Started

This guide outlines the protocol for setting up the environment, compiling the binaries, executing an interactive chat session, and running an autonomous agent task.

System Requirements

Before installation, verify that the host environment conforms to the following operational parameters:

Component	Specification Requirement
Operating System	Linux, macOS, or Windows. Native Windows coding requires the explicit reduced-isolation option; use Windows Subsystem for Linux when OS sandbox isolation is required.
Rust Compiler	Version 1.97.1 or later (required for building from source)
Terminal Emulator	Modern console supporting UTF-8 encoding and 256-color escape sequences
Network Link	Required for cloud LLM API communication (unnecessary for local Ollama deployments)

Provider Access Configuration

Perspt requires access to an external model oracle. You must define and export the appropriate API key as an environment variable. The system inspects the environment and maps the configuration according to a deterministic detection priority:

```
Vertex AI > Gemini > OpenAI > Anthropic > Groq > Cohere > XAI > DeepSeek > Ollama
```

Set the key for your selected provider:

```
# Example configurations
export OPENAI_API_KEY="sk-..."
export ANTHROPIC_API_KEY="sk-ant-..."
export GEMINI_API_KEY="..."
```

For offline execution using local models, start the Ollama service:

```
ollama serve
ollama pull llama3.2
```

Quick Installation

From Source (Recommended)

To compile the release binary directly from the source repository:

```
git clone https://github.com/eonseed/perspt.git
cd perspt
cargo build --release
./target/release/perspt --version
```

Cargo Install

To install the package into your Cargo binary path:

```
cargo install perspt
perspt --version
```

Binary Archive

To download and deploy the precompiled release archive:

```
curl -L https://github.com/eonseed/perspt/releases/latest/download/perspt-linux-x86_64.tar.gz
| tar xz
chmod +x perspt && sudo mv perspt /usr/local/bin/
```

Interactive Dialogue Session

The terminal user interface (TUI) is the default interactive environment. To launch the TUI:

```
perspt
```

Upon initiation, the system establishes a session using the detected API key.

- **Input Entry**: Enter your dialogue prompt and press **Enter** to stream the response.
- **Scrollback**: Navigate the conversation scrollback window using **Up/Down** or **Page Up/Page Down**.
- **Exit**: Press **Esc** to terminate the TUI session.

For non-interactive pipelines or shell-script piping, use the simple chat command:

```
perspt simple-chat
# Optionally record the session output
perspt simple-chat --log-file session.txt
```

Type `exit` or enter `Ctrl+D` to terminate the simple chat process.

Autonomous Agent Execution

Agent mode compiles a task charter into a state graph of modules and executes them under a closed-loop stabilizer.

To execute an autonomous coding task:

```
perspt agent -w ./demo-calculator \
  "Create a Python calculator package with add, subtract, multiply, divide. Include type hints
and pytest tests."
```

Operational Execution Steps

During execution, the SRBN engine performs the following operations:

1. **System Detection**: The program identifies Python as the target workspace language, and registers the corresponding LSP verifier and pytest environments.
2. **Graph Planning**: Planning is a governed architect turn that revises the work graph through the `update_graph` tool. Each node lists its declared file footprint, and the dispatcher schedules ready nodes by footprint conflict (no two concurrent nodes may touch the same files).
3. **Stabilization Loop**: For each dispatched node, the Actuator issues typed tool calls against a reversible candidate overlay; the deterministic kernel admits each call before it is applied. The system then computes the Lyapunov energy on the realized candidate:
   - Syntactic energy (V_{syn}): Diagnostics from the LSP.
   - Logical energy (V_{log}): Test failures from the test runner.
   - Build energy (V_{boot}): Exit codes of environment setups.

   If $V(x) > \varepsilon$, the engine compiles the error diagnostics into a correction prompt and retries. This loops until the node converges ($V(x) \leq \varepsilon$) or the retry cap is reached.
4. **Interactive Review**: In interactive mode, the TUI displays the proposed file changes (unified diffs) and verifier states for approval before commit.
5. **Merkle Commit**: Stable nodes are written to the Merkle ledger and committed to the active workspace.

Verifying Output Structures

Upon task completion, inspect the workspace directory to verify the generated files:

```
ls demo-calculator/
# Expected structure:
# pyproject.toml  src/  tests/  uv.lock
```

To run the verification suite locally:

```
cd demo-calculator && uv run pytest -v
```

Headless Mode

For non-interactive environments, such as automated build pipelines, use the `--yes` flag to bypass the interactive review gate:

```
perspt agent --yes -w ./rust-csv-converter "Build a Rust CLI tool that converts CSV to JSON"
```

Exploration-Only Mode

To survey a repository under a strictly read-only capability, run only the exploration phase. Every call passes the kernel, mutation attempts are recorded denials, and nothing is mutated or promoted:

```
perspt agent --exploration-only -w ./rust-csv-converter "Summarize how CSV parsing is
structured"
```

Perspt also accepts `--allow-experimental-prompts` to substitute validated `[prompts]` bundle sections live; such overrides remain experimental until a change record passes paired evaluation.

Parameterizing Models per Role

The agent runtime routes model calls by role. You can allocate different models to these roles depending on the complexity of the work:

- **Actuator** (`--model` / `--actuator-model`): Proposes the governed coding tool calls.
- **Explorer** (`--explorer-model`): Optional cheaper read-only repository exploration.
- **Adjudicator** (`--adjudicator-model`): Optional no-tool conjunctive diff veto.

To run the agent with customized model selections:

```
perspt agent \
  --actuator-model gemini-2.5-flash \
  --explorer-model gemini-2.5-flash \
  --adjudicator-model gemini-2.5-pro \
  -w ./project "Task description"
```

The `[models]` table in `config.toml` additionally routes architect, actuator, verifier, speculator, and adjudicator turns as fully qualified `provider::model` values; when present it takes precedence over the flat `*_model` configuration fields.

The actuator route also supports ordered fallback models in case the primary oracle returns a rate limit or API error. The flag is repeatable, and failover is sticky:

```
perspt agent \
  --actuator-model gemini-2.5-pro \
  --fallback-model gemini-2.5-flash \
  --fallback-model gemini-2.5-flash-lite \
  -w ./project "Task description"
```

Next Steps

Guide	Summary
Tutorials	Step-by-step learning guides.

Guide	Summary
Configuration Guide	Providers, models, and preferences.
Agent Mode Tutorial	Master autonomous coding.
Architecture	Understand the fourteen-crate design.

Tutorials

Step-by-step guides to master Perspt.

Learning Path

Order	Tutorial	Outcome
1	First Chat	Understand basic TUI interaction
2	Local Models with Ollama	Set up Ollama for privacy
3	Agent Mode Tutorial	Master multi-file autonomous coding
4	Headless Mode	Run agent without interactive prompts
5	Custom Workflows	Automate with scripts and CI/CD
6	Example: Python ETL Pipeline	Build a real Python project from scratch
7	Example: Rust CLI Tool	Build a real Rust project from scratch
8	Example: Scientific Computing	Apply agent mode to scientific computing
9	Monitoring Agent Execution with the Dashboard	Monitor agent sessions via the web dashboard

First Chat

Start your first conversation with an LLM using Perspt.

Prerequisites

- Perspt installed (see Installation)
- An API key from any provider, or Ollama running locally

Step 1: Set Your API Key

```
export GEMINI_API_KEY="your-key"
```

Step 2: Launch the TUI

```
perspt
```

Perspt auto-detects the provider from the environment variable and launches the chat TUI. You will see a status bar showing the provider and model.

Step 3: Chat

Type a message and press **Enter**. Perspt streams the response in real time with markdown formatting (code blocks, headers, lists, bold, italic).

```
You: Explain Rust's ownership model in 3 sentences.

Assistant: Rust's ownership model ensures each value has exactly one owner at a
time. When the owner goes out of scope, the value is automatically dropped.
Borrowing rules allow temporary references without transferring ownership, and
the compiler enforces these rules at compile time to prevent data races and
dangling pointers.
```

Step 4: Navigate

Key	Action
Enter	Send message
Ctrl+J	Insert newline in input
Page Up / Down	Scroll chat history
Ctrl+Up / Down	Scroll one line
Esc or **Ctrl+Q**	Exit
/save	Save conversation to file

Step 5: Try Simple CLI Mode

For a minimal interface without the TUI:

```
perspt simple-chat
# Or with logging:
perspt simple-chat --log-file session.txt
```

Type your question, get a streamed text answer. Type `exit` or `Ctrl+D` to quit.

Next Steps

- Agent Mode Tutorial - Try autonomous multi-file coding
- Local Models with Ollama - Use Ollama for offline conversations

Agent Mode Tutorial

Master autonomous multi-file code generation with the experimental SRBN engine.

Overview

Agent mode lets Perspt plan, write, test, and commit multi-file projects autonomously. The SRBN runtime (extended by the dependency-aware mutable work graph) decomposes tasks into a graph of nodes — each revision acyclic — with each node owning specific output files, verified by real LSP diagnostics and test runners. Since v0.6.6 the engine underneath is the PSP-9 governed tool loop — capability-scoped typed tool calls, measured energy descent, and a Merkle-chained ledger — extended by the PSP-10 typed prompt and context mechanisms.

> **Experimental Feature**
> Agent mode implements the SRBN theoretical framework. The engine is functional and usable, but has not yet been benchmarked. Results may vary depending on model capability and task complexity.

Prerequisites

- Perspt v0.6.6+
- An API key for a capable model
- For Python projects: `uv` and `python3` installed
- For Rust projects: `cargo` and `rustc` installed

Basic Usage

```
# Plan and build a project in a new directory
perspt agent -w ./my-project "Create a Python calculator package"

# Auto-approve all changes (headless)
perspt agent -y -w ./my-project "Create a REST API in Rust"

# Use specific models per route
perspt agent \
  --actuator-model gemini-3.1-pro \
  --explorer-model gemini-3.5-flash \
  -w ./project "Build an ETL pipeline"
```

Step-by-Step: Python Calculator

5.3.4.1 Step 1: Start the Agent

```
mkdir calc-demo && cd calc-demo
perspt agent -w . \
  "Create a Python calculator package with add, subtract, multiply,
   divide operations. Include type hints, a pyproject.toml with
   build-system, and comprehensive pytest tests."
```

5.3.4.2 Step 2: Watch the Governed Loop

The agent proceeds through the governed loop phases:

Detection - Perspt inspects the workspace and selects a domain package (here `coding`); the selected domain is printed at startup. Pass `--domain` to pick one explicitly.

Exploration - Read-only repository mapping runs before any mutation is proposed. A cheaper model can serve this route via `--explorer-model`.

Planning - By default the session runs a single work-graph node. With `--max-parallel-nodes` above 1 (which requires `--yes`), one governed architect turn is forced to call the privileged `update_graph` tool. The proposal declares nodes with output targets and edges and is validated (acyclic, complete) before acceptance; an invalid or empty proposal falls back to the single-node graph, and the fallback is recorded in the ledger.

Each node's declared output targets form its write footprint, so the scheduler keeps concurrent nodes from touching the same files.

Execution - A closed-loop scheduler (utilizing the mutable work graph) re-evaluates the graph each round and dispatches *ready* nodes whose footprints do not conflict — not a precomputed topological walk. Reworked nodes are re-picked and inserted nodes are executed on later rounds. For each node:

1. The actuator proposes typed tool calls (file edits, commands, reads)
2. Effects are applied inside an isolated candidate workspace
3. Compiler, test, and lint sensors measure the checkpoint's energy
4. A checkpoint is accepted only when measured energy descends by at least `--rho-gate`; non-descending attempts draw down the shared `--rejection-budget`

Nodes run one at a time by default; `--max-parallel-nodes` raises the concurrent dispatch bound (above 1 requires `--yes`).

5.3.4.3 Step 3: Review Changes

The agent runs inside a TUI with three tabs — **Dashboard**, **Tasks**, and **Diff** — cycled with `Tab`/`Shift-Tab` or selected directly with `1`, `2`, `3`. `Up`/`k` and `Down`/`j` navigate, `p` pauses and resumes, `a` opens the approval modal, and `q` quits.

In interactive mode, the review modal presents the pending change for approval. Keys inside the modal:

- **y** - Approve
- **n** - Reject
- **c** - Send correction feedback to the agent
- **e** - Edit
- **d** - View diff
- **s** - Skip
- **Left**/**Right** - Move between actions; **Enter** confirms the selected action
- **Esc** - Close the modal

5.3.4.4 Step 4: Inspect Results

After all nodes converge and pass sheaf validation:

```
ls -la
# pyproject.toml  src/  tests/  uv.lock

# Run the tests
cd calc-demo && uv run pytest -v

# Check the ledger
perspt ledger --recent
```

Model Route Configuration

Assign specialized models to each route:

```
perspt agent \
  --actuator-model gemini-3.1-pro \
  --explorer-model gemini-3.5-flash \
  --adjudicator-model gemini-3.1-pro \
  --fallback-model gemini-3.5-flash \
  -w ./project "Build a web server"
```

Flag	Purpose	Recommendation
`--actuator-model`	Proposes governed coding tool calls (alias: `--model`)	Strong coding model (e.g., Gemini Pro, Claude Sonnet)
`--explorer-model`	Cheaper read-only repository exploration	Fast model (e.g., Gemini Flash)
`--adjudicator-model`	No-tool conjunctive diff adjudication	Analytical model (e.g., Gemini Pro)
`--fallback-model`	Ordered sticky actuator fallback; repeat the flag to add routes	Reliable alternate provider

The `[models]` table in the configuration file can pin the same routes per role (`actuator`, `speculator` for the explorer route, `adjudicator`); its `architect` role supplies the higher-capability handoff route used by the recovery ladder.

Descent Gating

Tune how much measured progress each accepted checkpoint must show:

```
# Require steeper measured descent per accepted checkpoint
perspt agent --rho-gate 1.0 -w . "Add tests"

# Accept shallower descent for exploratory work
perspt agent --rho-gate 0.1 -w . "Quick prototype"
```

Each accepted checkpoint must lower the measured energy (compiler, test, and lint sensor readings) by at least `--rho-gate` (default 0.5). Checkpoints that fail the gate are refused and draw down the shared `--rejection-budget`.

Approval

Approval is interactive by default: the agent pauses at promotion and asks in the review modal. Pass `-y`/`--yes` to approve final promotion automatically. A run without a terminal (CI, piped output) requires `--yes`; otherwise it fails fast rather than silently escalating.

```
perspt agent --yes -w . "Modify database schema"
```

Loop Bounds

```
# Allow more model turns per node (default 12)
perspt agent --max-turns 20 -w . "Large refactor"

# Tighten the per-turn tool-call bound (default 8)
perspt agent --max-calls-per-turn 4 -w . "Small fix"

# Shrink the shared non-descending and recovery budget (default 4)
perspt agent --rejection-budget 2 -w . "Iterative improvement"
```

Exploration and Experimental Prompts

```
# Read-only exploration: deterministic map plus an explorer tool
# loop; nothing is mutated or promoted
perspt agent --exploration-only -w . "Map the module structure"
```

```
# Substitute validated [prompts] bundle sections live (Gate AE:
# experimental until a change record passes paired evaluation)
perspt agent --allow-experimental-prompts -w . "Refactor the parser"
```

Managing Sessions

```
# Show session status: lifecycle counts, energy breakdown, escalations
perspt status

# Abort the current session
perspt abort

# Resume the last interrupted session with trust context
perspt resume --last
```

The `status` command shows per-node lifecycle counts (queued, running, verifying, retrying, completed, failed, escalated), the latest energy breakdown, total retry count, and recent escalation reports.

The `resume` command displays trust context before resuming: escalation count, last energy state, and total retries across all nodes.

Auditing Sessions

Every model turn and tool effect is recorded in the Merkle-chained ledger; no logging flag is needed:

```
# Deterministic, credential-free replay of a session
perspt replay <SESSION_ID>

# The prompt programs a session actually compiled, with digests
perspt prompts explain-session --db-path <PATH> <SESSION_ID>

# A session's recorded context events (compactions, refusals)
perspt context explain-turn --db-path <PATH> <SESSION_ID>

# Ledger statistics
perspt ledger --stats
```

Best Practices

1. **Start with a clear task description** - Include language, package structure, and testing requirements in the prompt
2. **Use workspace directories** - Always specify `-w <dir>` for clarity
3. **Set loop bounds** - Use `--max-turns` and `--rejection-budget` to bound runaway sessions
4. **Review before committing** - In interactive mode, inspect diffs carefully
5. **Use per-route models** - Match model capabilities to each route with `--actuator-model`, `--explorer-model`, and `--adjudicator-model`
6. **Track changes** - Use `perspt ledger` to review and rollback

Troubleshooting

Agent stuck in retry loop:

- Check LSP is working: `ty check file.py` or `cargo check`
- Relax the descent gate: `--rho-gate 0.1`

- Raise the shared recovery budget: `--rejection-budget 8`

High energy despite clean code:

- Check test failures: `uv run pytest -v`
- Review LSP diagnostics
- Replay the session to see what the sensors measured: `perspt replay <SESSION_ID>`

Plugin not detected:

- Ensure required binaries are installed (`uv`, `cargo`, `node`, etc.)
- Check `perspt status` for active plugins

See Also

- Headless Mode - Fully autonomous operation
- SRBN Architecture - SRBN technical details
- Agent Options Reference - Full CLI reference

Headless Mode

Run Perspt's experimental SRBN agent without interactive prompts. This is designed for CI/CD pipelines, batch code generation, and automated workflows.

Overview

In headless mode (`--yes` flag), the agent auto-approves final promotion, skipping the interactive review modal. Combined with `--workdir`, it enables bounded autonomous project generation. It does not bypass capability grants, verification gates, budgets, the ledger, or workspace confinement.

```
perspt agent --yes -w /tmp/output "Create a Python ETL pipeline"
```

When to Use Headless Mode

- **CI/CD pipelines** - Generate boilerplate or scaffold projects in automation
- **Batch processing** - Run multiple agent tasks in sequence from a script
- **Rapid prototyping** - Skip review when iterating quickly
- **Testing the agent** - Validate agent behavior without manual intervention

When NOT to use headless mode:

- **Production codebases** - Always review changes before committing
- **Security-sensitive projects** - Manual review catches policy violations
- **Learning** - Interactive mode teaches you how SRBN works

Basic Headless Run

```
export GEMINI_API_KEY="your-key"

# Create a project autonomously
perspt agent --yes -w /tmp/my-project \
  "Create a Python data validation library using Pydantic.
   Include src layout, pyproject.toml, and pytest tests."
```

The agent will:

1. Detect language plugins
2. Plan the work graph — a single node by default; with `--max-parallel-nodes` above 1, one governed architect turn may decompose the task
3. Execute all nodes within their capability grants
4. Run verification (LSP + tests) on each node
5. Commit stable nodes to the ledger
6. Print a summary

Key Flags

Flag	Description
`--yes` / `-y`	Auto-approve final promotion (headless mode)
`-w, --workdir <DIR>`	Working directory for the project
`--max-turns <N>`	Maximum model turns per node (default 12)
`--max-calls-per-turn <N>`	Maximum model-issued and nested tool calls per turn (default 8)
`--rejection-budget <N>`	Shared non-descending and recovery budget (default 4)
`--output-summary <FILE>`	Write the terminal session summary as JSON

> **Headless requirements**
> A run without a terminal (CI, piped output) requires `--yes`: promotion approval cannot be prompted without a terminal, so the run fails fast instead of silently escalating. `--max-parallel-nodes` above 1 also requires `--yes`. A run that ends in escalation or incomplete work exits with a nonzero status, so CI never reads a stopped node as success.

Reading Structured Progress

In headless mode, Perspt prints its startup banner (task, workspace, detected domain), then streams ledger narration lines and per-node status changes as the run progresses. When the run finishes it prints the outcome, session id, turns used, ledger head hash, and any promoted paths.

For a machine-readable record, pass `--output-summary`:

```
perspt agent --yes -w /tmp/out --output-summary summary.json \
  "Build a CLI tool in Rust that converts CSV to JSON"
```

The JSON summary contains the session id, node id, outcome, turns used, ledger head, and promoted paths.

Checking Session Status

After a headless run, inspect the results:

```
# Session status
perspt status

# Recent ledger entries
perspt ledger --recent

# Ledger statistics
```

```
persp ledger --stats

# Deterministic, credential-free replay of a session
persp replay <SESSION_ID>

# Resume a failed session
persp resume --last
```

Scripting Multiple Tasks

Run multiple agent tasks from a shell script:

```
#!/bin/bash
set -e
export GEMINI_API_KEY="your-key"

tasks=(
  "Create a Python CSV parser library"
  "Create a Rust JSON validator CLI"
  "Create a Python REST API with FastAPI"
)

for i in "${!tasks[@]}"; do
  dir="/tmp/project-$i"
  mkdir -p "$dir"
  persp agent --yes --max-turns 8 -w "$dir" "${tasks[$i]}"
  echo "=== Project $i complete ==="
done
```

Safety Recommendations

1. **Always set loop bounds** - `--max-turns` and `--rejection-budget` bound runaway sessions
2. **Use disposable directories** - Point `-w` to a fresh directory
3. **Review after generation** - Inspect the output before using it in production
4. **Audit with the ledger** - `persp replay <SESSION_ID>` reconstructs what the agent did, no logging flag required
5. **Check the exit status** - A nonzero exit means escalation or incomplete work, not verified success

See Also

- Agent Mode Tutorial - Interactive agent mode tutorial
- SRBN Architecture - SRBN technical details
- Agent Options Reference - Full agent CLI reference

Local Models with Ollama

Run Persp with local models for privacy and offline usage.

Prerequisites

- Ollama installed
- Sufficient RAM for your chosen model (7B models: 8GB+, 70B models: 64GB+)

Setup

```
# Install Ollama (macOS)
brew install ollama

# Start the Ollama service
ollama serve

# Pull a model
ollama pull llama3.2
ollama pull codellama  # For coding tasks
```

Using Ollama with Perspt

Perspt auto-detects Ollama if no cloud API keys are set:

```
# Unset any cloud keys
unset OPENAI_API_KEY ANTHROPIC_API_KEY GEMINI_API_KEY

# Launch Perspt - auto-detects Ollama
perspt

# Or specify a model explicitly
perspt chat --model llama3.2
```

Agent Mode with Local Models

```
perspt agent --model qwen2.5-coder -w ./my-project "Create a Python utility"
```

> **Performance Note**
>
> Local models are slower and less capable than cloud models for complex agent tasks. For best results with agent mode, use a capable model (70B+ parameters) or use a cloud model for the actuator route and keep the cheap local model on read-only exploration:
>
> ```
> export GEMINI_API_KEY="your-key"
> perspt agent \
>   --actuator-model gemini-3.1-pro \
>   --explorer-model qwen2.5-coder \
>   -w ./project "Create a utility"
> ```

Available Models

Popular Ollama models for use with Perspt:

Model	Size	Best For
`llama3.3`	8B/70B	General chat
`qwen2.5-coder`	7B/32B	Code generation
`deepseek-coder-v2`	16B/236B	Code generation
`mistral`	7B	General purpose
`phi4`	14B	Reasoning and lightweight tasks

Custom Workflows

Integrate Perspt into scripts, CI/CD pipelines, and automation.

Simple CLI for Scripting

The `simple-chat` command provides a Unix-friendly interface:

```
# Interactive
perspt simple-chat

# With session logging
perspt simple-chat --log-file session.txt
```

Batch Agent Runs

Run multiple agent tasks from a script:

```
#!/bin/bash
set -e
export GEMINI_API_KEY="your-key"

perspt agent --yes --max-turns 8 -w /tmp/proj1 "Create a Python CSV parser"
perspt agent --yes --max-turns 8 -w /tmp/proj2 "Create a Rust CLI calculator"
```

CI/CD Integration

Use headless mode in CI/CD pipelines:

```
# GitHub Actions example
name: Generate Boilerplate
on:
  workflow_dispatch:
    inputs:
      task:
        description: 'Task description'
        required: true

jobs:
  generate:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4
      - name: Install Perspt
        run: cargo install perspt
      - name: Run Agent
        env:
          GEMINI_API_KEY: ${{ secrets.GEMINI_API_KEY }}
        run: |
          perspt agent --yes --max-turns 12 \
            -w ./generated "${{ inputs.task }}"
      - name: Commit Results
        run: |
          git add generated/
          git commit -m "Generated: ${{ inputs.task }}"
```

Post-Run Analysis

After an agent run, use the management commands:

```
# Session status
perspt status

# Audit a session from the ledger
perspt replay <SESSION_ID>
perspt prompts explain-session --db-path <PATH> <SESSION_ID>
perspt context explain-turn --db-path <PATH> <SESSION_ID>

# Ledger history
perspt ledger --recent
perspt ledger --stats

# Resume incomplete sessions
perspt resume --last
```

Example: Python ETL Pipeline

This tutorial demonstrates building a complete Python ETL (Extract, Transform, Load) pipeline using Perspt's agent mode.

Task Description

We will ask the agent to create a data pipeline that:

- Reads CSV files with validation
- Transforms data using Pydantic models
- Handles missing values and edge cases
- Includes comprehensive pytest tests

Running the Agent

```
export GEMINI_API_KEY="your-key"

perspt agent --yes -w /tmp/etl-demo \
  --actuator-model gemini-3.1-pro \
  --explorer-model gemini-3.5-flash \
  "Create a Python ETL pipeline package. It should:
   1. Read CSV files into Pydantic models for validation
   2. Transform records (clean nulls, normalize strings, compute derived fields)
   3. Write validated records to a new CSV
   4. Include a CLI entry point in src/main.py
   5. Use uv for dependency management with pandas and pydantic
   6. Include comprehensive pytest tests for each module"
```

Expected Output

The agent produces a project with `src/` layout:

```
/tmp/etl-demo/
+-- pyproject.toml          # [build-system] with uv_build backend
+-- uv.lock
+-- src/
|   +-- etl_pipeline/
|   |   +-- __init__.py
|   |   +-- core.py         # Main pipeline logic
|   |   +-- validator.py    # Pydantic models
|   |   +-- transformer.py  # Data transformations
```

```
|   +-- main.py             # CLI entry point
+-- tests/
    +-- test_core.py
    +-- test_validator.py
    +-- test_transformer.py
```

Verification

The engine verifies every checkpoint with real sensors: the `ty` LSP server and pytest measure the energy of each candidate, and a checkpoint is accepted only when that measured energy descends by at least the `--rho-gate` margin. When the run finishes, the terminal summary reports the outcome, the session id, turns used, the ledger head hash, and the promoted paths.

After completion, verify manually:

```
cd /tmp/etl-demo
uv run pytest -v
# Expected: 15-20 tests passing
```

Key Observations

- The agent uses `uv init --lib` to create proper `src/` layout with `[build-system]` in `pyproject.toml`
- Ownership closure ensures no two nodes write to the same file
- The `ty` LSP server provides real-time type checking
- pytest provides test verification
- Work is promoted only after the run's final checkpoint passes every hard predicate (a `HardPass` outcome)

See Also

- Agent Mode Tutorial - Agent mode tutorial
- Example: Rust CLI Tool - Rust project example

Example: Rust CLI Tool

Build a Rust command-line tool using Perspt's agent mode.

Task Description

We will ask the agent to create a CLI tool that converts CSV files to JSON.

Running the Agent

```
export GEMINI_API_KEY="your-key"

perspt agent --yes -w /tmp/csv2json \
  --actuator-model gemini-3.1-pro \
  --explorer-model gemini-3.5-flash \
  "Create a Rust CLI tool called csv2json that:
   1. Reads a CSV file from a path argument
   2. Converts each row to a JSON object
   3. Outputs JSON array to stdout or a file (--output flag)
   4. Handles errors gracefully with anyhow
   5. Uses clap for argument parsing
   6. Includes unit tests and integration tests"
```

Expected Output

```
/tmp/csv2json/
+-- Cargo.toml
+-- src/
|   +-- main.rs        # Entry point, clap args
|   +-- lib.rs          # Core conversion logic
|   +-- converter.rs    # CSV -> JSON conversion
+-- tests/
    +-- integration.rs  # Integration tests
```

Verification

The agent uses the `rust` plugin:

- **LSP**: `rust-analyzer` for V_syn
- **Tests**: `cargo test` for V_log
- **Init**: `cargo init` for V_boot

```
cd /tmp/csv2json
cargo test
cargo run -- --help
```

Key Observations

- The `rust` plugin selects `rust-analyzer` as the LSP server
- `cargo init` bootstraps the project structure (V_boot)
- `cargo test` runs both unit and integration tests (V_log)
- Ownership closure assigns each source file to exactly one node

See Also

- Agent Mode Tutorial - Agent mode tutorial
- Example: Python ETL Pipeline - Python project example

Example: Scientific Computing

This tutorial demonstrates using Perspt's headless agent mode for scientific computing tasks: CFD simulation setup and wind tunnel data analysis.

> **Experimental**
> These examples demonstrate advanced prompt patterns for scientific computing. Results depend on model capability and may require iteration. Treat them as starting points rather than guaranteed production workflows.

CFD Simulation Setup (Python)

Create a computational fluid dynamics simulation scaffolding:

```
export GEMINI_API_KEY="your-key"

perspt agent --yes -w /tmp/cfd-sim \
  --actuator-model gemini-3.1-pro \
```

```
  --explorer-model gemini-3.5-flash \
  "Create a Python CFD simulation package that:
   1. Defines a 2D grid mesh using numpy arrays
   2. Implements a simple Navier-Stokes solver (lid-driven cavity)
   3. Uses finite difference method for spatial discretization
   4. Includes a Jacobi iterative pressure solver
   5. Outputs velocity and pressure fields as numpy arrays
   6. Includes visualization with matplotlib (contour plots)
   7. Has pytest tests validating conservation laws
   8. Uses Pydantic for simulation parameters (Re, dt, grid_size)"
```

Expected output structure:

```
/tmp/cfd-sim/
+-- pyproject.toml
+-- src/
|   +-- cfd_sim/
|   |   +-- __init__.py
|   |   +-- mesh.py          # Grid generation
|   |   +-- solver.py        # Navier-Stokes solver
|   |   +-- pressure.py      # Jacobi pressure solver
|   |   +-- visualization.py # matplotlib plotting
|   |   +-- parameters.py    # Pydantic config
|   +-- main.py
+-- tests/
    +-- test_mesh.py
    +-- test_solver.py
```

Wind Tunnel Data Analysis (Python)

Analyze experimental wind tunnel data:

```
perspt agent --yes -w /tmp/wind-tunnel \
  --actuator-model gemini-3.1-pro \
  --explorer-model gemini-3.5-flash \
  "Create a Python wind tunnel data analysis package that:
   1. Reads pressure tap data from CSV files
   2. Computes lift and drag coefficients (Cl, Cd) from pressure distributions
   3. Implements trapezoidal integration for force computation
   4. Generates polar plots of pressure coefficient (Cp) distribution
   5. Supports multiple angle-of-attack sweeps
   6. Includes Pydantic models for AirfoilData and TestConditions
   7. Exports results as JSON and matplotlib figures
   8. Has pytest tests with known NACA 0012 reference data"
```

Finite Element Solver (Rust)

Build a simple FEM solver in Rust:

```
perspt agent --yes -w /tmp/fem-solver \
  --actuator-model gemini-3.1-pro \
  --explorer-model gemini-3.5-flash \
  "Create a Rust finite element package that:
   1. Defines 2D triangular mesh elements
   2. Assembles global stiffness matrix for heat conduction
   3. Applies Dirichlet boundary conditions
   4. Solves using Gauss elimination (nalgebra)
   5. Outputs nodal temperatures
   6. Includes tests with known analytical solutions"
```

Tips for Scientific Tasks

1. **Be specific about algorithms** - Name the numerical method, discretization scheme, and solver type in the prompt
2. **Include validation criteria** - Reference known analytical solutions or benchmark data for tests
3. **Relax the descent gate** - Scientific tests often depend on numerical tolerances that need tuning after initial generation: `--rho-gate 0.1`
4. **Raise the loop bounds** - Scientific code often requires more iterations for convergence: `--max-turns 20 --rejection-budget 8`
5. **Review carefully** - Numerical correctness requires domain expertise beyond what the LSP can verify

See Also

- Headless Mode - Headless mode guide
- Agent Mode Tutorial - Interactive agent mode

Monitoring Agent Execution with the Dashboard

This tutorial walks through using the Perspt web dashboard to observe an agent session in real time.

Prerequisites

- Perspt installed (`cargo install --path crates/perspt-cli`)
- A project directory to work in

Step 1: Start an Agent Session

```
perspt agent -w ./myproject "Create a REST API in Rust"
```

The agent will begin planning and executing tasks. Leave this terminal running.

Step 2: Launch the Dashboard

Open a **new terminal** and run:

```
perspt dashboard
```

You should see:

```
Perspt dashboard listening on http://127.0.0.1:3000
```

Step 3: Open the Overview

Navigate to `http://localhost:3000` in your browser. The Overview page shows your active session with:

- **Status badge** - “running” while the agent works
- **Node count** - stable/total nodes projected from the session’s latest graph revision
- **Events** - how many ledger events the session has recorded

Step 4: Open the Session Detail

Click your session to open `/sessions/{id}`. The detail page shows the task, working directory, status, and detected toolchain, plus node totals, event and measurement counts, the last and average energy, and a per-node summary table.

Step 5: Explore the Topology

Click **Topology** (`/sessions/{id}/topology`). The page shows the work graph as the ledger recorded it:

- **Revision lineage** - every graph revision, newest marked “latest”
- **Nodes of the latest revision** - each node with its state badge (stable, running, stopped, blocked, retired)
- **Edge table** - parent -> child relationships

Step 6: Watch Energy Convergence

Click **Energy** to see the trajectory of measured candidates, in ledger order: one row per measurement with its node, generation, energy, and whether it was a hard pass. The summary bar shows the count, average, minimum, and maximum energy and how many measurements passed hard. Watch values decrease as accepted checkpoints descend.

Step 7: Check the Backlog

Click **Backlog** for the conditional-capacity diagnostics: node state counts, how many nodes sit in the backlog (and how many of those are still unmeasured), the backlog potential $\Phi(W)$, its drift, and a per-node energy table.

Step 8: Understand Decisions

The **Decisions** page is the raw PSP-9 event trace: a flat, Merkle-chained table with one row per ledger record — its sequence number, event kind, a truncated payload summary, and the short hash that chains it to its predecessor. This is the same record that `perspt replay` reconstructs.

Step 9: Review Governance

The **Governance** page shows the session’s authority epoch and grant status, recent calibration epochs, adjudication verdicts, and delayed audit samples still pending a label.

Step 10: Monitor Live Updates

The dashboard receives Server-Sent Events (SSE) from the server every 2 seconds. You can keep the browser open and watch as the agent progresses through its task.

When the agent completes, the session status changes to “completed” (green badge) on the Overview page.

User Guide

In-depth coverage of every Perspt feature.

Guide	Coverage
Basic Usage	TUI navigation, chat commands, and session management.
Agent Mode	Governed tool loop, work graph, measured verification, and review.
Advanced Features	Per-role model routing, acceptance gate, budgets, ledger, and resume.
Providers	Configure OpenAI, Anthropic, Gemini, Ollama, and more.
Model Context Protocol (MCP)	Connect governed external tools to agent mode and the chat TUI.
Troubleshooting	Diagnose and fix common issues.
Dashboard	Real-time web monitoring of agent sessions.

Basic Usage

Perspt offers three primary modes of interaction: the **TUI** (rich interactive terminal UI), **simple-chat** (plain-text streaming for pipeline integration), and **Agent** (self-stabilizing autonomous multi-agent coding workflows).

Launching the TUI

```
# Auto-detect provider from env keys
perspt

# Explicit model (provider comes from config or env detection)
perspt chat --model claude-fable

# Use a specific config file
perspt --config ./config.toml chat --model gpt-5.5
```

The TUI provides:

- Markdown rendering (code blocks, headers, lists, bold, italic)
- Premium ASCII table rendering with multi-line cell wrapping
- Inline LaTeX math transpilation to styled Unicode equations (bold and cyan)
- Real-time response streaming
- Scroll navigation
- Status bar (provider, model, streaming indicator, and reasoning toggles)

Keyboard Shortcuts

Key	Action
Enter	Send message

Key	Action
Ctrl+J	Insert newline in input
Page Up / Down	Scroll chat history by 10 lines
Shift+Up / Down	Scroll chat history by 1 line
Ctrl+Up / Down	Scroll chat history by 1 line
Home / End	Go to start / end of input line
Ctrl+C / **Ctrl+Q**	Quit
Ctrl+R	Toggle inner reasoning process display
Ctrl+A / **Ctrl+E**	Go to start / end of input line
Ctrl+B / **Ctrl+F**	Move cursor left / right
Ctrl+D / **Ctrl+H**	Delete / Backspace character
Ctrl+K / **Ctrl+U**	Kill to end / start of line
Ctrl+W	Delete word before cursor

Slash Commands (Directives)

Both CLI and TUI modes support slash commands typed directly in the input box:

Command	Description
`/exit` or `/quit`	Exit the application session
`/clear`	Reset the active conversation history
`/model <name>`	Switch the active LLM model on the fly
`/save <path>` (TUI only)	Export full conversation history to a markdown file
`/mcp` (TUI only)	Show MCP configuration state, discovery failures, policy rejections, and admitted operation names without contacting the model
`/mcp accept [JSON]`, `decline`, or `cancel` (TUI only)	Answer a pending MCP elicitation request
`/help`	Print the menu of available slash commands

The chat TUI can use latest MCP tools, resources, prompts, completions, sampling, elicitation, subscriptions, and tasks from any `[[external_tools]]` server entry in `config.toml` with `modes = ["chat"]` or `modes = ["agent", "chat"]` is considered for read-only admission. Every allowed remote tool also needs an exact local policy table. Tool activity is shown as a concise transient status only while a call is active. MCP protocol events, arguments, results, discovery notices, and subscription traces do not become conversation or reasoning content. The selected model chooses an admitted tool automatically when needed; `/mcp` is the explicit diagnostic surface. Tool results are labeled as untrusted content before returning to the model. **Ctrl+R** continues to control genuine model reasoning, including tool-aware turns. Unicode and multiline terminal clipboard paste is inserted at the input cursor. See Model Context Protocol (MCP) for configuration and the security boundary.

Simple CLI Mode

For a minimal text interface suitable for piping and logging:

```
perspt simple-chat
perspt simple-chat --model gpt-4o-mini
perspt simple-chat --log-file session.txt
```

Type your message, press Enter. Responses stream to stdout. Type `exit` or press `Ctrl+D` to quit.

Autonomous Agent Mode

For complex software engineering and coding jobs that require self-stabilizing, multi-file edits:

```
perspt agent "Implement a parsing logic for custom CSV formats in src/csv.rs"
```

This starts the multi-agent orchestration loop. The system automatically plans, executes, and verifies the changes using local verification commands. You can specify custom models and configure validation parameters to match your codebase. Refer to Agent Mode for details.

Provider Auto-Detection

Perspt checks environment variables in this priority order:

Env Variable	Provider	Default Model
`VERTEX_PROJECT_ID`	Vertex AI	`vertex::gemini-2.5-flash`
`GEMINI_API_KEY`	Gemini	`gemini-3.1-flash-lite-preview`
`OPENAI_API_KEY`	OpenAI	`gpt-4o-mini`
`ANTHROPIC_API_KEY`	Anthropic	`claude-3-5-sonnet-20241022`
`GROQ_API_KEY`	Groq	`llama-3.1-8b-instant`
`COHERE_API_KEY`	Cohere	`command-r-plus`
`XAI_API_KEY`	xAI	`grok-beta`
`DEEPSEEK_API_KEY`	DeepSeek	`deepseek-chat`
(none)	Ollama (local)	`llama3.2`

See Providers for full provider details.

Agent Mode

The agent command runs the governed PSP-9 tool loop for autonomous multi-file code generation. Agent mode is the current coding-domain implementation of the stability contracts described in Stability Papers in Agent Mode.

Launching Agent Mode

```
perspt agent -w <DIR> "<task>"

# Examples
```

```
perspt agent -w ./my-project "Create a Python REST API"
perspt agent -y -w /tmp/demo "Build a Rust CLI tool"
```

A run without a terminal (CI, pipes) requires `--yes`: promotion approval cannot be prompted without one, so the CLI fails fast instead of silently escalating a fully verified run. Any outcome other than a hard pass exits with a nonzero process status, so scripts never read a stopped node as success.

Core Workflow

The agent runs a closed loop in which every model proposal is governed and every acceptance is measured:

1. **Planning** - The default single-node run starts directly from a deterministic one-node graph. When `--max-parallel-nodes` is above one, one governed architect turn, restricted to `update_graph`, may decompose the task into a typed work graph; an empty or invalid proposal safely falls back to the single node.
2. **Proposal** - The actuator proposes typed tool calls (file writes, edits, commands) against a disposable candidate overlay. The real workspace is never touched by a proposal.
3. **Admission** - Every proposal passes the deterministic five-clause kernel — authority, contract, effect scope, barrier increment, and risk budget — before it takes effect. A denied call is recorded in the ledger and costs the model a correction, never the workspace a mutation.
4. **Measurement** - Admitted mutations are measured on the realized filesystem by the language plugins (compilers, tests, linters, LSP diagnostics), which produce the scalar energy $V(y)$. Tests evolve with the implementation by default. The optional `backward-compatible` policy adds a historical regression view, while `external-oracle` adds separately protected acceptance evidence; see Configuration Guide.
5. **Acceptance** - A candidate is accepted when it is a hard pass, or when it achieves the required measured descent $V(y) \leq V(\text{best}) - \rho_{\text{gate}}$ (`--rho-gate`, default `0.5`). A rejected candidate restores the best accepted checkpoint.
6. **Ledger** - Every proposal, denial, measurement, gate decision, and checkpoint is appended to a hash-chained ledger *before* it is used.

On a gate failure the runtime can open a bounded search forest: a small number of eager branches with exact-keyed no-goods, pre-action reservations that settle against observed actuals, and exactly one candidate committed through the ordinary gate. See Advanced Features for the `[exploration]` configuration.

Test Evidence Policies

Tests are part of the project being developed, not an automatically immutable artifact. The coding gate always runs the resulting project suite. The `[verification] test_policy` setting controls whether another test-evidence view is also required:

Policy	When to use it
`evolving` (default)	Feature work, refactoring, API migrations, and corrected behavior where implementation and tests legitimately evolve together. New tests run; changed or obsolete historical expectations are not resurrected.

Policy	When to use it
`backward-compatible`	Additive changes and bug fixes that promise recognized pre-existing test files will continue to pass. The resulting suite and the restored regression view must both pass.
`external-oracle`	Work with independently maintained acceptance evidence. Perspt copies the configured overlay onto a private candidate view and runs its command as an additional blocking test stage. Put withheld material outside the workspace.

The first policy says the project's configured suite passed for its resulting contract; it does not turn tests authored by the actuator into independent proof. The external policy supplies that independence when the user, CI, or a release process owns the acceptance suite. Configuration and complete examples are in Configuration Guide.

Windows Process Isolation

On native Windows, TUI chat and `simple-chat` require no special option. Coding remains strict by default and refuses to start verifier or inspection processes because Perspt does not yet implement a restricted-token/ACL sandbox. Use `--allow-unisolated` for a single acknowledged run, or configure `[verification] allow_unisolated = true` when durable resume is required. The candidate overlay, deterministic admission kernel, measurement gates, ledger, and promotion approval remain active, but child processes can access anything the host user can access. WSL provides the Linux sandbox path.

Multi-Node Dispatch

By default the work graph executes one node at a time. `--max-parallel-nodes` raises the number of concurrent nodes; values above 1 require `--yes` because promotion approval cannot be prompted per node while other nodes run. The dispatcher schedules by footprint conflict — two nodes whose declared file footprints overlap never run concurrently — and each node writes into content-addressed staging. Conflict detection is dependency-aware (downstream refinements win by edge precedence), and one global integration gate measures the merged result before anything is promoted.

Node Classes

Each work-graph node carries a class describing the kind of work it performs:

Class	Description
Explore	Read-only investigation of the workspace.
Plan	Task decomposition and graph revision.
Implement	Code changes toward the node goal.
Verify	Measurement-only verification work.
Test	Test authoring and execution.
Integrate	Merging staged work across nodes.
Repair	Recovery from a failed or rejected node.
Interface	Public API definitions, type signatures, traits.

The planner currently emits `Implement` nodes; the full taxonomy is the SDK contract that dispatch and the ledger record.

Review Modal (Interactive)

When running without `--yes`, the review modal presents the affected files, the verification gate results (syntax, build, tests, lint, plus test pass/fail counts), the scalar energy `V(x)`, and a degraded flag listing any sensors that could not run.

- **y** - Approve and promote
- **n** - Reject and restore the best checkpoint
- **c** - Send feedback for correction
- **e** - Open files in your editor
- **d** - Toggle full diff view
- **s** - Skip this review

`Left`/`Right` move the button selection, `Enter` confirms the selected decision, and `Esc` dismisses the modal.

Session Management

```
# Check session state (ledger, measurements, energy, gate, denials)
perspt status

# Abort the current session by revoking its authority epoch
perspt abort

# Resume the most recent session
perspt resume --last
```

`perspt status` prints the ledger event count and head, measurement and denial counts, the last energy and gate decision, search forest, branch, and no-good counts, adjudicator verdicts, the capacity $\Phi(W)$, and validator independence statistics. `perspt resume` prints the session id, task, working directory, and status, then verifies the ledger chain before continuing; live authority is never serialized, so resume always mints fresh capabilities.

Domains

`--domain` selects the domain package (`coding`, `research`); by default the best-matching domain is detected from the workspace, with `coding` as the fallback.

Exploration-Only Runs

`--exploration-only` runs only the read-only exploration phase: a deterministic repository map plus an interactive explorer tool loop. Nothing is mutated or promoted, and the run never prompts, so it works without `--yes` even in non-interactive contexts.

Headless Summary

In headless mode the agent prints the terminal summary lines — outcome, session id, turns used, ledger head, and promoted paths:

```
Outcome: HardPass
Session: 019820f3-...
Turns: 6
Ledger head: 4f2c9a1b8d7e6a50
Promoted paths: src/api.py, tests/test_api.py
```

`--output-summary <FILE>` writes the same summary as JSON for CI integration.

Dashboard Monitoring

While an agent session runs, you can observe it in a browser:

```
# In a separate terminal
perspt dashboard
```

Open `http://localhost:3000` to see the Overview, Session detail, Topology, Backlog, Energy, Decisions, and Governance pages. The Decisions page is a flat Merkle-chained event trace. The dashboard reads the session store in read-only mode so it never interferes with the running agent.

See Dashboard for full details.

Live Dashboard Monitoring

Use `--dashboard` to start the web monitoring dashboard alongside the agent:

```
perspt agent --dashboard "Build a REST server"
# Open http://127.0.0.1:3000 in a browser
```

The embedded dashboard reads through the same database connection the agent writes to, providing real-time views of work-graph topology, energy convergence, and the ledgered decision trace. Use `--dashboard-port` to change the port. The server stops when the agent exits.

> **Note**
> The embedded dashboard runs unauthenticated on `127.0.0.1`. It is intended for local monitoring only; do not expose the port beyond the local machine.

See Advanced Features for per-role model routing, the acceptance gate, budgets, and search configuration, and Agent Options Reference for the full flag reference.

Advanced Features

Per-Role Model Routing

The runtime routes each role through the `[models]` table of the config file. Values are fully qualified `provider::model` routes so identity, calibration, and replay never depend on an ambient default provider:

```
[models]
architect = "vertex::gemini-3.1-pro"
actuator = "vertex::gemini-3.5-flash"
verifier = "vertex::gemini-3.1-pro"
speculator = "vertex::gemini-3.5-flash-lite"
adjudicator = "vertex::gemini-3.1-pro"
```

The `speculator` entry is consumed only as the fallback route for the explorer role. On the command line, three roles can be overridden per run:

```
persp agent \
  --actuator-model vertex::gemini-3.5-flash \
  --explorer-model vertex::gemini-3.5-flash-lite \
  --adjudicator-model vertex::gemini-3.1-pro \
  -w ./project "Task description"
```

`--model` is an alias for `--actuator-model`. `--fallback-model` adds an ordered sticky failover route for the actuator; repeat the option to add more routes. Role-specific models are never silently reused as actuator fallbacks:

```
persp agent \
  --actuator-model vertex::gemini-3.5-flash-lite \
  --fallback-model vertex::gemini-3.5-flash \
  --fallback-model vertex::gemini-3.1-pro \
  -w ./project "Task"
```

The adjudicator, when configured, is a no-tool validator of the realized diff. Its verdict is an uncalibrated conjunctive veto, never an energy measurement.

The Acceptance Gate

A candidate is accepted when it is a hard pass — every sensor clean — or when it achieves the required measured energy descent

$$V(y) \leq V(\text{best}) - \rho_{\text{gate}}.$$

The energy is measured on the realized filesystem by the language plugins (compilers, tests, linters, LSP diagnostics), never estimated by a model. `--rho-gate` sets the required descent per accepted checkpoint:

```
# Default: rho = 0.5
persp agent --rho-gate 0.5 -w . "Precise task"

# Accept smaller measured improvements
persp agent --rho-gate 0.25 -w . "Incremental refactor"
```

A rejected candidate restores the best accepted checkpoint, so the workspace never regresses past a measured state.

Budgets

The loop is finite by construction. Each budget is a hard bound, and exhausting one escalates with a residual certificate rather than looping:

Flag	Description
`--max-turns <N>`	Maximum model turns per node (default 12).
`--max-calls-per-turn <N>`	Maximum direct and nested tool calls per turn (default 8).
`--rejection-budget <N>`	Shared non-descending and recovery budget (default 4).
`--max-parallel <N>`	Maximum independent compiler, test, and lint sensors (default 4).
`--max-parallel-nodes <N>`	Concurrent work-graph nodes (default 1; above 1 requires `--yes`).

Bounded Search

On a gate failure the runtime can open a bounded search forest instead of retrying blindly. Branches are sequential and eager, with a hard cap of three identities per forest; every dead end is recorded as an exact-keyed no-good so it is never re-proposed, every branch takes pre-action reservations that settle against observed actuals, and exactly one candidate is committed through the ordinary acceptance gate. The `[exploration]` config block bounds the search:

```
[exploration]
initial_branches = 1     # branches opened before any expansion trigger
max_branches = 3         # branch identities per forest (hard cap 3)
distinct_family = true   # prefer a distinct model family on expansion
max_workspace_files = 2000
max_workspace_bytes = 50000000
```

`distinct_family` is a prior, never a certificate. The workspace caps bound the cumulative eager-copy file and byte reservations. Search activity appears in `perspt status` as forest, branch, committed, and no-good counts.

Prompt Programs

Every model call is compiled from typed prompt sections into a per-call prompt program. The program's route, dialect, section provenance, and tool-surface hash are ledgered, so a session's exact prompts can be audited after the fact:

```
# List every compiled section: id, version, stage, role, hash
perspt prompts list

# Show the programs a session actually compiled, with digests
perspt prompts explain-session --db-path <FILE> <SESSION_ID>
```

`--allow-experimental-prompts` substitutes validated `[prompts]` bundle sections live. This is Gate AE and remains experimental until a change record passes paired evaluation.

Resident Context

The conversation the model sees is paged: content is stored as content-addressed pages, and the transported conversation carries tombstones in place of folded content. The composed request is checked against the route's input allowance and dialect byte limit before any call, and the model recalls folded pages only through the governed `context_recall` tool. Compactions and refusals are recorded in the ledger:

```
# Show a session's recorded context events (compactions, refusals)
perspt context explain-turn --db-path <FILE> <SESSION_ID>
```

Merkle Ledger

Every stable node is committed to a content-addressed Merkle ledger stored in DuckDB. This provides:

- **Auditability** - Full trace of what each node produced
- **Rollback** - Restore to any point in the session
- **Resume** - Continue interrupted sessions with verified context

```
perspt ledger --recent
perspt ledger --stats
```

Run Summaries

`--output-summary <FILE>` writes the terminal run summary as JSON (session id, node id, outcome, turns used, ledger head, promoted paths) for CI integration. Any outcome other than a hard pass exits with a nonzero process status:

```
perspt agent -y --output-summary summary.json -w . "Task"
cat summary.json
```

Resume and Replay

```
# Resume the most recent session
perspt resume --last

# Credential-free audit replay of a session
perspt replay <SESSION_ID>
```

Resume verifies the ledger chain, rebuilds staging by folding the ledger, and re-enters dispatch. Live authority is intentionally never serialized: resume always rechecks the durable authority epoch and mints fresh capabilities. `--persistent-grants` on the original run stores signed grant intent so an unattended resume can complete bracketed promotion intents.

Exploration-Only Runs

`--exploration-only` runs only the read-only exploration phase: a deterministic repository map plus an interactive explorer tool loop. Nothing is mutated or promoted:

```
perspt agent --exploration-only -w . "Map the request handling path"
```

Governed Dependency Mutation

Dependency changes (`cargo add`, `uv add`, `npm install`) are an external effect and are denied by default. `--allow-dependency-mutation` grants them, still subject to the kernel and the ledger:

```
perspt agent --allow-dependency-mutation -w . "Add pandas and use it"
```

Providers

Perspt supports multiple LLM providers through the `genai` Rust client crate, which provides unified access to all major commercial, open-source, and cloud-provider model APIs.

Supported Providers and Adapters

Perspt accepts nine provider identifiers, each routed to the matching `genai` adapter (any other value is rejected as unsupported):

Provider / Adapter	Adapter Kind ID	Reference Model	Notes
OpenAI	`openai`	`gpt-5.5`	SOTA GPT-5.5, GPT-5-mini
Anthropic	`anthropic`	`claude-fable`	Claude Fable, Opus 4.8, Sonnet 4.6, Haiku 4.6
Google Gemini	`gemini`, `google`	`gemini-3.5-flash`	Gemini 3.5 Flash, 3.1 Pro, 3.1 Flash-Lite
Google Vertex AI	`vertex`	`vertex::gemini-3.5-flash`	Google Cloud Vertex platform
Groq	`groq`	`llama-3.3-70b-specdec`	Ultra-low latency Llama/Gemma on LPU
Cohere	`cohere`	`command-a-plus`	Command A+, North Mini Code
xAI	`xai`	`grok-4`	Grok 4 family
DeepSeek	`deepseek`	`deepseek-v4`	DeepSeek v4 models (Chat, Coder)
Ollama	`ollama`	`llama3.3`	Local offline models

Configuration Methods

1. Environment Variables (recommended):

```
export GEMINI_API_KEY="your-key"
perspt
```

2. CLI Flags:

```
perspt chat --model gpt-5.5
```

3. Config File (`config.toml`):

```
provider = "anthropic"
model = "claude-fable"
```

Agent mode can additionally bind several credentials and routes at once through the `[providers.<id>]` and `[models]` tables, which hold fully qualified `provider::model` routes for multi-route portfolios. See Agent Mode for details.

Provider-Specific Notes

OpenAI

```
export OPENAI_API_KEY="sk-xxx"
perspt chat --model gpt-5.5
```

Azure OpenAI (via OpenAI Compatible)

Azure OpenAI requires configuring the base URL override and the API key:

```
export OPENAI_API_KEY="your-azure-key"
export OPENAI_BASE_URL="https://your-resource.openai.azure.com/openai/deployments/your-
```

```
deployment"
persp chat --model gpt-5.5
```

Anthropic

```
export ANTHROPIC_API_KEY="sk-ant-xxx"
persp chat --model claude-fable
```

Google Gemini

```
export GEMINI_API_KEY="AIza..."
persp chat --model gemini-3.5-flash
```

Google Vertex AI

Vertex AI requires your Google Cloud project ID and a location (optional, defaults to `global`). Authentication is typically handled via Google Application Default Credentials (ADC); setting `VERTEX_API_KEY` to a bearer token overrides the ADC token.

```
export VERTEX_PROJECT_ID="my-gcp-project-123"
export VERTEX_LOCATION="us-central1"
# Run using Vertex model prefix
persp chat --model vertex::gemini-3.5-flash
```

Ollama (Local)

```
ollama serve
ollama pull llama3.3
persp chat --model llama3.3
```

No API key required. Persp auto-detects Ollama as the fallback provider.

Model Context Protocol (MCP)

Persp is an optional MCP client for the governed coding agent and the interactive chat TUI. The implementation uses the official Rust MCP SDK and speaks **MCP 2026-07-28 only**. It starts with `server/discover` and does not fall back to `initialize` or an older protocol version. A server that does not advertise 2026-07-28 is rejected.

The client supports stdio and stateless Streamable HTTP, including:

- tools, resources, resource templates, prompts, and completions;
- explicit roots disclosure;
- opt-in model sampling and user elicitation;
- progress, logging, cancellation, catalog-change, resource-update, task, and custom notifications;
- `subscriptions/listen` with list-change and resource filters;
- multi-round tool results and the tasks extension for long-running calls.

MCP is absent when no `[[external_tools]]` table is configured. It does not replace Persp's coding-domain tools, planner, verifier, candidate workspace, or acceptance gate.

Authority and Trust

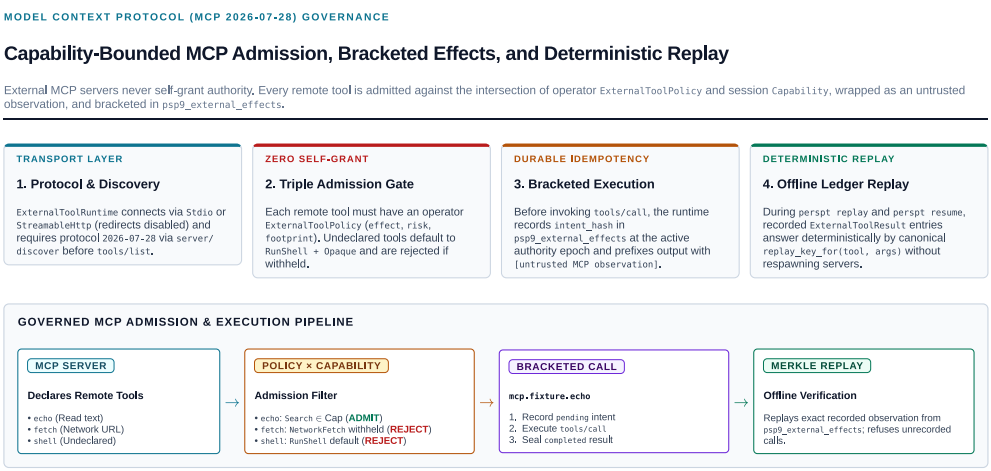


Fig. 4 MCP 2026-07-28 strict discovery handshake, local policy admission, and write-ahead bracketed tool execution.

A server description, schema, prompt, resource, notification, or result is untrusted input. A server cannot grant itself authority. Each remote tool must match a user-owned policy assigning its effect, risk, and scheduling footprint. Undeclared tools fail closed. Agent calls still pass the capability kernel and are write-ahead bracketed; replay uses the recorded result and never reconnects.

Resources and prompts are exposed through namespaced, locally defined read-only operations only when the server advertises the capability:

```
mcp.<server>._perspt_resources_list
mcp.<server>._perspt_resource_templates_list
mcp.<server>._perspt_resource_read
mcp.<server>._perspt_prompts_list
mcp.<server>._perspt_prompt_get
mcp.<server>._perspt_complete
```

Their output is returned as untrusted data and is never silently inserted into the system prompt. Remote names beginning with `_perspt_` are reserved and rejected, preventing collisions with these client operations. `_perspt_prompt_get` accepts optional, equally sized `argument_names` and `argument_values` string arrays; entries at the same index form one MCP prompt argument. Each array is capped at 64 entries.

Quick Setup

This stdio example enables independent agent and chat lifecycles, discloses one root, permits sampling and elicitation, and subscribes to one resource:

```toml
[[external_tools]]
id = "docs"
transport = "stdio"
command = ["company-docs-mcp", "--stdio"]
modes = ["agent", "chat"]
timeout_ms = 30000
```

```
max_result_bytes = 1048576
max_stderr_bytes = 65536
max_task_wait_ms = 300000
env_from_env = { DOCS_TOKEN = "COMPANY_DOCS_TOKEN" }

roots = [{ uri = "file:///absolute/path/to/project", name = "Project" }]
sampling = true
max_sampling_tokens = 4096
elicitation = true
subscriptions = true
resource_subscriptions = ["file:///absolute/path/to/project/guide.md"]
tasks = true

[external_tools.tools.search]
effect = "search"
risk = "low"
footprint = { selectors = [{ kind = "scoped_argument", family = "company-docs", field =
"query", access = "read" }] }
```

The policy key must exactly match the remote name. A footprint field must exist in the advertised input schema. After admission the model-facing tool is `mcp.docs.search`.

For Streamable HTTP use these transport fields:

```
[[external_tools]]
id = "records"
transport = "streamable_http"
url = "https://tools.example.test/mcp"
modes = ["agent", "chat"]
headers_from_env = { Authorization = "RECORDS_AUTHORIZATION" }

[external_tools.tools.lookup]
effect = "data_read"
risk = "low"
footprint = { selectors = [{ kind = "scoped_argument", family = "records", field = "id",
access = "read" }] }
```

Set `RECORDS_AUTHORIZATION` to the complete server value, for example `Bearer ...`. Non-loopback endpoints require HTTPS and redirects are disabled. Streamable HTTP is stateless under 2026-07-28; Perspt does not recover by negotiating a legacy session.

Chat TUI

Run `perspt --config ./config.toml chat` and type `/mcp`. It reports discovery failures, policy rejections, and admitted operations. `/help` includes MCP and elicitation commands. The model calls tools automatically; `/mcp` performs no remote call.

Normal chat starts without discovery or policy messages in the conversation. When the model selects an admitted operation, the TUI shows one transient, human-readable activity label and then the final answer. Raw MCP lifecycle events, tool arguments, results, subscription notifications, and server logs stay out of both the conversation and the reasoning panel. Use `/mcp` when you explicitly want discovery and admission diagnostics. **Ctrl+R** remains the reasoning control and displays only reasoning emitted by the selected model; tool-aware Qwen turns use the same reasoning stream.

Chat has a fixed read-only authority ceiling. Mutating, shell, network-fetch, dependency, graph, and policy effects are rejected even when the same tool is admissible in the governed agent. `perspt simple-chat` intentionally creates no MCP lifecycle.

For form or URL elicitation the input remains active and the TUI shows the request as JSON. Respond explicitly with:

```
/mcp accept {"field":"value"}
/mcp accept
/mcp decline
/mcp cancel
```

Perspt does not open an elicitation URL automatically. Form responses are JSON objects. Perspt advertises client-side schema validation as disabled, leaving the server responsible for validating the requested schema.

Terminal bracketed paste is enabled in both chat input paths. Normal terminal paste shortcuts (Command-V, Ctrl-Shift-V, or Shift-Insert, depending on the terminal) insert Unicode and multiline clipboard text at the cursor. The mode is disabled again on TUI exit.

Agent Behavior

Agent servers are discovered when a session assembles its node catalog. External schemas use the ordinary deferred tool-search path. Proposals, arguments, effects, grants, execution, results, and uncertain completion are governed like the rest of the coding tool plane.

Sampling uses the already selected local route. MCP model preferences cannot redirect credentials or choose an unconfigured provider. Perspt forwards `maxTokens`, temperature, stop sequences, tools, and tool choice. It advertises tool-aware sampling but not `includeContext`; requests for implicit MCP-server context are rejected.

The command-line agent has no interactive MCP form surface, so an enabled elicitation request receives `decline` immediately. Chat is interactive. SDK hosts may install their own `McpElicitationProvider`.

Subscriptions and Dynamic Catalogs

With `subscriptions = true` (the default), Perspt intersects its filter with server capabilities before opening `subscriptions/listen`. Tool-list changes invalidate the SDK cache; the shared runtime removes old bindings and repeats local admission before a chat model turn. A removed tool cannot remain callable. Resource and prompt changes, progress, logging, tasks, cancellation, and custom extension notifications are typed `McpServerEvent` values for product display or SDK consumption.

`resource_subscriptions` contains exact URIs and should be narrow. An empty list still permits supported tools/prompts/resources list-change filters.

Multi-Round Results and Tasks

Tool calls accept complete results, bounded multi-round input-required responses, or asynchronous task handles. Inputs may request sampling, elicitation, or roots only when locally enabled; unknown kinds fail closed.

Every wire request uses `timeout_ms`. `max_task_wait_ms` is an overall deadline across task polling and input rounds (default five minutes), and poll hints are clamped. A failed, cancelled, malformed, or expired task becomes an uncertain external completion for agent reconciliation.

Configuration Reference

Table 25 `[[external_tools]]` fields

Field	Meaning
`id`	Required ASCII id used in `mcp.<id>.<operation>`.
`transport`	Exactly `stdio` or `streamable_http`.
`command` / `url`	Direct argv for stdio or HTTP endpoint; mutually exclusive.
`modes`	`["agent"]`, `["chat"]`, or both. Omission means agent only.
`env_from_env`	Stdio child variable to source-variable mapping. Children otherwise receive only `PATH` and Windows `SystemRoot`.
`headers_from_env`	HTTP header to source-variable mapping; values stay out of TOML.
`timeout_ms`	Per-request and discovery deadline; default 30000.
`max_result_bytes`	Maximum accepted serialized result and raw SSE event; default 1 MiB.
`max_stderr_bytes`	Rolling stdio stderr tail; default 64 KiB.
`roots`	Explicit absolute `file://` roots. Empty means unadvertised.
`sampling` / `max_sampling_tokens`	Opt in to server model calls and cap one request (default 4096).
`elicitation`	Opt in to form/URL elicitation; interactive in chat, declined by the non-interactive agent.
`subscriptions` / `resource_subscriptions`	Enable the listen stream (default true) and exact resource filters.
`tasks` / `max_task_wait_ms`	Enable task/MRTR handling (default true) and its overall deadline (default 300000).
`tools`	Local policy tables keyed by exact remote tool name.

Each policy accepts `effect`, `risk`, `footprint`, and optional `proposal_bindings`. Omitting effect or footprint becomes a high-risk opaque shell declaration and is not admitted by default.

Security Notes

A configured stdio command is user-selected code, and an HTTP server acts outside the candidate workspace. Admission constrains calls but does not sandbox the server. Trust its implementation/operator and grant the narrowest policy.

Result objects are size-checked after protocol decoding and SSE events before decoding. Stderr capture, subscription filters, MRTR rounds, task duration, sampling tokens, and chat tool rounds are bounded. The official SDK owns framing, correlation, cancellation, and dispatch; Perspt owns transport security, capabilities, admission, product interaction, and replay policy.

Troubleshooting

Symptom	Check
`/mcp` says disabled	Add `"chat"` to `modes` and restart chat.
Protocol is incompatible	Upgrade the server to 2026-07-28; there is no legacy fallback.
No remote tools are admitted	Add exact policies and schema-valid footprints. Resource/prompt operations may still appear when advertised.

Symptom	Check
Stdio program or credential is missing	Verify direct argv and map variables with `env_from_env`.
HTTP is refused	Use HTTPS outside localhost and check header source variables.
Sampling is rejected	Enable it, stay under the token cap, and omit unadvertised context.
Elicitation is declined in agent mode	Use chat or install an SDK host provider.
A task times out	Increase the overall deadline deliberately or fix the server.

Troubleshooting

API Key Issues

Symptom: "No API key found" error.

Solutions:

1. Check env var is exported: `echo $GEMINI_API_KEY`
2. Verify spelling: `ANTHROPIC_API_KEY` (not `ANTHROPIC_KEY`)
3. Store it in the config: `perspt config --set api_key="your-key"`
4. Check config file: `~/.config/perspt/config.toml`

Connection Errors

Symptom: "Connection refused" or timeouts.

Solutions:

1. Check internet connectivity
2. For Ollama: ensure `ollama serve` is running
3. Check firewall/proxy settings
4. Try a different provider

Agent Mode Issues

Agent stuck in retry loop:

1. Check tool prerequisites: `which uv`, `which cargo`, `which node`
2. Check LSP is functioning: `ty check .` or `cargo check`
3. Relax the descent gate: `--rho-gate 0.3`
4. Raise the retry allowance: `--rejection-budget 8`
5. Check `perspt status` for gate decisions and denials

High energy despite clean code:

1. Run tests manually: `uv run pytest -v` or `cargo test`
2. Check for LSP diagnostics: `ty check .`
3. Bound runaway turns: `--max-turns 8 --max-calls-per-turn 4`
4. Verify contract compliance

Plugin not detected:

1. Ensure required binaries are installed in PATH

2. Check workspace has expected marker files (`Cargo.toml`, `pyproject.toml`)
3. Run `perspt status` to see ledger counters, last energy, and gate state

TUI Rendering Issues

Symptom: Garbled output, incorrect colors.

Solutions:

1. Ensure terminal supports 256 colors: `echo $TERM`
2. Try a different terminal emulator
3. Fallback to simple CLI: `perspt simple-chat`
4. Check for conflicting terminal multiplexer settings

Degraded Verification

When tool binaries (`ty`, `cargo`, `pytest`) are missing, checkpoints may be flagged as **degraded** in the review modal, with the reasons listed alongside the flag. The agent keeps working, but with lower verification confidence.

To restore full verification, install the required tools:

```
# Python projects
pip install ty pytest

# Rust projects
rustup component add rust-analyzer

# Node.js projects
npm install -g typescript
```

Session Recovery

If a session is interrupted:

```
# Check what's in progress
perspt status

# Resume the last session (shows trust context first)
perspt resume --last

# Or abort and start fresh
perspt abort
```

If the session store itself refuses to open because of a poisoned DuckDB write-ahead log, back it up and quarantine it (the WAL is never deleted):

```
perspt db repair --db-path <path-to-db> --discard-wal
```

Getting Help

```
perspt --help
perspt agent --help
perspt chat --help
```

For more details, see:

- CLI Reference - Full CLI reference
- Advanced Troubleshooting - Advanced troubleshooting

Dashboard

The Perspt web dashboard provides real-time browser-based monitoring of agent execution. Launch it alongside a running agent session to observe DAG topology, energy convergence, backlog diagnostics, governance state, and decision traces.

Launching the Dashboard

In a separate terminal from your agent session:

```
perspt dashboard
```

This starts an Axum web server on `http://127.0.0.1:3000`. Open that URL in your browser. The dashboard reads the DuckDB session store in read-only mode, so it can run safely alongside the agent.

To use a different port:

```
perspt dashboard --port 8080
```

Dashboard Pages

Overview - Lists all recent sessions with status badges (running, completed, failed), node completion counts, and budget consumption. Click any session to drill into its sub-pages.

DAG Topology - Shows node cards colored by state (green for committed/verified, red for failed, blue for running). Displays the task graph edges below. Useful for understanding the agent's decomposition of work.

Backlog - Conditional-capacity diagnostics over the PSP-9 ledger: counts of latest-revision nodes per state and the last measured energy of each backlog node. Every number is a diagnostic projection, not a stability claim.

Energy Convergence - Displays the measured energy trajectory: one row per candidate measurement with its sequence, node, generation, energy value, and hard-pass flag. A summary shows the average, minimum, and maximum energy plus the hard-pass count.

Decision Trace - The raw PSP-9 event trace: a flat table of Merkle-chained ledger records, one row per event with its sequence number, kind, summary, and short hash.

Governance - Calibration epochs with their state, target rho, threshold, sample count, and model route, plus validator verdicts and pending delayed audits.

Authentication

The dashboard binds to `127.0.0.1` only and runs without authentication. This is safe for local development: the server is not reachable from other machines, and it opens the DuckDB session store in read-only mode.

Using with a Running Agent

The typical workflow:

1. Start an agent session:

   ```
   perspt agent -w ./myproject "Create a REST API server"
   ```

2. In another terminal, launch the dashboard:

```
perspt dashboard
```

3. Open `http://localhost:3000` in your browser.
4. The dashboard updates via Server-Sent Events (SSE) every 2 seconds, showing live node state changes as the agent works.

Viewing Historical Sessions

The Overview page lists sessions 20 per page, with pagination controls to browse further back. Sessions from past agent runs remain in the DuckDB database and can be browsed even after the agent has stopped.

Concepts

Understanding the foundations of Perspt.

Concept	Scope
SRBN Architecture	The theoretical framework and its experimental implementation in Perspt.
Stability Papers in Agent Mode	How the three Stability papers shape PSP-8, PSP-9 (Final), PSP-10, and agent mode.
Workspace Crates	The fourteen-crate modular architecture.
PSP Process	Perspt Specification Proposals for design and development.

SRBN Architecture

The **Stabilized Recursive Barrier Network (SRBN)** is the idea behind Perspt's experimental autonomous coding agent. It comes from the *Stability is All You Need* paper series.

The idea is plain to state. A language model is good at proposing changes, but a proposal is only a guess. Left unchecked, a long run of guesses drifts: small mistakes pile up until the work is broken. SRBN refuses to trust a guess on its word. It measures every proposed change, keeps the change only when the measurement shows progress, and writes the kept result to a permanent record. Perspt's coding agent implements this work via a reusable software-development-kit (SDK) platform (originally specified in the references section under PSP-8).

> **Theory vs. Implementation**
>
> This page describes both the SRBN paper series and how Perspt's runtime implements it. Where a claim comes from the papers' formal proofs, it is noted as a **paper result**. Where the developer team makes engineering choices that approximate or extend the theory, those are noted as **implementation details**. The theoretical framework is mature; empirical benchmarks on Perspt's implementation have not yet been published.

Overview

The SRBN paper models coding tasks as a directed acyclic graph (DAG) of nodes with a sheaf structure that enforces consistency across shared boundaries. The system establishes the following concrete mechanics: each node owns a set of output files (ownership closure), runs a governed tool loop of typed tool calls against an isolated candidate, and must pass the measured acceptance gate on deterministic verifier evidence. Only then is the node's winner promoted and committed to the Merkle ledger.

The runtime environment structures execution in two primary ways:

- **Quadratic energy.** Acceptance is gated on the quadratic residual energy $V(x) = \sum_e w_e \parallel r_e \parallel^2$ (see Lyapunov Energy below).
- **Mutable work graph.** Rather than walking a precomputed topological order, a closed-loop scheduler re-evaluates a dependency-aware ready queue each round and may requeue, split, insert, or replan nodes as verifier evidence arrives. Each individual graph *revision* stays acyclic. Bounded parallelism shipped in v0.6.6: the multi-node dispatcher behind `--max-parallel-nodes` runs ready nodes concurrently, and two nodes with conflicting file footprints never run at the same time.

The core concepts of the control system — ownership closure, typed tool calls, the verifier profiles, node classes, and the Merkle ledger — are described below.

The Control Loop

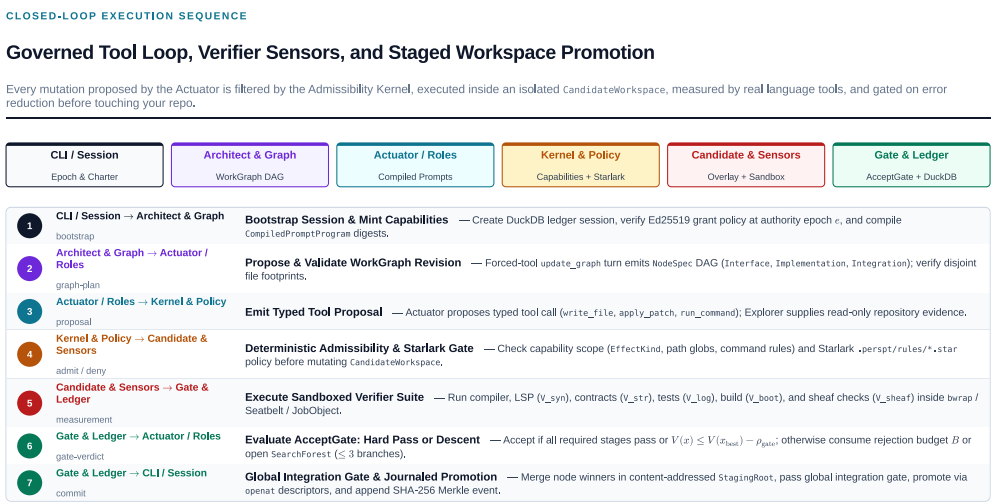


Fig. 5 Sequence of the PSP-9 / PSP-10 governed candidate control loop across Architect, Actuator, Admissibility Kernel, Candidate Overlay, Deterministic Verifier, Accept Gate, and Merkle Ledger.

Bootstrap and the architect turn run once at the start; the remaining phases run **per node inside the closed loop**. The dispatcher re-evaluates the mutable work graph, fills free slots with ready nodes, and runs the governed tool loop for each. A node that fails its gate can open a bounded search forest and be re-attempted, so these phases are *not* a single topological pass:

#	Phase	Description
1	**Session Bootstrap**	Create the durable session, detect the domain package and language plugins, mint epoch-bound capabilities from the grant policy, and compile the prompt programs for the resolved routes and dialects.
2	**Architect Turn**	A governed, forced-tool-choice model turn restricted to the privileged `update_graph` tool proposes the work-graph revision. Acyclicity and completeness are validated deterministically before the revision is accepted; a single-node run keeps one node without an architect call.
3	**Work-Graph Dispatch**	Behind `--max-parallel-nodes`, the dispatcher fills free slots with ready nodes. Conflicting file footprints never run concurrently, and each node attempt is

#	Phase	Description
		bound to a generation seeded from the current staging root, never from unstaged sibling state.
4	**Governed Tool Loop**	Per node, model turns issue typed tool calls. Each call becomes a proposal; the deterministic admissibility kernel decides whether it may affect the candidate; admitted effects mutate an isolated candidate overlay; sandboxed deterministic verifiers re-measure the candidate.
5	**Accept Gate**	A candidate is accepted on a hard verifier pass, or on a measured energy descent of at least ρ_{gate} below the best accepted state. Non-descending candidates consume the shared rejection budget; the gate is evaluated on the re-measured candidate, never on the model's account of it.
6	**Bounded Search** (optional)	On gate failure with budget remaining, a bounded search forest runs up to three isolated branches with distinct strategies against the accepted root. Every branch action reserves its cost before it runs, exact no-goods suppress repeats, and exactly one selected candidate is committed through the same gate.
7	**Staging & Integration**	Each node's winner lands in a content-addressed staging root instead of the user workspace. When dispatch settles, the merged state passes one global integration gate; failure restores the prior staging root.
8	**Promotion**	The integrated winner is written to the user workspace through journaled, descriptor-relative promotion and committed to the hash-chained ledger, from which `perspt replay` and `perspt resume` reconstruct the session.

Lyapunov Energy

The stability of generated code is measured using a Lyapunov energy function, adapted from the paper's sheaf-theoretic formulation into concrete verification barriers. The system evaluates a canonical **quadratic residual energy**: each sensor e emits a residual with a non-negative magnitude $r_e(x)$, and the energy is a weighted sum of their squares.

Let $\mathcal{E}$ be the set of active sensors monitoring the system. For a proposed state x, the total Lyapunov energy $V(x)$ is defined by the quadratic form:

$$V(x) = \sum_{e \in \mathcal{E}} w_e \parallel r_e(x) \parallel^2$$

where $w_e > 0$ represents the positive weight assigned to the residual class of sensor e.

The five component readouts V_{syn}, V_{str}, V_{log}, V_{boot}, and V_{sheaf} are **derived rollups** of this same energy, grouped by component type, such that the total is the sum of the rollups:

$$V(x) = V_{\text{syn}} + V_{\text{str}} + V_{\text{log}} + V_{\text{boot}} + V_{\text{sheaf}}$$

In this formulation, the rollups themselves carry the squared, weighted residual contributions:

$$V_{\text{comp}} = \sum_{e \in \text{ comp}} w_e \parallel r_e(x) \parallel^2$$

There is no secondary $\alpha/\beta/\gamma$ aggregation pass. The per-class weights w_e are fixed by the domain package's energy model, leaving the core mathematical engine as a pure sum of pre-weighted squares.

7.1.3.1 Components

We enumerate the five component categories and the specific residual classes mapped to them from the *perspt-coding* domain package:

Component	Mapped Residual Classes (Weights)	Operational Semantics
V_syn	• `Syntax` (weight: 4.0) • `Type` (weight: 3.0) • `Build` (weight: 3.0)	Captures syntax errors, build-time compilation failures, and LSP diagnostic warnings. Compiler warnings or type-check diagnostics produce residuals of class `Type`, where the magnitude is the raw diagnostic count. A build failure raises a blocking residual of class `Build`.
V_str	• `ImportGraph` (weight: 2.0) • `SymbolMismatch` (weight: 2.0) • `InterfaceMismatch` (weight: 2.5) • `OwnershipViolation` (weight: 2.0) • `Policy` (weight: 1.0) • `Dependency` (weight: 1.5) • `Manifest` (weight: 1.5) • `Format` (weight: 0.25)	Measures structural contract adherence. If a required symbol defined by a node's public interface signature is absent from the generated output files, the `GoalPresence` sensor registers a blocking `SymbolMismatch` residual, preventing convergence. Policy and format irregularities are likewise squared into this component.
V_log	• `TestFailure` (weight: 2.0) • `Runtime` (weight: 2.0) • `Regression` (weight: 3.0)	Evaluates behavioral outcomes. Unit test failures yield a `TestFailure` residual with a magnitude matching the count of failing test cases. Post-build runtime smoke probes detect process crashes, tracebacks, or numeric anomalies, adding a `Runtime` residual.
V_boot	• `SensorUnavailable` (weight: 1.0) • `ToolFailure` (weight: 1.0)	Registers system infrastructure state. If a required sensor (e.g., an LSP server or test runner) is degraded or unavailable, it raises a `SensorUnavailable` residual, forcing the energy high so the system knows its stability status is indeterminate.
V_sheaf	• `SheafInconsistency` (weight: 2.0)	Evaluates global structural consistency across nodes. Cross-node import/dependency mismatches or signature differences raise a `SheafInconsistency` residual.

7.1.3.2 Convergence Criterion

The system is considered stable if and only if the candidate achieves a **hard pass**: every required verifier and every hard policy constraint passes. A domain may additionally declare an analytic energy floor; a candidate at or below the floor is checkpointed as a classified terminal state but is never reported as verified success. If the state is not stable, the accept-gate evaluates descent. The single runtime acceptance knob is `--rho-gate`. A state x can be provisionally accepted during convergence if:

$$V(x) \leq V(x_{\text{best}}) - \rho_{\text{gate}}$$

where x_{best} is the best previously accepted state, and ρ_{gate} is the minimum descent gate (default 0.50), ensuring non-trivial progress.

7.1.3.3 Spectral and Independence Diagnostics

The system supports an energy-slope constant $\mu = 2\,\lambda^{+}_{\min}(A)$ — twice the algebraic connectivity (Fiedler value) of the verification graph built from the quadratic energy $V(x) = x^{\top}Ax$. It measures how strongly the verifier ensemble drives the code toward consensus, and its sensitivity to a candidate verifier edge distinguishes an *independent* verifier (which raises the spectral gap) from a *redundant* one. A companion measure, ρ_{eff}, captures effective verifier independence from observed miss correlations.

> **Note**
> `mu` and ρ_{eff} are **diagnostics, not acceptance-gate inputs** — by design they are computed off the critical path and never block a node. The independence statistics are **live**: `independence::compute` folds the `psp9_verdicts` ledger into per-validator miss rates and matched-stratum pair statistics, and ρ_{eff} is reported only when every pair meets the matched-sample floor (a Hoeffding upper confidence bound backs the certified value). `perspt status` and the dashboard's Governance page surface these figures. The spectral eigensolver (`spectral::VerificationGraph`) remains an offline `perspt-sdk` diagnostic.

Node Classes

The system utilizes three node classes that govern execution order and verification:

Class	Description
Interface	Define exported signatures, schemas, and seals. Must be committed before dependent Implementation nodes can proceed. Produces an interface-seal digest.
Implementation	Operate on owned files using sealed interfaces from parent nodes. The bulk of code generation happens here.
Integration	Reconcile cross-owner boundaries after all dependent nodes converge. Used for multi-language projects or cross-module wiring.

Ownership Closure

The **ownership closure** rule is a fundamental invariant of PSP-5:

> *Each output file appears in exactly one node's* `output_files` *list.*

This prevents conflicting writes. In the current runtime the guarantee is enforced at dispatch through footprints: two nodes whose declared file footprints conflict never run concurrently, and each node attempt is seeded from the staging root rather than from unstaged sibling state.

Model Roles

The runtime uses one required and two optional model roles. Verification is **not** a model role: the verifiers are deterministic sandboxed sensors (compilers, test runners, linters), never a model call.

Role	Purpose	CLI Flag
Actuator	Proposes governed coding tool calls; the only role that can mutate the candidate. Also serves the architect's `update_graph` turn.	`--model` / `--actuator-model`
Explorer	Optional cheaper read-only repository exploration.	`--explorer-model`
Adjudicator	Optional no-tool conjunctive diff adjudication before promotion.	`--adjudicator-model`

Configure per-role routes via CLI:

```
perspt agent \
  --actuator-model <provider::model> \
  --explorer-model <provider::model> \
  --adjudicator-model <provider::model> \
  --fallback-model <provider::model> \
  "Build a REST API"
```

`--fallback-model` is repeatable and defines the ordered sticky actuator failover route. Persistent per-role routes live in the `[models]` table of `config.toml` as fully qualified `provider::model` values, so identity, calibration, and replay never depend on an ambient default provider. No model name is hard-coded in the runtime.

PSP-10: Model-Conditioned Prompt Programs

PSP-10 replaces free-form prompt templates with compiled prompt programs: typed sections, deterministic composition per model route, and ledgered digests for every model call.

7.1.7.1 Typed Prompt Sections

Prompt text lives in versioned section files under `crates/perspt-core/prompts/<stage>/NN_name.md`, one directory per stage (`session_bootstrap`, `graph_plan`, `repository_explore`, `adjudicate`, `evidence_summarize`), beside a committed `manifest.toml`. The `perspt-prompt-macros` crate compiles every section at build time from the owning crate's `build.rs`: it parses the front matter into a typed `SectionSchema` (id, version, role, priority, size bound, declared variables), runs the codegen validation list, and emits typed section structs into `OUT_DIR`. A malformed section fails `cargo build` with an error naming the offending file — never a session.

7.1.7.2 Stage Composition and Digests

At runtime a `StageComposition` composes one stage's rendered sections for the resolved model route and dialect into a `CompiledPromptProgram`: the ordered messages with per-message rendered hashes, any sections dropped by deterministic budget fitting, the tool-spec hash, and a `program_digest` computed over canonical bytes of the identity-bearing fields. One model call binds a platform program (the universal envelope) and a domain program; the pair forms a `CompiledPromptInvocation` with its own invocation digest.

7.1.7.3 Provenance

Every model call is ledgered: a `prompt_program_compiled` event records each program's sections and digest, and a `prompt_program_invoked` event records the actor, turn, platform digest, domain digest, invocation digest, and tool-spec hash. A session's prompts are therefore reconstructible

and diffable after the fact. Validated replacement bundles may substitute section bodies live only behind `--allow-experimental-prompts`; external bodies obey exactly the validation rules the build does.

7.1.7.4 The `perspt prompts` CLI

```
perspt prompts list                    # every compiled section: id, version, stage, role, hash
perspt prompts render <stage>          # compose one stage with fixture variables
perspt prompts lint --bundle <dir>     # validate an external bundle directory
perspt prompts manifest <dir>          # regenerate a library's manifest.toml
perspt prompts explain-session --db-path <db> <session-id>
                                       # the programs a session actually compiled, with digests
```

`perspt context explain-turn` is the companion command for a session's recorded resident-context events (compactions and refusals).

Rejection Budget

There is no per-error-type retry table. The loop runs against a single shared, non-replenishing **rejection budget** B (default 4, `--rejection-budget`): non-descending candidates and recovery attempts consume it, while accepted descents do not. Together with the descent gate this yields the finite-decision bound $\lfloor V_0/\rho_{\text{gate}} \rfloor + B + 1$ on gate decisions per node. When the budget is exhausted, the gate issues a **residual certificate** enumerating the remaining errors and the node terminates as a classified failure rather than looping. In headless mode (`--yes`) the session exits with a non-zero code when the outcome is not verified success.

Typed Tool Proposals

The Actuator does not emit artifact bundles. Every mutation is an individual typed tool call: the call becomes an effect proposal, the deterministic admissibility kernel checks it against the node's capability (path patterns, command patterns, effect kinds), and only an admitted effect executes — against the isolated candidate overlay, never the user workspace. Proposed commands additionally pass the deterministic `sanitize_command` and workspace-bound guards plus the Starlark policy engine, and normally run inside the process sandbox. Explicit native Windows reduced-isolation mode retains those agent checks but runs child processes with host-user authority. The candidate journals a pre-image of every file it touches, so any attempt can be discarded without residue.

Plugin-Driven Verification

Language plugins determine the verification toolchain:

Plugin	LSP Server	Test Runner	Init Command	Required Binaries
Rust	`rust-analyzer`	`cargo test`	`cargo init`	`cargo`, `rustc`
Python	`ty` or `pyright`	`pytest`	`uv init --lib`	`uv`, `python3`
JavaScript	`typescript-language-server`	`npm test`	`npm init -y`	`node`, `npm`
Go	`gopls`	`go test`	`go mod init`	`go`

The plugin is selected automatically during the Detection phase based on existing project files or the task description. Multi-language projects activate multiple plugins simultaneously.

Degraded Verification

When a verification tool is unavailable (e.g., `ty` not installed), the SRBN engine falls back to degraded mode:

- **Sensor fallback**: If the primary LSP server is not found, try a secondary (e.g., `pyright` instead of `ty`). Emit a `SensorFallback` event.
- **Degraded stages**: If no LSP server is available at all, V_syn is set to 0.0 and the stage is marked degraded. Energy convergence proceeds without that component.
- **Stability blocked**: If too many stages degrade, the node cannot converge and is escalated.

Merkle Ledger

All changes are recorded in a DuckDB-backed Merkle ledger:

- **Integrity** - Each commit has a cryptographic hash chaining to its parent
- **Rollback** - Roll back a session's newest completed promotion via `perspt ledger --rollback` (the argument is a session id prefix)
- **Resume** - `perspt resume` continues an interrupted session from its newest durable checkpoint by folding the ledger (see Crash Resume)
- **Audit** - Complete trail of AI-generated changes with energy breakdowns

```
perspt ledger --recent     # View recent commits
perspt ledger --rollback abc123
perspt ledger --stats      # Session statistics
```

Isolation and Resume

Nothing speculative ever touches the user workspace. Three layers keep observed work separated from accepted work, and crash resume preserves the separation.

> **Key Invariant**
> Model-proposed work reaches the user workspace only through the accept gate, the staging root, and the global integration gate — in that order. Everything else is discarded without residue.

7.1.13.1 Candidate Overlays

Each node attempt runs in a `CandidateWorkspace` overlay. Admitted effects mutate the overlay; every touched file's pre-image is journaled; verifiers measure the overlay, not the workspace. A rejected attempt is dropped wholesale.

Test evidence follows an explicit policy. The ordinary `evolving` policy measures the resulting implementation and resulting tests together, which is necessary when a task intentionally changes a contract. A `backward-compatible` run adds a regression view with recognized historical test files restored. An `external-oracle` run adds a separately supplied acceptance overlay that the actuator does not promote. These are additional test-stage views of the same realized candidate, not alternate paths around the syntax, build, or test requirements.

7.1.13.2 Search Forest Branches

A bounded search forest opens at most three branch identities, one branch attempt at a time. Every branch runs the ordinary governed loop in an isolated eager-copy workspace against the same immutable accepted root, and its internal states stay private to the forest. Exactly one

selected candidate is committed through a single gate submission, and the committed decision must equal the preview or the forest fails closed. Exact no-goods learned from failed branches suppress equivalent later attempts, and every branch action reserves its cost against the declared limit vector before it runs.

7.1.13.3 Staging and the Integration Gate

With `--max-parallel-nodes` above 1, node winners land in a content-addressed staging root instead of the user workspace. Disjoint footprints are enforced at dispatch; the merged state must pass one global integration gate before promotion, and a gate failure restores the prior staging root. The integration gate uses the same session-bound test policy as the node gates: resulting tests for ordinary evolving development, an additional historical regression view for explicitly backward-compatible work, or a separately protected external acceptance overlay.

7.1.13.4 Crash Resume

`perspt resume` folds the durable ledger rather than deserializing live state. A single-node session re-enters its loop from the newest durable candidate checkpoint with exactly the remaining budgets, its accepted candidate rebuilt from content-addressed artifacts and a fresh capability minted at the current authority epoch (a bumped epoch refuses the resume). A multi-node session resumes through graph dispatch: the staging root is rebuilt from the ledger and every winner still reaches the user workspace only through the global integration gate.

Stability Papers in Agent Mode

The execution of autonomous agents in Perspt is governed by control-theoretic stability contracts derived from the *Stability is All You Need* paper series. We do not assume that a language model's proposal is correct. Instead, the agent system measures the proposed state, determines its residual errors, and admits changes only when they satisfy a formal convergence constraint.

We formulate this stabilization process using three distinct contracts.

The Three Contracts

Let a session consist of a sequence of proposed states $x_1, x_2, \ldots$ and a corresponding sequence of accepted states $s_1, s_2, \ldots$.

7.2.1.1 Paper I: The Single-Agent Stability Contract

Paper I (*Stability is All You Need: Lyapunov-Guided Hierarchies for Long-Horizon LLM Reliability*) establishes the concept of a state-space manifold and the Lyapunov energy function.

Let x be a candidate state. The system computes a vector of residual errors $r(x)$. We define the Lyapunov energy $V(x)$ as a measure of the distance from x to the verified manifold $V(x) = 0$. The single-agent contract asserts that the state converges if each step reduces $V(x)$.

In the Perspt runtime, the candidate state corresponds to a modification proposed by the actuator model. The governed tool loop validates the proposal through the admissibility kernel and uses domain-specific sensors to calculate the energy.

7.2.1.2 Paper II: The Harness Contract

Paper II (*Stability is All You Need II: SRBN-Control Beyond Harness Engineering*) describes the measurement harness. The harness must observe every candidate, partition them into observed and accepted trajectories, and guarantee that the system terminates with a formal certificate of remaining errors if convergence is not achieved.

In the Perspt runtime, this contract is implemented by the tracking engine. Every verification run, retry attempt, and error message is persisted in the ledger, culminating in either a committed state or an explicit escalation report.

7.2.1.3 Paper III: The Platform Contract

Paper III (*Stability is All You Need III: The SRBN Platform Contract*) lifts the single-agent harness to a multi-domain platform. The platform must isolate stochastic proposals from deterministic side-effects. It mandates that models propose actions, while a deterministic capability kernel enforces security bounds and records budgets.

In the Perspt runtime, this is implemented via the SDK-first separation. The core SRBN engine (in `perspt-sdk`) manages scheduling and gating, while domain packages (such as `perspt-coding`) define domain-specific sensors and capabilities.

The Byzantine Generals Metaphor

To illustrate the stabilization protocol, we map the closed-loop convergence system to the Byzantine Generals Problem formulation (Lamport, Shostak, and Pease, 1982).

Let the Actuator agent be a commanding general whose loyalty is unknown; the commander's objective is to propose a series of operations (file modifications, deletions, or command executions) to reach a stable repository state. Let the verification sensors (the compiler, the type-checker, the linter, and the configured test evidence) be the loyal lieutenants. Tests evolve with the implementation in the default development policy; historical or externally protected evidence participates only when the corresponding verification policy is selected.

The protocol proceeds in rounds:

1. **Proposal**: The commander (Actuator) issues a proposed workspace change.
2. **Observation**: The loyal lieutenants (Sensors) independently inspect the proposal and report their findings as a vector of residual events.
3. **Consensus**: The gatekeeper (the governed tool loop and its admissibility kernel) aggregates the lieutenants' reports using the quadratic energy function. If the loyal lieutenants consensus reveals no critical residual errors (i.e., $V(x) \leq \varepsilon$), the proposal is committed.
4. **Correction or Escalation**: If the energy exceeds the threshold, the lieutenants' specific error reports are compiled into a feedback message. The commander is instructed to repair the state. If the commander fails to reach stability within the allocated round budget, the lieutenants reject further proposals and execute an escalation protocol to alert the operator.

The Energy Formulation

Let E be the set of active sensors. Each sensor $e \in E$ evaluates the candidate state x and emits a residual magnitude $r_e(x) \geq 0$.

The Lyapunov energy $V(x)$ is the weighted sum of the squared residuals:

$$V(x) = \sum_{e \in E} w_e \parallel r_e(x) \parallel^2, \qquad w_e > 0$$

We group the individual residuals into five component rollups:

- **Syntactic energy** (V_{syn}): Derived from compiler syntax errors and language server diagnostics.
- **Structural energy** (V_{str}): Derived from interface contract violations.
- **Logical energy** (V_{log}): Derived from test suite failures.
- **Bootstrap energy** (V_{boot}): Derived from initialization and build environment failures.
- **Sheaf energy** (V_{sheaf}): Derived from cross-node import and dependency contradictions.

The total energy is the sum of these component rollups:

$$V(x) = V_{\text{syn}} + V_{\text{str}} + V_{\text{log}} + V_{\text{boot}} + V_{\text{sheaf}}$$

A candidate state x is admitted to the accepted trajectory if and only if it satisfies the gating condition:

$$\text{accept}(x) \iff V(x) \leq \varepsilon \quad \vee \quad V(x) < V(x_{\text{best}}) - \rho_{\text{gate}}$$

where ε is the convergence threshold (default 0.10), x_{best} is the best previously accepted state in the current node generation, and ρ_{gate} is the minimum required descent step.

Observed vs. Accepted Trajectories

We maintain a strict boundary between the actions the model proposes and the actions the system commits.

Let T_{obs} be the set of all observed candidate states:

$$T_{\text{obs}} = \{x_t \mid t \geq 1\}$$

Let T_{acc} be the set of accepted states:

$$T_{\text{acc}} = \{x \in T_{\text{obs}} \mid \text{accept}(x) = \text{true}\}$$

The Merkle ledger commits only elements of T_{acc}. Unaccepted proposals in $T_{\text{obs}} \setminus T_{\text{acc}}$ are recorded as telemetry for correction feedback, but they never modify the workspace files.

7.2.4.1 The GoalPresence Verification Sensor

A primary check performed by the structural sensor (lieutenant) is the validation of required symbol presence. Let $\mathcal{S}_{\text{req}}$ be the set of symbols (functions, structures, classes, or modules) that the node is required to implement, as declared by its behavioral contract interface signature and goal text. Let $\mathcal{S}_{\text{obs}}$ be the set of symbols actually defined in the generated source code.

The goal-presence sensor computes the missing required symbols set $\mathcal{M}$:

$$\mathcal{M} = \mathcal{S}_{\text{req}} \setminus \mathcal{S}_{\text{obs}}$$

If $\mathcal{M} \neq \varnothing$, the sensor emits a blocking residual of class `SymbolMismatch`:

$$r_{\text{symbol}}(x) = \mid \mathcal{M} \mid$$

This forces the structural energy component V_{str} to remain high, preventing the state from satisfying the acceptance gate until all required symbols are present in the workspace.

Operational Mechanisms

The platform achieves stabilization through ten mechanisms:

1. **Closed-Loop Scheduler**: Re-evaluates the ready state of the graph dynamically. If a node fails to converge, the scheduler registers a repair action as a graph revision event (e.g., splitting a node or inserting an interface node).

2. **Capability Filtering**: All proposals must pass through the admissibility kernel. A proposal requesting a file edit or command run must possess a valid capability token.
3. **Structured Prompt Compilation**: When a proposal is rejected, the system compiles the residual diagnostics into a correction prompt.
4. **Event-Sourced Ledger**: Every transaction (proposals, rejections, commits, rollbacks) is written to a Merkle ledger to enable complete session replay.
5. **Domain Abstraction**: The core engine processes abstract residuals and energy models. Domain-specific behavior is relegated to domain packages.
6. **Bounded Search Forests**: On a gate failure, up to three isolated branches attempt distinct strategies against the same accepted root. Exact no-goods learned from failed branches suppress equivalent later attempts.
7. **Pre-Action Budget Reservations**: Every search-branch action reserves its cost against a shared limit vector before it runs; a refused reservation aborts the branch at no cost.
8. **Resident-Context Paging**: The conversation is partitioned into immutable content-addressed pages. Evicted pages leave in-place tombstones and remain recallable through the `context_recall` tool.
9. **Prompt-Program Provenance**: Every model call binds a compiled prompt program; the program and invocation digests are recorded in the ledger.
10. **Staging and Integration Gate**: Multi-node winners land in a content-addressed staging root and reach the user workspace only through one global integration gate.

Workspace Crates

Perspt is organized as a fourteen-crate Rust workspace under the `crates/` directory. The program separates user interaction, core execution, security policy, persistence, and domain specifications into distinct libraries.

Eight crates form the running executable program; three crates define the reusable platform SDK and its target domains; `perspt-prompt-macros` is the build-time compiler for the prompt section libraries; `perspt-benchmark` is an optional, manually run evaluation suite (feature-gated in `perspt-cli`, never part of normal validation); one meta-crate re-exports the libraries.

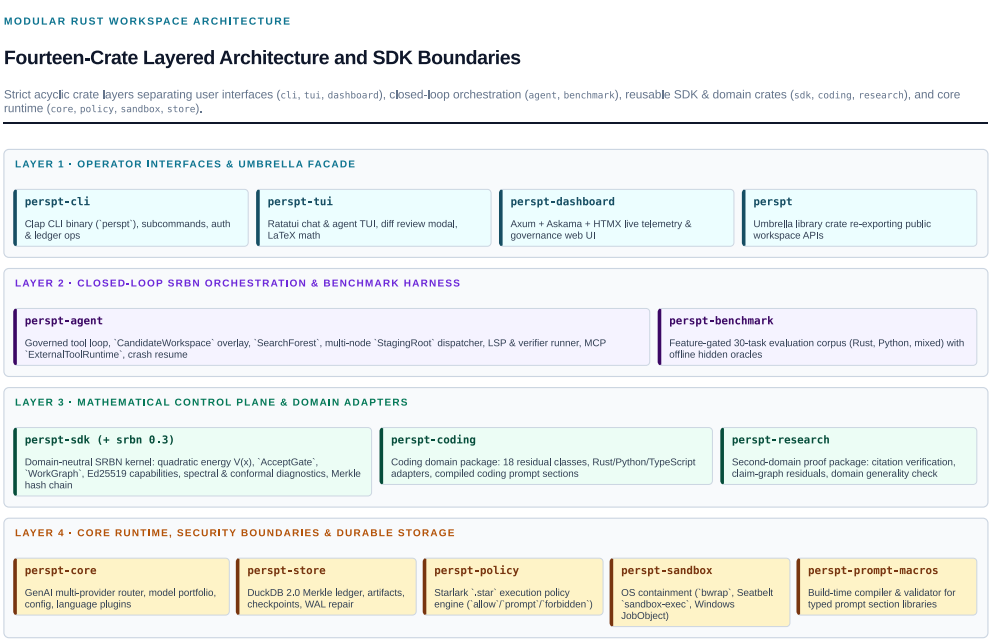


Fig. 6 Layered architecture of the fourteen-crate Perspt workspace across user surfaces, governed runtime & security, domain-neutral SRBN SDK, and evaluation harness.

Crate Summary

Crate	Operational Purpose	Primary Structural Types
perspt-cli	Parses command-line inputs and dispatches system commands.	`Cli`, `Commands`
perspt-core	Formulates shared data structures, configurations, and provider connections.	`GenAIProvider`, `Config`, `TaskPlan`, `NodeClass`, `AgentEvent`
perspt-agent	Executes the governed PSP-9 tool loop and work-graph runtime.	`Psp9AgentRuntime`, `CandidateWorkspace`, `AgentTools`, `GenAiTransport`
perspt-tui	Implements the terminal user interface for interactive review.	`ChatApp`, `AgentApp`, `DiffViewer`, `ReviewModal`
perspt-store	Manages the DuckDB persistence layer for session logs.	`SessionStore`, `SessionRecord`, `Psp9LedgerRow`
perspt-policy	Executes Starlark policies to validate proposed command structures.	`PolicyEngine`, `sanitize_command`
perspt-sandbox	Runs external commands inside isolated subprocess environments.	`SandboxedCommand`, `CommandResult`
perspt-dashboard	Implements a read-only browser-based monitoring engine.	`AppState`, Axum router, SSE stream
perspt-sdk	Defines domain-neutral SRBN stability contracts and gate models.	`ResidualEvent`, `EnergyModel`, `AgentDomainPackage`, `ResidualCertificate`

Crate	Operational Purpose	Primary Structural Types
perspt-coding	Implements the coding-domain adapters, sensors, and mappers.	`CodingDomain`, `LanguageId`, `CodingAdapterRegistry`
perspt-research	Implements the research-domain adapter to demonstrate SDK reuse.	`ResearchDomain`
perspt-prompt-macros	Compiles prompt section files into typed sections at build time.	`StageDecl`, `PromptBuildError`
perspt-bench-mark	Runs the optional, manually invoked PSP-10 evaluation ladder.	`BenchmarkSuite`, `BenchmarkRunOptions`

7.3.1.1 perspt-cli

The entry-point command-line utility. It parses command-line flags and subcommands using the `clap` library and dispatches execution control to corresponding subsystems in other workspace crates.

Key structural types include:

- `Cli`: The root structure declaring command-line configuration arguments.
- `Commands`: The enum representing the sixteen subcommands (plus a feature-gated seventeenth):
  - `chat`: Starts the interactive terminal chat mode (the default).
  - `simple-chat`: Simple CLI chat mode without the TUI.
  - `agent`: Runs the governed PSP-9 agent on a task.
  - `init`: Initializes project memory and Starlark policy rules.
  - `config`: Modifies runtime configuration values in `config.toml`.
  - `ledger`: Commands to query, inspect, or rollback Merkle ledger sessions.
  - `status`: Shows current agent status.
  - `audit`: Delayed audit labels and conformal activation.
  - `providers`: Prints the provider capability matrix.
  - `replay`: Deterministic, credential-free audit replay of a session.
  - `abort`: Aborts a PSP-9 session by revoking its authority epoch.
  - `resume`: Resumes a paused or crashed session.
  - `dashboard`: Launches the monitoring Axum server and opens the browser.
  - `db`: Inspects and repairs the local DuckDB store.
  - `prompts`: Inspects and maintains the compiled prompt section libraries.
  - `context`: Explains a session's recorded resident-context events.
  - `benchmark` (`--features benchmark` only): Runs the optional model-backed evaluation tooling.

7.3.1.2 perspt-core

The foundation crate. It defines the central types, data models, and configurations shared globally across all crates. Among workspace crates it depends only on `perspt-sdk` and, at build time, on the `perspt-prompt-macros` compiler.

Key components:

- **Core Types**: Structures representing the execution parameters, such as:

```rust
pub enum NodeClass {
    /// Defines exported signatures, schemas, ownership manifests
    Interface,
    /// Operates on node-owned files plus adjacent sealed interfaces
    Implementation,
    /// Reconciles cross-owner or cross-plugin boundaries
    Integration,
}
```

- **Event Plane**: Defines the event system carrying over 40 lifecycle events. The `AgentEvent` enum is transmitted across channels to update the TUI and log database records.
- **Provider Gate**: Integrates with the `genai` crate to handle provider-agnostic LLM requests, supporting token streaming, prompt assemblies, and backoff limits.
- **Prompt Libraries**: Hosts the versioned prompt section files under `crates/perspt-core/prompts/` (`session_bootstrap`, `graph_plan`, `repository_explore`, `adjudicate`, `evidence_summarize`), compiled at build time by `perspt-prompt-macros` against the `perspt-sdk` prompt module.

7.3.1.3 perspt-agent

The authoritative PSP-9 runtime executor. It houses the governed tool loop, the work-graph dispatcher, tool definitions, and system sensors. Prompt text does not live here: the section libraries live under `crates/perspt-core/prompts/`.

- **Governed Tool Loop**: The loop in `crates/perspt-agent/src/toolloop/` composes the `perspt-sdk` admissibility kernel, the measured acceptance gate, and the durable ledger. Every model-issued tool call becomes a typed proposal; admissible effects mutate an isolated `CandidateWorkspace` overlay; the `CodingCandidateMeasurer` re-measures the candidate under the configured test-evidence policy; and the gate is evaluated on the measurement, never on the model's account of it. Ordinary development uses the resulting tests, while historical regression and protected external acceptance suites are explicit opt-ins.
- **Work-Graph Dispatcher**: `Psp9AgentRuntime` drives multi-node execution behind `--max-parallel-nodes`. A governed architect turn proposes graph revisions through the privileged `update_graph` tool; the dispatcher fills free slots with ready nodes whose file footprints do not conflict, stages each node's winner into a content-addressed staging root, and promotes the merged result through one global integration gate. `perspt-agent` depends on `perspt-sdk` directly; there is no bridge layer.

7.3.1.4 perspt-tui

The terminal user interface built using the `ratatui` library. It provides high-performance terminal panels optimized for real-time human-in-the-loop review.

- **ChatApp**: An interactive chat client that supports streaming LLM dialogue, rendering tables, and parsing markdown.
- **AgentApp**: Displays active workspace executions. It visualizes the task graph DAG, displays active Lyapunov energy gauges, and opens a `ReviewModal` with unified code diffs for manual validation.

7.3.1.5 perspt-store

The local database storage library built on DuckDB. It manages all persistent schemas for sessions, LLM requests, energy convergence logs, and ledger commits.

Key interface structures:

- `SessionStore`: Encapsulates connection pools to the DuckDB file database.
- **Schema Definitions**: It creates and maintains eleven tables:
  - `sessions`: Top-level session tracking.
  - `schema_migrations`: Applied schema versions.
  - `psp9_ledger_events`: The durable hash-chained canonical event stream.
  - `psp9_artifacts`: Content-addressed artifact bytes keyed by hash.
  - `psp9_authority_epochs`: Per-session authority epoch counters.
  - `psp9_grant_policies`: Persisted signed grant intent.
  - `psp9_external_effects`: External effect intents keyed by idempotency key.
  - `psp9_context_checkpoints`: Durable context checkpoints per covered event root.
  - `psp9_verdicts`: Per-validator verdicts feeding the independence statistics.
  - `psp9_calibration_epochs` and `psp9_calibration_samples`: Conformal calibration state and delayed-audit sample labels.

7.3.1.6 perspt-policy

The security policy engine running the Starlark scripting runtime. It checks proposed mutations before execution to guarantee safety bounds.

Key API signatures:

```rust
pub enum PolicyDecision {
    Allow,
    Prompt(String),
    Deny(String),
}

impl PolicyEngine {
    pub fn evaluate(&self, command: &str) -> PolicyDecision;
    pub fn is_safe(&self, command: &str) -> bool;
}
```

Each Starlark policy is a single function that receives the command string and returns `"allow"`, `"prompt"`, `"deny"`, or a boolean; the engine folds multiple policies by taking the strictest decision. The crate also exports the deterministic guards `sanitize_command`, `validate_workspace_bound`, and `validate_artifact_mutation`, which run before any Starlark evaluation.

7.3.1.7 perspt-sandbox

Bounded command-execution and process-isolation library. It uses Bubblewrap on Linux and `sandbox-exec` on macOS, fails closed when required isolation is unavailable, and exposes an explicit best-effort mode for externally isolated embedders and acknowledged native Windows operation.

- `SandboxedCommand`: Wraps standard library commands, enforcing memory ceilings, processor timeouts, environment sanitization, and redirecting stdout/stderr to files for analysis.

7.3.1.8 perspt-dashboard

A read-only local dashboard built using the Axum web framework. It reads execution history directly from the DuckDB store and streams live progress to a local browser page.

- **Server Routing**: Sets up REST routes for historical session analysis.
- **SSE Stream**: Sends live event frames (graph state changes, energy rollups) using Server-Sent Events (SSE).

7.3.1.9 perspt-sdk

The core platform library defining the reusable SRBN stability protocol specifications.

- **Domain-Neutral SDK**: Implements the mathematical formulations for the Measured Acceptance Gate. It has no knowledge of language rules, compilers, or project layouts. It operates purely on mathematical residuals and domain contracts.
- **Domain Integration Trait**: The core trait that custom domain extensions must implement:

```rust
pub trait AgentDomainPackage: Send + Sync {
    fn domain_id(&self) -> DomainId;
    fn detect(&self, workspace: &WorkspaceSnapshot) -> DomainDetection;
    fn residual_schema(&self, scope: &DomainScope) -> ResidualSchema;
    fn energy_model(&self, scope: &DomainScope) -> EnergyModel;
    fn correction_directions(&self, residuals: &[ResidualEvent]) ->
Vec<CorrectionDirection>;
    // PSP-9 agent contract (required):
    fn tool_entries(&self, scope: &DomainScope) -> Vec<ToolEntry>;
    fn hard_gate_policy(&self, scope: &DomainScope) -> HardGatePolicy;
    fn verifier_suite(&self, scope: &DomainScope) -> VerifierSuiteSpec;
    fn safety_barrier(&self, scope: &DomainScope) -> SafetyBarrierDisposition;
}
```

7.3.1.10 perspt-coding

The coding-domain package, implementing the `AgentDomainPackage` trait.

- **Coding Domain Model**: Translates code verification outcomes (compiler syntax diagnostics, type warnings, failed tests, linter reports, and missing symbols) into SDK `ResidualEvent` representations.
- **Language Adapters**: Implements adapters for Rust, Python, and TypeScript to parse raw toolchain output, associate them with residual classes, and supply directed correction instructions (e.g. telling the Actuator what type error was detected and how to resolve it). Adapters are registered in the `LanguageId`-keyed `CodingAdapterRegistry`, so adding a language means registering an adapter, never editing the runtime.

7.3.1.11 perspt-research

An experimental verification domain package.

- **Academic Manuscript Domain**: Implements the SDK traits to manage research document production. It treats LaTeX syntax errors, missing citations, and bibliography inconsistencies as residual events, driving convergence towards a fully validated scientific manuscript. It serves as a proof of concept for the SDK's domain-agnostic separation of concerns.

7.3.1.12 perspt-prompt-macros

The build-time compiler for prompt section files (PSP-10). Called from an owning crate's `build.rs`, it parses every `prompts/<stage>/NN_name.md` file, runs the codegen validation list, and emits typed section structs into `OUT_DIR`. A malformed section fails `cargo build` with an error naming the offending file, never a session. The same validation functions serve the runtime bundle scanner and `perspt prompts lint`.

7.3.1.13 perspt-benchmark

Optional PSP-10 evaluation tooling, independent of the runtime mechanism tests. It exposes the `perspt benchmark` subcommand only when `perspt-cli` is built with `--features benchmark`; live runs require configured routes and credentials and never run in CI or normal validation. Suites (`smoke`, `adaptive`, `full`) run a cumulative ladder of arms over matched tasks and publish paired differences with a seeded bootstrap.

PSP Process

A complex software system cannot be modified by arbitrary whim. We must frame every major modification as a formal proposal, detailing the design, the motivation, and the proof of correctness. We call such a proposal a **Perspt Specification Proposal (PSP)**.

PSPs serve as the primary mechanism for:

- Proposing major features before implementation.
- Documenting design decisions and trade-offs.
- Providing a historical record of architectural evolution.
- Enabling community review of significant changes.

PSP Lifecycle

A proposal passes through four distinct states. First, it is written as a **Draft**. Once accepted for implementation, it becomes **Active**. When the implementation is complete and verified, the proposal becomes **Final**. If a subsequent proposal modifies the same system, the earlier proposal may become **Superseded**.

Status	Meaning
Draft	Proposal is under active design and review.
Active	Proposal is accepted and under active implementation.
Final	Implementation is complete and fully verified.
Superseded	Replaced or invalidated by a newer specification proposal.

Key PSPs

We tabulate the specifications that govern the architecture of the system.

PSP	Title	Status
PSP-1	Core Chat Interface	Final
PSP-2	Multi-Provider Support	Final
PSP-3	Simple CLI Mode	Final
PSP-4	SRBN Agent Mode	Superseded by PSP-5
PSP-5	Multi-File Coding UX and Repo-Native Verification	Final
PSP-6	Web Dashboard and Real-Time Monitoring	Final

PSP	Title	Status
PSP-7	Robust Typed Correction Loops and Plugin-Aware Prompt Contracts	Final
PSP-8	SRBN Agent SDK, Coding Domain Package, and Mutable Work Graph	Final
PSP-9	**Governed Agentic Platform: SRBN Tool Harness and Model Portfolio**	**Final**
PSP-10	**Bounded Search Trajectories and Model-Conditioned Prompt Programs**	**Final**
PSP-11	**Domain-Neutral Runtime Inversion, Multi-Actor Supervisor, and Replayable Human Interaction Platform**	**Discussion**

7.4.2.1 PSP-5: The Core Lifecycle

PSP-5 is the historical specification of the original multi-file agent execution; the operative platform is now defined by PSP-9 and PSP-10. It supersedes PSP-4 and established:

- **Project-first execution** - The task is modeled as a directed acyclic graph (DAG) of nodes instead of a single-file task.
- **Ownership closure** - Each file is owned by exactly one node.
- **Artifact bundle protocol** - Structured operations containing writes, diffs, and commands.
- **Node classes** - Division of nodes into Interface, Implementation, and Integration classes.
- **Five-component energy** - The Lyapunov energy model decomposed into syntactic, structural, logical, bootstrap, and sheaf components.
- **Plugin-driven verification** - Selection of language-specific verification tools.
- **Provisional branches** - Speculative execution isolated from the main workspace until parent nodes commit.
- **Interface seals** - SHA-256 digests of exported signatures to enforce cross-node contracts.

See the full specification at `docs/psps/records/000005-srbn-multi-file-coding-ux.rst`.

7.4.2.2 PSP-7: The Correction Loop

PSP-7 builds upon the foundation of PSP-5 to enforce robust convergence under non-deterministic failures. It introduces:

- **Typed parse pipeline** - A five-layer fail-closed parsing protocol that recovers from malformed model outputs.
- **Retry classification** - Explicit categorization of verification failures to determine whether to retarget, replan, or retry.
- **Prompt compiler with provenance** - Systematic generation of correction prompts using structural error evidence.
- **Manifest policy enforcement** - Protection of root manifests from implicit mutation.
- **Strict budget exhaustion** - Interception of execution before LLM invocation when steps, revisions, or cost limits are exceeded.

See the full specification at `docs/psps/records/000007-robust-typed-correction-loops.rst`.

7.4.2.3 PSP-8: The Reusable Platform

PSP-8 defines the architecture for separating domain-neutral control mechanisms from domain-specific features. It introduces:

- **SDK-First design** - Movement of scheduling, energy gating, capability checks, and replay ledgers to a reusable SDK.
- **Quadratic energy** - Adoption of the quadratic residual energy formula $V(x) = \sum_e w_e \parallel r_e \parallel^2$.
- **Mutable work graph** - A scheduler that modifies, splits, or inserts nodes in the queue as verification evidence arrives.
- **Capability kernel** - Scoped admissibility check over proposed effects to guarantee sandbox bounds.

See the full specification at `docs/psps/records/000008-srbn-agent-sdk-work-graph.rst`.

7.4.2.4 PSP-9: The Governed Platform

PSP-9 replaces bundle generation with a governed tool loop and defines the operative agent runtime. It introduces:

- **Governed tool loop** - Every model-issued tool call becomes a typed proposal; a deterministic admissibility kernel decides whether it may affect the candidate.
- **Measured acceptance gate** - A candidate is admitted on a hard verifier pass or a measured energy descent of at least ρ_{gate} below the best accepted state.
- **Model portfolio** - Per-role routes (actuator, explorer, adjudicator) with an ordered sticky fallback chain, configured via `[models]` or CLI flags.
- **Durable event ledger** - A hash-chained event stream enabling deterministic, credential-free replay and crash resume.
- **Open language registry** - A `LanguageId`-keyed adapter registry replaces closed per-language enum dispatch.

See the full specification at `docs/psps/records/000009-governed-agentic-platform.rst`.

7.4.2.5 PSP-10: Search and Prompt Programs

PSP-10 extends the governed platform with bounded search trajectories and model-conditioned prompt programs. It introduces:

- **Typed prompt sections** - Prompt libraries compiled at build time from versioned section files, with program and invocation digests recorded in the ledger.
- **Bounded search forests** - On gate failure, isolated branches run against the accepted root under pre-reserved budgets, with exact no-good learning.
- **Resident-context paging** - The conversation becomes content-addressed pages; evicted pages leave tombstones and remain recallable via `context_recall`.
- **Multi-node dispatch** - Concurrent work-graph nodes behind `--max-parallel-nodes`, staged into a content-addressed root and promoted through one global integration gate.

See the full specification at `docs/psps/records/000010-bounded-search-trajectories.rst`.

7.4.2.6 PSP-11: Domain-Neutral Runtime Inversion & Multi-Actor Supervisor

PSP-11 specifies the inversion of the governed candidate engine into `perspt-sdk`, a multi-actor supervisor over shared durable stores, and replayable human interaction gates.

See the full specification at `docs/psps/records/000011-domain-neutral-runtime-inversion.rst`.

Writing a PSP

A developer wishing to propose an architectural modification must adhere to the following protocol:

1. Create a branch from the repository root.
2. Run `guidedog dsc new "Your PSP Title" --type=standards-track --dir=docs/psps` to instantiate a numbered record in `docs/psps/records/`.
3. Define the Abstract, Motivation, Specification, Rationale, and Reference Implementation.
4. Run `guidedog dsc index --dir=docs/psps && guidedog dsc check --dir=docs/psps` and submit the proposal for review.
5. Promote and transition the record via `guidedog dsc promote` / `guidedog dsc state` when consensus is reached.

How-To Guides

Task-oriented recipes for common operations.

Guide	Summary
Agent Options Reference	Full reference for `perspt agent` flags.
Configuration	Config files, env vars, and precedence.
Set Up Providers	Set up each LLM provider.
Security and Policy Rules	Starlark policies, sandboxing, and security.
Dashboard Setup	Configure and launch the web dashboard.

Agent Options Reference

`perspt agent` runs the governed PSP-9 tool loop, extended by PSP-10 with bounded search forests, resident-context paging, prompt programs, and multi-node dispatch. The CLI exposes only controls consumed by that runtime.

Core Options

Flag	Description
`<TASK>`	Task description.
`-w, --workdir <DIR>`	Workspace root.
`-y, --yes`	Approve final promotion within already-granted authority. It does not grant network, shell, policy, or out-of-workspace effects. A non-interactive run requires `--yes`: promotion approval cannot be prompted without a terminal. `--max-parallel-nodes` above `1` also requires it.
`--model <PROVIDER::MODEL>`	Override the primary actuator route.
`--domain <DOMAIN>`	Domain package to run (e.g. `coding`, `research`); default: detect.
`--exploration-only`	Run only the read-only exploration phase: deterministic map plus an interactive explorer tool loop; nothing is mutated or promoted.
`--allow-dependency-mutation`	Grant governed dependency mutation (`cargo add`, `uv add`, `npm install`).
`--allow-unisolated`	Explicitly permit verifier, inspection, and LSP child processes to run without an OS sandbox. Intended for acknowledged native Windows use or an embedder that isolates the complete Perspt process. Candidate, capability, gate, ledger, and approval con-

Flag	Description
	trols remain active, but the child has the host user's filesystem and network authority.
`--allow-experimental-prompts`	Substitute validated `[prompts]` bundle sections live (Gate AE: experimental until a change record passes paired evaluation).
`--db-path <PATH>`	Path to the PSP-9 ledger database (defaults to the platform data directory).
`--output-summary <FILE>`	Write the terminal PSP-9 run summary as JSON.
`--persistent-grants`	Store signed grant intent. Resume always rechecks the durable authority epoch and must mint a fresh live capability.

Finite Harness

Flag	Description
`--rho-gate <VALUE>`	Required measured energy descent per accepted checkpoint (default `0.5`).
`--max-turns <N>`	Maximum model turns per node (default `12`).
`--max-calls-per-turn <N>`	Maximum direct and nested tool calls per turn (default `8`).
`--rejection-budget <N>`	Shared non-descending and recovery budget (default `4`).
`--max-parallel <N>`	Maximum independent compiler, test, and lint sensors (default `4`).
`--max-parallel-nodes <N>`	Concurrent work-graph nodes (default `1`). Values above `1` require `--yes`.

Model Portfolio

Flag	Description
`--actuator-model <M>`	Primary implementation route.
`--explorer-model <M>`	Cheap, no-tool repository-map summarizer.
`--adjudicator-model <M>`	Optional no-tool validator of the realized diff. Its verdict is an uncalibrated conjunctive veto, never an energy measurement.
`--fallback-model <M>`	Add an ordered sticky actuator failover route. Repeat the option to add more routes. Role-specific models are never silently reused as actuator fallbacks.

Dashboard

`--dashboard` starts the read-only monitoring server alongside the agent. `--dashboard-port <PORT>` selects its port (default `3000`).

Examples

```
persept agent -y -w /tmp/proj "Create a Python calculator"

persept agent \
  --actuator-model vertex::gemini-3.5-flash-lite \
  --explorer-model vertex::gemini-3.5-flash-lite \
  --fallback-model vertex::gemini-3.5-flash \
  --rho-gate 0.25 \
  --max-turns 16 \
  --max-calls-per-turn 8 \
  --rejection-budget 5 \
  --max-parallel 4 \
  -w ./project "Build a web server"

persept agent --dashboard --dashboard-port 8080 -w ./myapp "Add unit tests"

# Native Windows reduced-isolation mode
persept agent --allow-unisolated -w . "Fix the parser"
```

Configuration

Config File Location

Perspt searches for configuration in this order:

1. `--config <path>` (CLI flag)
2. `~/.config/perspt/config.toml` (Linux), `~/Library/Application Support/perspt/config.toml` (macOS), `%APPDATA%\perspt\config.toml` (Windows)
3. Environment variables
4. Auto-detection

Config File Format

The file is TOML. All fields are optional. `provider` accepts the aliases `provider_type` and `default_provider`; `model` accepts the alias `default_model`.

```
provider = "gemini"
model = "gemini-3.5-flash"
api_key = "AIza..."

# Optional endpoint override for OpenAI-compatible / local / proxy servers
# base_url = "http://localhost:8000/v1"

# Optional per-role routes for `perspt agent`; values stay fully
# qualified as provider::model
# [models]
# architect = "openai::gpt-5.5"
# actuator = "openai::gpt-5-mini"
# verifier = "openai::gpt-5-mini"
# speculator = "openai::gpt-5-mini"
# adjudicator = "openai::gpt-5.5"
```

Agent Runtime Sections

`perspt agent` reads further optional TOML tables, each validated at startup.

`[providers.<id>]` lets several provider credentials coexist in one process. `api_key_env` names an environment variable; Vertex entries may omit a static key and use the per-request ADC token resolver:

```
[providers.anthropic]
api_key_env = "ANTHROPIC_API_KEY"

[providers.local]
adapter = "ollama"
base_url = "http://localhost:11434"
```

`[exploration]` bounds the PSP-10 search forest: sequential eager branches with a hard branch cap of 3. The former `[ensemble]` section was removed by PSP-10 - a present block fails validation with a migration error pointing here:

```
[exploration]
initial_branches = 1
max_branches = 3
distinct_family = true
```

`[[external_tools]]` declares shared latest-only MCP 2026-07-28 servers. The official Rust SDK supplies tools, resources, prompts, completions, roots, sampling, elicitation, subscriptions, MRTR, and tasks. Secret values are never stored here: environment maps contain destination names and source variable names. Omitted `modes` means agent-only; chat must be explicit. Every allowed remote tool needs a local policy, including a schema-valid footprint:

```
[[external_tools]]
id = "docs"
transport = "stdio"
command = ["company-docs-mcp", "--stdio"]
modes = ["agent", "chat"]
env_from_env = { DOCS_TOKEN = "COMPANY_DOCS_TOKEN" }
roots = [{ uri = "file:///absolute/path/to/project", name = "Project" }]
sampling = true
max_sampling_tokens = 4096
elicitation = true
subscriptions = true
tasks = true
max_task_wait_ms = 300000

[external_tools.tools.search]
effect = "search"
risk = "low"
footprint = { selectors = [{ kind = "scoped_argument", family = "company-docs", field =
"query", access = "read" }] }
```

The policy key must equal the remote tool name and its `field` must exist in the advertised input schema. In chat, `/mcp` explains failures and lists the admitted namespaced operations. Older protocol versions have no fallback. See Model Context Protocol (MCP) for both transports, every capability and bound, agent/chat behavior, clipboard paste, and security guidance.

`[prompts]` pins external prompt replacement bundle directories and the paired activation bounds (Gate AE):

```toml
[prompts]
bundles = ["./prompt-bundles/coding"]
activation_min_tasks = 30
```

`[context]` sets reserves and working-set bounds for the paged resident context; every configured reserve must be positive:

```toml
[context]
working_set_turns = 8
output_reserve_tokens = 4096
```

`[verification]` sets the test-evidence policy, optional `format` sensor, process-isolation fallback, and per-stage timeouts. `evolving` is the iterative-development default; `backward-compatible` additionally runs recognized historical test files; `external-oracle` additionally runs a protected overlay configured under `[verification.external_oracle]`:

```toml
[verification]
test_policy = "evolving"
allow_unisolated = false
require_format = true
stage_timeout_secs = 180
```

Keep `allow_unisolated` false for fail-closed execution. Native Windows users may set it true to run compilers, tests, inspection commands, and LSP servers without an OS sandbox; Perspt retains its agent-level gates but child processes have the host user's authority.

Environment Variables

Variable	Provider	Priority
`VERTEX_PROJECT_ID`	Vertex AI	Highest
`GEMINI_API_KEY`	Gemini	2
`OPENAI_API_KEY`	OpenAI	3
`ANTHROPIC_API_KEY`	Anthropic	4
`GROQ_API_KEY`	Groq	5
`COHERE_API_KEY`	Cohere	6
`XAI_API_KEY`	xAI	7
`DEEPSEEK_API_KEY`	DeepSeek	8
(none)	Ollama	Fallback

> **Note**
> When multiple keys are set, the highest-priority provider is used. Override with the `provider` field in the config or by setting only the key you want detected.

CLI Flag and Config Override

Manage the config file from the CLI:

```
# Show the effective config (api_key masked)
perspt config --show

# Set values (structured TOML write)
perspt config --set provider=openai
perspt config --set default_model=gpt-5.5

# Edit in $EDITOR
perspt config --edit

# Override the model per run
perspt chat --model gpt-5.5

# Use a specific config file
perspt --config /path/to/config.toml chat
```

Logging Configuration

```
# Default: error-level logging only (avoids TUI noise)
perspt

# Enable debug logging with RUST_LOG
RUST_LOG=debug perspt simple-chat

# Audit a finished agent run (deterministic, credential-free replay)
perspt replay <SESSION_ID>
```

Set Up Providers

OpenAI

```
export OPENAI_API_KEY="sk-xxx"
perspt chat --model gpt-5.5
```

Supported models: `gpt-5.5`, `gpt-5-mini`, `o1`, and any other model your provider account exposes.

Anthropic

```
export ANTHROPIC_API_KEY="sk-ant-xxx"
perspt chat --model claude-fable
```

Supported models: `claude-fable`, `opus-4.8` and others.

Google Gemini

```
export GEMINI_API_KEY="AIza..."
perspt chat --model gemini-3.5-flash
```

Supported models: `gemini-3.1-pro`, `gemini-3.5-flash`, `gemini-2.5-pro`, and others.

Google Vertex AI

Vertex AI is supported through Google Cloud authentication. You must configure your Google Cloud project ID and optionally the region.

```
export VERTEX_PROJECT_ID="my-gcp-project-id"
export VERTEX_REGION="us-central1"
# Run using the Vertex model prefix
perspt chat --model vertex::gemini-3.5-flash
```

Supported models: `vertex::gemini-3.1-pro`, `vertex::gemini-3.5-flash`, and other models enabled in your Vertex project.

Groq

```
export GROQ_API_KEY="gsk_xxx"
perspt chat --model llama-4-70b
```

Groq provides ultra-fast inference for open-source models.

Cohere

```
export COHERE_API_KEY="xxx"
perspt chat --model command-r7
```

xAI

```
export XAI_API_KEY="xxx"
perspt chat --model grok-4
```

DeepSeek

```
export DEEPSEEK_API_KEY="xxx"
perspt chat --model deepseek-v4
```

Ollama (Local)

No API key needed. Ollama is the fallback when no cloud keys are set.

```
# Start Ollama
ollama serve

# Pull a model
ollama pull llama4

# Use with Perspt
perspt chat --model llama4
```

Multiple concurrent models are supported. Use `ollama list` to see installed models.

Security and Policy Rules

Perspt provides multiple layers of security for agent mode.

Starlark Policies (perspt-policy)

The `perspt-policy` crate evaluates Starlark scripts against every command line the agent proposes. Policies can:

- **Deny shell commands** - Block dangerous commands (`rm -rf`, etc.)
- **Require confirmation** - Gate risky commands behind a user prompt

Each `.star` file in the rules directory (`rules/` under the config directory) must define `evaluate`, a function that receives the full command line and returns `"allow"`, `"prompt"`, or `"deny"` (a bool is also accepted). The builtins are `matches_pattern(command, pattern)` (substring match) and `log_policy(message)`:

```
# Example Starlark policy
def evaluate(command):
    for pattern in ["rm -rf", "sudo", "chmod 777"]:
        if matches_pattern(command, pattern):
            log_policy("blocked: " + command)
            return "deny"
    if matches_pattern(command, "git push"):
        return "prompt"
    return "allow"
```

When several files are loaded, the strictest decision wins, and a policy that fails to evaluate denies the command.

Sandbox Isolation (perspt-sandbox)

The `perspt-sandbox` crate provides bounded command execution and OS-backed isolation where a backend is registered:

- **Filesystem scoping** - Commands run in a restricted view of the filesystem
- **Process limits** - Wall-clock timeout and bounded captured output
- **Network control** - Optional network access restriction

Configuration:

```
# Agent with sandbox enabled
perspt agent -w ./project "Task"

# The sandbox restricts commands to the working directory
```

Linux uses Bubblewrap and macOS uses `sandbox-exec`. Native Windows fails closed by default. `--allow-unisolated` is an explicit reduced-isolation override: the agent-level capability and verification machinery stays active, but child processes retain the host user's filesystem and network authority.

Authority Model

PSP-9 authority is capability-based:

- Live capabilities are minted from grants at session start and are bound to the session's authority epoch
- `perspt abort` revokes the epoch durably: in-flight promotions and stale resume intents are refused, and the workspace is untouched
- `--allow-dependency-mutation` is the explicit grant for governed dependency mutation (`cargo add`, `uv add`, `npm install`)
- `--persistent-grants` persists signed grant intent only, never live capabilities: resume must mint fresh ones

Footprint Scheduling

The multi-node dispatcher schedules by write footprints:

- Each node declares the files it writes (a node that declares none holds an opaque whole-workspace footprint)
- Nodes with conflicting footprints never run concurrently
- Promotion is single-flight: it runs only in the dispatcher's completion arm

This prevents conflicting edits and provides a clear audit trail.

Review Modal

In interactive mode (without `--yes`), every node's changes must be manually approved. The review modal shows:

- Full diff of all changes
- Verification results (syntax, build, test, lint), the measured energy, and a degraded flag with reasons when any sensor was skipped
- Options to reject, correct, or edit

For security-sensitive projects, always use interactive mode.

Merkle Ledger

Every committed change is recorded in a content-addressed Merkle tree stored in DuckDB. This provides:

- **Tamper detection** - Hash chain integrity
- **Full auditability** - Every node's input/output is recorded
- **Rollback capability** - Restore to any point in the session

```
perspt ledger --recent
perspt ledger --stats
```

Best Practices

1. **Use interactive mode for production code** - Always review diffs
2. **Replay sessions** - `perspt replay <SESSION_ID>` is a deterministic, credential-free audit replay
3. **Use workspace directories** - `-w <dir>` scopes agent writes
4. **Label delayed audit samples** - `perspt audit` lists pending samples and records safe/unsafe labels
5. **Inspect prompt and context provenance** - `perspt prompts explain-session` and `perspt context explain-turn` show what a session compiled and compacted
6. **Review ledger after headless runs** - `perspt ledger --recent`

Dashboard Setup

Configure and launch the Perspt web dashboard for monitoring agent sessions.

Locate the Config File

The Perspt configuration file lives in your platform's config directory:

- **Linux**: `~/.config/perspt/config.toml`
- **macOS**: `~/Library/Application Support/perspt/config.toml`

If migrating from the legacy `~/.perspt/` layout, Perspt will warn and fall back to the old location automatically.

Understand the Access Model

The dashboard binds to `127.0.0.1` only and runs unauthenticated. It is a read-only view of the local session database, suitable for localhost-only use; do not forward the port to other machines.

Change the Default Port

The default port is `3000`. Override via CLI flag:

```
perspt dashboard --port 8080
```

Build from Source

```
cargo build -p perspt-dashboard --release
```

The dashboard binary is built as part of the `perspt` CLI. Running `cargo build --release` from the workspace root includes it.

Launch Locally

```
perspt dashboard
```

The server binds to `127.0.0.1:3000` by default. Open `http://localhost:3000` in your browser.

Point to a Specific Database

By default, the dashboard reads from the platform data directory (`~/.local/share/perspt/perspt.db` on Linux). To use a different database file:

```
perspt dashboard --db-path /path/to/perspt.db
```

Embed in Agent Mode

Instead of running the dashboard as a separate process, you can start it alongside the agent using the `--dashboard` flag:

```
perspt agent --dashboard "Create a REST API"
```

This spawns the dashboard as a background task within the agent process. It opens a separate read-only DuckDB connection to the same database file the agent writes to - DuckDB supports one writer plus concurrent readers.

To use a custom port:

```
perspt agent --dashboard --dashboard-port 8080 "Add tests"
```

The embedded dashboard provides the same interface as the standalone `perspt dashboard` command: overview, session detail, topology, backlog, energy, decisions, and governance pages. It stops automatically when the agent process exits.

> **Note**
> The embedded dashboard is especially useful for headless CI runs (`--yes --dashboard`) where you want a browser view of progress without a separate terminal.

Configuration Guide

Perspt supports zero-config auto-detection, environment variables, a TOML config file, and command-line flags. They are applied in this priority order (highest first):

1. **Command-line arguments**
2. **Configuration file** (`config.toml`)
3. **Environment variables**
4. **Auto provider detection**
5. **Built-in defaults**

Automatic Provider Detection

Set any supported API key environment variable and run `perspt` with no arguments:

Prior-ity	Provider	Environment Variable	Default Model
1	Google Vertex AI	`VERTEX_PROJECT_ID`	`vertex::gemini-2.5-flash`
2	Google Gemini	`GEMINI_API_KEY`	`gemini-3.1-flash-lite-preview`
3	OpenAI	`OPENAI_API_KEY`	`gpt-4o-mini`
4	Anthropic	`ANTHROPIC_API_KEY`	`claude-3-5-sonnet-20241022`
5	Groq	`GROQ_API_KEY`	`llama-3.1-8b-instant`
6	Cohere	`COHERE_API_KEY`	`command-r-plus`
7	XAI	`XAI_API_KEY`	`grok-beta`
8	DeepSeek	`DEEPSEEK_API_KEY`	`deepseek-chat`
9	Ollama	*(none - auto-detected)*	`llama3.2`

```
# Example: set a key and run
export GEMINI_API_KEY="your-key"
perspt               # auto-detects Gemini, uses gemini-3.1-flash-lite-preview
perspt chat --model gemini-3.1-pro   # override model
```

Advanced Enterprise Provider Configurations

Unlike standard API-key based providers, enterprise platforms like **Google Agent Platform (formerly Vertex AI)** require multi-part configurations and secure credentials to function.

Google Agent Platform (formerly Vertex AI)

Google Agent Platform/Vertex AI uses secure OAuth2 Bearer Tokens rather than standard static API keys. You must supply your Google Cloud Project ID and regional location alongside the access token.

Required Environment Variables:

Variable	Description
`VERTEX_API_KEY`	Your OAuth2 Bearer access token (generate dynamically via gcloud CLI).
`VERTEX_PROJECT_ID`	Your Google Cloud Platform (GCP) Project ID.
`VERTEX_LOCATION`	The GCP region hosting Vertex AI resources (e.g., `us-central1` or `europe-west3`).

Token Generation Quickstart:

Since OAuth2 access tokens are short-lived (usually expiring in 1 hour), you can export the token dynamically in your shell before running Perspt:

```
# 1. Authenticate with Google Cloud CLI
gcloud auth login

# 2. Configure variables and inject your access token
export VERTEX_PROJECT_ID="your-gcp-project-123"
export VERTEX_LOCATION="us-central1"
export VERTEX_API_KEY=$(gcloud auth print-access-token)

# 3. Launch Perspt using a Vertex AI model
perspt chat --model gemini-3.5-flash
```

Supported Models & Naming Conventions

In version 0.6.6, a fully qualified model name resolves its own provider: a `provider::model` prefix (`openai`, `anthropic`, `gemini`/`google`, `vertex`, `groq`, `cohere`, `ollama`, `xai`, or `deepseek`) selects that provider directly, and the model part is passed through verbatim. A bare model name uses the configured or auto-detected provider instead.

```
perspt chat --model vertex::gemini-3.1-pro   # provider from the namespace
perspt chat --model gemini-3.1-pro           # provider from config/detection
```

Configuration File

Perspt reads `config.toml` from the platform config directory, or from an explicit path:

1. Path given via `perspt --config <PATH>`
2. `~/.config/perspt/config.toml` (Linux)
3. `~/Library/Application Support/perspt/config.toml` (macOS)
4. `%APPDATA%\perspt\config.toml` (Windows)

All fields are optional. `provider` accepts the aliases `provider_type` and `default_provider`; `model` accepts the alias `default_model`.

Minimal example:

```
provider = "gemini"
model = "gemini-3.1-pro"
api_key = "your-key"
```

Full example:

```
provider = "openai"
model = "phi-4-npu-ov"
api_key = "your-key"
# Override the endpoint for OpenAI-compatible / local / proxy servers
base_url = "http://localhost:8000/v1"

# Optional per-role routes for `perspt agent`, as fully qualified
# `provider::model` values. When present, the [models] table takes
# precedence over the flat *_model fields.
[models]
architect = "openai::gpt-5.5"
actuator = "openai::gpt-5-mini"
verifier = "openai::gpt-5-mini"
speculator = "openai::gpt-5-mini"
adjudicator = "openai::gpt-5.5"
```

> **Note**
> `base_url` overrides the endpoint for the active provider. This is useful for Azure OpenAI, proxy servers, local OpenAI-compatible servers, or self-hosted endpoints. You can also set the provider's `*_BASE_URL` environment variable (`OPENAI_BASE_URL`, `OLLAMA_BASE_URL`, …).

> **Note**
> Custom model names that genai does not recognize (for example `phi-4-npu-ov`) are routed to the configured `provider`. You can also target an adapter inline with namespacing, e.g. `openai::phi-4-npu-ov`.

Agent Configuration Blocks

The agent runtime reads four optional TOML blocks. All fields are optional; invalid values fail at startup.

Bounded search (`[exploration]`):

```
[exploration]
initial_branches = 1        # Branches opened before any expansion trigger
max_branches = 3            # Branch identities per forest (hard cap 3)
distinct_family = true      # Prefer a distinct model family on expansion
max_workspace_files = 2048  # Cumulative eager-copy file reservation cap
max_workspace_bytes = 134217728  # Cumulative eager-copy byte reservation cap
```

Prompt bundles (`[prompts]`):

```toml
[prompts]
bundles = ["./prompt-bundles/tuned"]  # External bundles, pinned at session start
activation_min_tasks = 30    # Minimum paired activation tasks (floor 30; raise-only)
noninferiority_margin = 0.05 # Noninferiority margin epsilon in [0, 0.05]
```

Resident-context reserves (`[context]`):

```toml
[context]
working_set_turns = 8          # Verbatim turns kept in the working set
synopsis_frame_tokens = 2048   # Token reserve for the synopsis frame
output_reserve_tokens = 8192   # Token reserve for model output
guard_reserve_tokens = 1024    # Guard reserve against overflow
```

Verification acceptance and test evidence (`[verification]`):

```toml
[verification]
test_policy = "evolving"  # Default: resulting code, tests, and configuration
allow_unisolated = false # Explicit reduced-isolation process execution
require_format = false    # Declare the plugin format stage as an acceptance sensor
stage_timeout_secs = 180  # Wall-clock limit for every governed verifier stage
test_timeout_secs = 300   # Per-stage override (also syntax/build/lint/format)
```

The test policy defines which test evidence must pass in addition to the coding domain's required syntax and build stages:

`allow_unisolated` is `false` by default. Enable it only when the entire Perspt process is isolated externally or when native Windows coding is more important than OS process isolation. It permits verifiers, read-only inspection commands, and language servers to run with the host user's authority; it does not turn those processes into a sandbox. The CLI also offers the one-run `--allow-unisolated` switch. A config value is preferable when an interrupted native Windows session may need `perspt resume`.

Policy	Acceptance behavior
`evolving`	The default for iterative development. Perspt runs the resulting implementation, resulting project tests, and resulting configuration. Existing tests may be corrected, replaced, or removed when the task intentionally changes their contract; newly written tests participate in the same gate.
`backward-compatible`	Runs the resulting suite and a second regression view in which recognized pre-existing test files are restored. Select it only when the task promises compatibility with those historical expectations.
`external-oracle`	Runs the resulting suite and then overlays separately protected acceptance material onto a private candidate copy. Its configured command is an additional required test-stage verdict. This is intended for CI, security fixes, contractual acceptance, and other work with an independently maintained suite.

An external oracle is explicit and fail-closed:

```toml
[verification]
test_policy = "external-oracle"
test_timeout_secs = 600
```

```
[verification.external_oracle]
path = "/srv/project-acceptance"       # or relative to the agent workspace
command = "cargo test --test acceptance"
```

The directory is copied over a private copy of the candidate only at the measurement boundary. It can contain tests, manifests, runner configuration, or harness scripts, and none of those files are promoted. Keep the directory outside the workspace when its contents must be withheld from the actuator. Configuring the table without `test_policy = "external-oracle"`, or selecting that policy without the table, is a startup error rather than silently ignored configuration.

`evolving` does not claim that model-authored tests are independent proof of semantic correctness. It means the configured project verification suite passed for the new contract. Use protected acceptance evidence when an independent semantic oracle is required.

> **Note**
> The `[ensemble]` section was removed by PSP-10 and is now a hard startup error: the proposal ensemble is replaced by the bounded search forest, and the error message points to `[exploration]`.

Command-Line Flags

Global flags apply to all subcommands:

Flag	Description
`-v, --verbose`	Enable verbose logging
`-c, --config <PATH>`	Path to configuration file
`-h, --help`	Show help
`-V, --version`	Show version

Chat-specific:

```
perspt chat --model <MODEL>
```

Agent-specific (see Agent Options Reference for the full list):

```
perspt agent [OPTIONS] "<TASK>"

# Key options:
--model <MODEL>              # Primary actuator alias
--actuator-model <MODEL>     # Governed tool-call route
--explorer-model <MODEL>     # Optional cheap no-tool exploration
--adjudicator-model <MODEL>  # Optional no-tool diff veto
--fallback-model <MODEL>     # Repeatable sticky actuator fallback
-w, --workdir <DIR>          # Working directory
-y, --yes                    # Auto-approve (headless)
--rho-gate <V>               # Required measured descent
--max-turns <N>              # Finite model-turn budget
```

```
--max-calls-per-turn <N>     # Direct and nested call budget
--rejection-budget <N>       # Shared recovery/rejection budget
--max-parallel <N>           # Parallel verifier sensors
--max-parallel-nodes <N>     # Concurrent work-graph nodes (>1 needs --yes)
--exploration-only           # Read-only exploration; nothing mutated
--allow-experimental-prompts # Substitute validated [prompts] bundles live
--domain <ID>                # Domain package (coding, research); default: detect
--allow-dependency-mutation  # Grant governed dependency mutation
--persistent-grants          # Sign durable grant intent
--db-path <PATH>             # PSP-9 ledger database path
--dashboard                  # Start the web dashboard alongside the agent
--dashboard-port <N>         # Embedded dashboard port (default 3000)
--output-summary <FILE>      # Terminal session summary as JSON
```

Manage configuration interactively:

```
persp config --show    # Print the effective config (api_key masked)
persp config --edit    # Open in $EDITOR
persp config --set provider=gemini
persp config --set default_model=gemini-3.1-pro
```

Initialize project-level configuration:

```
persp init --memory --rules
```

Dashboard Configuration

The `persp dashboard` subcommand accepts these options:

Flag	Default	Description
`--port`	`3000`	HTTP port for the dashboard server
`--db-path`	Platform default	Path to the DuckDB database file

The dashboard opens the database in **read-only** mode and never writes to it. The server always binds to `127.0.0.1`; cookies are set without the `Secure` flag so plain HTTP works on localhost.

Reference

Precise, authoritative reference for Perspt's interface.

Reference	Scope
CLI Reference	Complete command-line interface documentation.
Key Bindings	TUI keyboard shortcuts and review modal controls.
Advanced Troubleshooting	Advanced diagnostics and known issues.

CLI Reference

```
perspt [OPTIONS] [COMMAND]
```

Global Options

Flag	Description
`-v, --verbose`	Enable verbose logging
`-c, --config <PATH>`	Path to the TOML configuration file
`-h, --help`	Print help
`-V, --version`	Print version

Commands

10.1.2.1 `chat` (default)

Launch the TUI chat interface.

```
perspt chat [--model MODEL]
```

10.1.2.2 `simple-chat`

Launch the plain-text CLI chat.

```
perspt simple-chat [--model MODEL] [--log-file FILE]
```

10.1.2.3 `dashboard`

Launch the real-time web monitoring dashboard.

```
perspt dashboard [--port PORT] [--db-path PATH]
```

- `--port` - HTTP port (default `3000`)

- `--db-path` - Path to DuckDB database file (default: platform data directory)

The server always binds to `127.0.0.1`.

See Dashboard Setup for configuration details.

10.1.2.4 `agent`

Run the SRBN autonomous coding agent.

```
perspt agent [OPTIONS] <TASK>
```

- `--dashboard` - Start the web monitoring dashboard alongside the agent
- `--dashboard-port <PORT>` - Port for the embedded dashboard (default `3000`)
- `--allow-unisolated` - Explicit reduced-isolation process execution; required for native Windows coding until a sandbox backend exists

See Agent Options Reference for full agent options.

10.1.2.5 `init`

Initialize project memory and policy rules.

```
perspt init [--memory] [--rules]
```

- `--memory` - Create the `PERSPT.md` project memory file
- `--rules` - Create default Starlark policy rules

10.1.2.6 `config`

View or edit Perspt configuration.

```
perspt config [--show] [--set KEY=VALUE] [--edit]
```

- `--show` - Print the effective config (`api_key` masked)
- `--set KEY=VALUE` - Set a value with a structured TOML write
- `--edit` - Open the config file in `$EDITOR`

10.1.2.7 `ledger`

Query the Merkle ledger.

```
perspt ledger [--recent] [--stats] [--rollback SESSION_PREFIX]
```

- `--recent` - Show recent commits
- `--stats` - Show ledger statistics
- `--rollback SESSION_PREFIX` - Undo the session's newest completed promotion and label it unsafe

10.1.2.8 `status`

Show current session status.

```
perspt status
```

Displays: ledger event count, ledger head, measurement count, last energy, last gate decision, denial count, search forest/branch/no-good counters, validator verdicts, the Φ(W) conditional-

capacity diagnostic, validator-independence statistics, pending external effects, and a `perspt replay` hint.

10.1.2.9 `abort`

Abort a PSP-9 session by revoking its authority epoch.

```
perspt abort [--force] [SESSION_ID]
```

- `-f, --force` - Force abort without confirmation
- `SESSION_ID` - Session to abort (defaults to the newest running PSP-9 session)

10.1.2.10 `resume`

Resume an interrupted session.

```
perspt resume [SESSION_ID] [--last] [--db-path PATH]
```

- `SESSION_ID` - Session ID to resume
- `--last` - Resume the most recent session
- `--db-path` - Database file to inspect (defaults to the standard store)

Displays the session id, task, working directory, and status before resuming, and verifies the ledger chain is valid.

10.1.2.11 `audit`

Delayed audit labels and conformal activation (PSP-9).

```
perspt audit [SAMPLE] [--safe] [--unsafe]
```

- `SAMPLE` - Sample id (or unique prefix) to label; omit to list pending samples
- `--safe` - Label the sample as safe
- `--unsafe` - Label the sample as unsafe

10.1.2.12 `providers`

Print the provider capability matrix (PSP-9).

```
perspt providers [--probe]
```

- `--probe` - Run live behavioral probes against every configured model route

10.1.2.13 `replay`

Deterministic, credential-free audit replay of a session (PSP-9).

```
perspt replay <SESSION_ID> [--db-path PATH]
```

- `SESSION_ID` - The session id to replay
- `--db-path` - Database file to inspect (defaults to the standard store)

10.1.2.14 `db`

Inspect and repair the local DuckDB store.

```
perspt db repair --db-path PATH [--discard-wal]
```

- `repair` - Quarantine a poisoned WAL after making durable backups
- `--db-path` - Database file to repair
- `--discard-wal` - Explicitly authorize WAL quarantine; the WAL is never deleted

10.1.2.15 `prompts`

Inspect and maintain the compiled prompt section libraries (PSP-10).

```
perspt prompts list
perspt prompts render <STAGE>
perspt prompts lint [--bundle DIR]
perspt prompts manifest <DIR>
perspt prompts explain-session --db-path PATH <SESSION_ID>
```

- `list` - List every compiled section: id, version, stage, role, hash
- `render <STAGE>` - Compose one stage with fixture variables and print it
- `lint` - Run the codegen validation list over an external bundle directory
- `manifest <DIR>` - Regenerate a prompt library's committed `manifest.toml` (explicit)
- `explain-session` - Show the programs a session actually compiled, with digests

10.1.2.16 `context`

Explain a session's recorded resident-context events (PSP-10).

```
perspt context explain-turn --db-path PATH <SESSION_ID>
```

- `explain-turn` - Show a session's recorded context events (compactions, refusals)

10.1.2.17 `benchmark` (optional)

Build `perspt-cli` with the Cargo feature `benchmark` to expose the separate evaluation runner. It is absent from the default CLI build.

```
perspt benchmark validate
perspt --config config.local.toml benchmark run --suite smoke --output report.json
perspt benchmark aggregate report-a.json report-b.json
```

`validate` is credential-free. `run` is explicit and credentialed; its `smoke`, `adaptive`, and `full` suites use production role resolution from the selected Perspt configuration and record the configured topology. `run --tasks <N>` overrides the suite's task count. Coding verification stays deterministic, so a configured verifier route is provenance rather than a model call. Model names and family labels are not benchmark arguments. The benchmark is manual-only and feature-gated — the `perspt-benchmark` crate is a separate optional crate — and a benchmark run is never part of normal validation or CI.

Key Bindings

TUI Chat Mode

Key	Action
Enter	Send message
Ctrl+J / **Ctrl+Enter**	Insert newline in input
Page Up	Scroll chat up 10 lines
Page Down	Scroll chat down 10 lines
Ctrl+Up / **Shift+Up**	Scroll up one line
Ctrl+Down / **Shift+Down**	Scroll down one line
Up / **Down**	Traverse prompt history (or move between input lines)
Home	Move input cursor to beginning of line
End	Move input cursor to end of line
Ctrl+R	Toggle reasoning display
Ctrl+C / **Ctrl+Q**	Quit

Review Modal (Agent Mode)

Key	Action
y	Approve node changes
n	Reject node changes
c	Send correction feedback
e	Open files in editor
d	Toggle diff view
s	Skip node
Left / **Right**	Move action selection
Enter	Confirm selected action
Esc	Dismiss modal

Agent Mode

Key	Action
q	Quit
p	Pause/resume the agent
Tab / **Shift+Tab**	Cycle tabs

Key	Action
1 / 2 / 3	Jump to Dashboard / Tasks / Diff tab
Up / k	Navigate up
Down / j	Navigate down
Page Up / Page Down	Page navigation
Space / Enter	Select item
a	Open the approval modal

Input Editor

Key	Action
Left / Right	Move cursor
Ctrl+B / **Ctrl+F**	Move cursor left / right
Ctrl+A	Move to beginning of line
Ctrl+E	Move to end of line
Ctrl+W	Delete word backward
Ctrl+U	Delete to beginning of line
Ctrl+K	Delete to end of line
Backspace	Delete character backward
Delete	Delete character forward
Ctrl+D / **Ctrl+H**	Delete / Backspace character

Advanced Troubleshooting

Diagnostic Commands

```
# Session status with per-node details
perspt status

# Deterministic, credential-free audit replay of a session
perspt replay <SESSION_ID>

# Ledger integrity check
perspt ledger --stats

# Enable verbose logging
RUST_LOG=debug perspt simple-chat 2>debug.log
```

Common Error Patterns

10.3.2.1 Missing API key

When no provider is configured, Perspt detects one from environment keys in a fixed order: Vertex AI project settings first, then `GEMINI_API_KEY`, `OPENAI_API_KEY`, `ANTHROPIC_API_KEY`, `GROQ_API_KEY`, `COHERE_API_KEY`, `XAI_API_KEY`, and `DEEPSEEK_API_KEY`, falling back to a local Ollama setup when none are present. Set the provider environment variable or store a key with `perspt config --set api_key=<KEY>`.

10.3.2.2 `Unsupported provider`

The configured `provider` is not a recognized adapter. Supported values: `openai`, `anthropic`, `gemini` (or `google`), `vertex`, `groq`, `cohere`, `ollama`, `xai`, `deepseek`.

10.3.2.3 Provider failover

When a model route fails, the agent fails over to the next configured fallback route. The failover is charged to the shared recovery pool, so repeated provider failures can exhaust the session's rejection budget.

10.3.2.4 Poisoned DuckDB WAL

If the local DuckDB store fails to open because of a poisoned WAL, run `perspt db repair --db-path <PATH> [--discard-wal]`. The repair makes durable backups first and quarantines the WAL; the WAL is never deleted.

Agent-Specific Issues

Footprint conflict:

Two nodes declared overlapping file footprints. The dispatcher's scheduler never runs conflicting nodes concurrently — the later node waits for the conflicting node to finish. If this persists, simplify the task description.

Global integration failure:

Two nodes may pass separately and fail together. Node winners enter a content-addressed staging root instead of the user workspace, and one global verifier gate runs the full domain suite and the test evidence selected by `[verification] test_policy` on the combined state. Only a hard-passing integration root is promoted, atomically; on failure the prior staging root is restored and the user workspace is left byte-identical.

Degraded verification:

When some checks could not run, the review modal marks a node's stability metrics as degraded and lists the reasons. Address the listed reasons to restore full verification.

Terminal Restoration

If Perspt crashes and leaves the terminal in raw mode:

```
reset
# or
stty sane
```

Perspt restores the terminal (raw mode off, leave alternate screen) only on the normal exit path; after a crash, `reset` will fix it.

Performance

Slow response streaming:

1. Check network latency to the provider
2. Try a faster model (e.g., `gemini-3.5-flash`)
3. Use Groq for the fastest inference

High memory usage in agent mode:

1. Large DAGs with many nodes consume more memory
2. Use `--max-turns` and `--max-calls-per-turn` to bound per-node work
3. Use `--rejection-budget` to bound recovery attempts

API & SDK Reference

Perspt is built as a modular 14-crate Rust workspace. In addition to the `perspt` CLI binary, you can embed ``**perspt-sdk**``, ``**perspt-coding**``, and ``**perspt-agent**`` directly into your own Rust applications to build verified AI workflows and custom domain agents.

Integrating the Perspt SDK

Add `perspt-sdk` (and optionally `perspt-coding` for software engineering verifiers) to your project's `Cargo.toml`:

```
[dependencies]
perspt-sdk = { git = "https://github.com/eonseed/perspt" }
perspt-coding = { git = "https://github.com/eonseed/perspt" }
```

You can then score verifier diagnostics with the canonical quadratic Lyapunov energy $V(x) = \sum_{e \in E} w_e \parallel r_e(x) \parallel^2$ and evaluate the acceptance gate in your own Rust code:

```
use perspt_coding::CodingDomain;
use perspt_sdk::{
    domain::{AgentDomainPackage, DomainScope},
    energy::score_candidate,
    gate::evaluate_gate,
};

fn check_candidate(scope: &DomainScope, residuals: &[perspt_sdk::residual::ResidualEvent],
best_v: f64) -> perspt_sdk::error::Result<bool> {
    let domain = CodingDomain;
    let model = domain.energy_model(scope);
    let score = score_candidate(residuals, &model)?;
    let decision = evaluate_gate(
        score.hard_violations.is_empty(),
        score.total,
        best_v,
        model.rho_gate,
    )?;
    Ok(decision.is_accepted())
}
```

> **Tip**
> For full Rustdoc-generated API documentation across all workspace crates, open the Rustdoc API Reference or run locally:
>
> ```
> cargo doc --open --workspace --no-deps
> ```

Crate	Responsibility
perspt-core	Types, config, LLM provider, events, plugins.
perspt-sdk	Domain-neutral SRBN control plane: energy, gates, ledger.
perspt-coding	Coding domain package: residuals, adapters, barriers.
perspt-research	Research domain package (second domain proof).
perspt-agent	Governed candidate runtime, tool loop, verifier, work-graph dispatch.
perspt-tui	Ratatui terminal UI (chat + agent).
perspt-cli	Clap CLI entry point and subcommands.
perspt-store	DuckDB session persistence.
perspt-policy	Starlark policy engine.
perspt-sandbox	Command sandboxing and isolation.
perspt-dashboard	Axum + Askama + HTMX web dashboard.
perspt-prompt-macros	Build-time compiler for typed prompt section libraries.
perspt-benchmark	Optional, feature-gated manual evaluation tooling.

`perspt-core`

The foundation crate providing all canonical types, configuration, LLM provider integration, event system, and language plugins.

Modules

Module	Description
`types`	Core shared types: NodeClass, ModelTier, EnergyComponents, TaskPlan, PlannedTask, PlannedContract, PlannedTest, TaskType, CommandContract, SensorStatus, StageOutcome, VerifierStrictness, StructuralDigest, SummaryDigest, ArtifactKind, ContextBudget, RestrictionMap, ContextPackage, ContextProvenance
`config`	`Config` - provider/model/API-key basics plus the `[providers]`, `[models]`, `[external_tools]`, `[verification]`, `[exploration]`, `[prompts]`, and `[context]` tables
`events`	`AgentEvent` (33 variants), `AgentAction`, `NodeStatus`, `ActionType`, channel types
`llm_provider`	`GenAIProvider`, `LlmResponse`, `EOT_SIGNAL`, streaming support
`plugin`	`LanguagePlugin` trait, `PythonPlugin`, `RustPlugin`, `JsPlugin`, `PluginRegistry`
`memory`	`ProjectMemory` from `.perspt/memory.toml`
`normalize`	Model and provider name normalization
`prompts`	`PlatformPromptLibrary` - the platform prompt section library, compiled at build time by `perspt-prompt-macros`

Module	Description
`portfolio`	`ModelPortfolio`, `ProviderHandle` - several concurrently live provider handles
`tools_driver`	`CoreToolCall`, `CoreToolSpec`, `CoreTurnOutput` - the `genai` tool-calling driver
`local_command`	`LocalCommand` - provider-independent commands handled entirely by frontends
`path`	Canonical path resolution for artifact paths
`paths`	Centralized platform-aware path helpers (config/data/project tiers)

Key Types

Config - The main configuration struct (excerpt):

```rust
pub struct Config {
    pub provider: Option<String>,
    pub model: Option<String>,
    pub api_key: Option<String>,
    pub base_url: Option<String>,
    // ... per-tier model overrides, plus the [providers], [models],
    // [external_tools], [verification], [exploration], [prompts],
    // and [context] tables
}
```

EnergyComponents - Lyapunov energy decomposition:

```rust
pub struct EnergyComponents {
    pub v_syn: f32,    // LSP diagnostics
    pub v_str: f32,    // Contract compliance
    pub v_log: f32,    // Test results
    pub v_boot: f32,   // Bootstrap commands
    pub v_sheaf: f32,  // Cross-node validation
}
```

AgentEvent - 33 lifecycle events:

`TaskStatusChanged`, `PlanGenerated`, `EnergyUpdated`, `Log`, `NodeCompleted`, `ApprovalRequest`, `Complete`, `Error`, `PlanReady`, `NodeSelected`, `FallbackPlanner`, `VerificationComplete`, `BundleApplied`, `SensorFallback`, `DegradedVerification`, `EscalationClassified`, `SheafValidationComplete`, `GraphRewriteApplied`, `BranchCreated`, `InterfaceSealed`, `BranchFlushed`, `DependentUnblocked`, `BranchMerged`, `ContextDegraded`, `ContextBlocked`, `StructuralDependencyMissing`, `ModelFallback`, `ProvenanceDrift`, `ToolReadiness`, `BudgetUpdated`, `PlanRevised`, `FileDeleted`, `FileMoved`

See Architecture for the complete type inventory.

`perspt-sdk`

The domain-neutral SRBN control plane (PSP-8): residual evidence, the canonical quadratic energy, the measured acceptance gate, capabilities, scheduling, and the replay ledger. Domain packages such as `perspt-coding` provide task-specific residual construction, weights, and correction directions on top.

Modules

Module	Description
`residual`	`ResidualEvent` - the measured reason a state is unsafe or incomplete
`energy`	`score_candidate`/`EnergyModel` - the canonical quadratic residual energy $V(x) = \sum_e w_e \parallel r_e \parallel^2$
`gate`	Measured acceptance gate and finite-decision bound
`kernel`	Adapter over the published `srbn` kernel crate
`domain`	`AgentDomainPackage` - the domain-package contract
`capability`	Capability-constrained admissibility kernel: proposals, never unmediated effects
`admissibility`	The five-clause admissibility kernel (PSP-9)
`command`	Typed command IR and governance tiers (no implicit `sh -c`)
`scheduler`	Dependency-aware parallel ready-queue scheduler
`workgraph`	Mutable, revisioned work graph
`ledger`	Append-only, Merkle-chained event stream with audit replay
`checkpoint`	Context checkpoints: compaction as recorded projection, never deletion
`conformal`	Calibrated risk budgets via conformal risk control
`recovery`	The recovery lattice for classified failures
`routing`	Phase-aware model routing (`ModelRoute`, `ModelTier`)
`model`	Provider-neutral model plane identity and contract types (pure data)
`prompt`	The model-conditioned prompt plane: typed sections, composition, digested programs (PSP-10)
`search`	The bounded search plane: forests, branches, witnesses, limits (PSP-10)
`toolset`	The typed tool catalog with effect/risk/footprint contracts
`independence`	Measured validator dependence, never assumed independence
`spectral`	Spectral energy-slope constant μ from the verification graph
`stability`	Analytic stability claims (`NotClaimed` for the coding domain)
`certificate`	`ResidualCertificate` - the honest stop on exhaustion
`goal`	Goal-presence sensor: the verifier that refuses false stability
`observability`	Read-only dashboard/TUI projections over the event ledger
`benchmark`	Mechanism-check benchmark harness and metrics
`canon`	Domain-tagged canonical byte encoding for replayable digests
`scope`	Glob-like scope patterns shared by capabilities and grant policies
`error`	`SdkError`/`Result` - fail-closed numerical contracts

The crate re-exports the published `srbn`, `srbn_ledger`, and `srbn_serde` kernel crates so consumers depend on one SRBN source.

Key Types

evaluate_gate - The measured acceptance gate:

```
pub fn evaluate_gate(
    hard_pass: bool,
    candidate_v: f64,
    best_accepted_v: f64,
    rho_gate: f64,
) -> Result<GateDecision>;

pub fn finite_decision_bound(
    baseline_energy: f64,
    rho_gate: f64,
    rejection_budget: u32,
) -> Result<u64>;
```

EnergyScore - The result of scoring a candidate:

```
pub struct EnergyScore {
    pub total: f64,                       // Total energy V
    pub components: EnergyComponents,     // Derived rollups
    pub dominant: Vec<ResidualEventRef>,  // Dominant residuals first
    pub hard_violations: Vec<ResidualClass>,
}
```

`perspt-coding`

The coding domain package built on `perspt-sdk`: it declares the coding residual schema, supplies the coding energy model (weights, ρ_{gate}, correction budget), and maps dominant residuals into coding correction directions.

Modules

Module	Description
`barrier`	`OperationalSafetyBarrier` - versioned operational safety channels; correctness evidence stays in `V`, never duplicated into `h`
`diag`	The structured verifier plane: adapters preserve as much source structure as each tool provides
`lang`	Language adapters (Rust, Python, TypeScript) as verifier suites
`prompts`	The coding prompt section library; `branch_correct` renders a typed `CorrectionPacket` for the model
`registry`	`CodingAdapterRegistry`/`LanguageId` - open language-adapter registry keyed by id, never enum dispatch
`runtime`	Generic runtime smoke-probe scheme for exercising built artifacts
`symbols`	Symbol extraction for the goal-presence sensor

Key Types

CodingDomain - The `AgentDomainPackage` implementation:

```
pub struct CodingDomain;

impl AgentDomainPackage for CodingDomain {
    fn domain_id(&self) -> DomainId;                // "coding"
    fn detect(&self, workspace: &WorkspaceSnapshot) -> DomainDetection;
    fn residual_schema(&self, scope: &DomainScope) -> ResidualSchema;
    fn energy_model(&self, scope: &DomainScope) -> EnergyModel;
    fn correction_directions(&self, residuals: &[ResidualEvent])
        -> Vec<CorrectionDirection>;
    // ... hard_gate_policy, verifier_suite, safety_barrier
}
```

LanguageId - Open, stable language identifier:

```
pub struct LanguageId(pub String);

pub struct CodingAdapterRegistry {
    adapters: BTreeMap<LanguageId, Box<dyn LanguageAdapter>>,
}
```

The coding domain operates on discrete verifier residuals (compiler, LSP, AST, tests), so its analytic constants remain `NotClaimed`; only the measured discrete gate and the spectral μ apply.

`perspt-research`

The research domain package (skeleton): the second-domain readiness check proving the `perspt-sdk` contracts admit another domain without forking the scheduler, admissibility kernel, residual model, replay ledger, or dashboard event model.

Core Type

ResearchDomain - The `AgentDomainPackage` implementation:

```
pub struct ResearchDomain;

impl AgentDomainPackage for ResearchDomain {
    fn domain_id(&self) -> DomainId;   // "research"
    // detect, residual_schema, energy_model, correction_directions,
    // hard_gate_policy, verifier_suite, safety_barrier
}
```

Residual Mapping

Research reuses the SDK residual classes:

Residual Class	Research Meaning
`TestFailure`	Unsupported / contradicted claim
`InterfaceMismatch`	Source mismatch
`ImportGraph`	Citation gap (missing source link)
`ContextDrift`	Stale evidence

Residual Class	Research Meaning
`SheafInconsistency`	Cross-source contradiction

Detection looks for bibliography markers (`references.bib`, `refs.bib`, `bibliography.bib`, `sources.md`). The verifier suite is a single `citation-check` stage; the safety barrier is `NotClaimed`. Verifier suites for source discovery, citation provenance, and claim extraction are out of scope for the skeleton — the point is contract conformance.

`perspt-agent`

The governed PSP-9 agent runtime: candidate workspaces, the SRBN tool loop, sandboxed verification, and work-graph dispatch.

Modules

Module	Description
`candidate`	`CandidateWorkspace` - reversible coding candidate overlay with compiler-backed measurement
`exploration`	Deterministic, read-only repository orientation for the runtime
`external_tools`	`ExternalToolRuntime`/`ChatToolSession` - official-SDK MCP 2026-07-28 client for agent and chat: admitted tools plus read-only resource/prompt/completion operations, roots, sampling, elicitation, subscriptions, tasks, typed server events, and rejection diagnostics
`grant`	Persistent grant signing-key resolution
`lsp`	`LspClient` - JSON-RPC stdio client for language servers (the sensor architecture)
`measure`	`CodingCandidateMeasurer` - full verifier suite at gate boundaries, cheap syntax-only pass at mutation boundaries, and the explicit evolving/backward-compatible/external-oracle test-evidence policies
`probe`	`probe_route`/`ProbeReport` - behavioral provider probes (Gate U)
`promote`	Descriptor-relative workspace promotion
`realize`	`SnapshotRealizer`/`ProjectionMismatch` - the gate is evaluated on the realized candidate workspace, never the model's account of it
`runtime`	`Psp9AgentRuntime` - the authoritative PSP-9 agent runtime
`toolloop`	`ToolLoop` - the SRBN tool loop; every model-issued tool call is a governed proposal through the admissibility kernel
`tools`	`AgentTools` - filesystem and search operations plus the open `CandidateHandlerRegistry` the governed candidate dispatches through
`transport`	`GenAiTransport` - the only adapter joining the SDK's provider-neutral contract and `perspt-core`'s `genai` driver
`turn`	Universal actor turn runner shared by every stochastic actor (PSP-10)

Module	Description
`verifier`	Governed verifier sandbox: compiler/test/lint processes run behind a deny-network profile and a read allow-list; an explicit reduced-isolation policy is available on hosts without a backend

Key Types

Psp9ModelRoutes - Explicit model-plane routes for one PSP-9 session:

```
pub struct Psp9ModelRoutes {
    pub primary: Option<String>,
    pub actuator: Option<String>,
    pub explorer: Option<String>,
    pub adjudicator: Option<String>,
    pub fallbacks: Vec<String>,
}
```

Psp9RunSummary - Terminal outcome of one run:

```
pub struct Psp9RunSummary {
    pub session_id: String,
    pub node_id: String,
    pub outcome: NodeTerminalOutcome,
    pub turns_used: u32,
    pub ledger_head: String,
    pub promoted_paths: Vec<String>,
}
```

Usage

```
let runtime = Psp9AgentRuntime::from_config(workdir, &config, routes, run_config)?
    .with_database_path(db_path);

let summary = runtime.run(task).await?;
```

`perspt-tui`

Ratatui-based terminal user interface with two application modes.

Applications

Type	Description
`ChatApp`	Interactive chat with markdown rendering and response streaming
`AgentApp`	Agent dashboard with DAG tree, energy display, and review modal

Widgets and Rendering Modules

Module	Description
`dashboard`	`Dashboard` - main agent dashboard layout with panels
`task_tree`	DAG visualization showing node states and energy

Module	Description
`review_modal`	`ReviewModal` - grouped diff viewer with approve/reject/correct controls
`diff_viewer`	`DiffViewer` - unified diff display with syntax highlighting
`simple_input`	Minimal input widget with full control over key handling
`markdown`	Markdown, table, and LaTeX-math rendering for the chat TUI
`latex`	LaTeX to Unicode transpilation for terminal math rendering
`chat_view`	Message-area and input-box rendering for the chat TUI
`chat_commands`	Chat-local slash commands and conversation export
`theme`	`Theme` - color scheme and styling

Entry Points

```
pub fn run_chat_tui(...) -> Result<()>;
pub fn run_agent_tui_with_runtime(...) -> Result<()>;
```

Terminal setup and teardown live in `tui_runner`:

```
pub fn init_terminal(config: &TuiRunnerConfig) -> io::Result<TuiTerminal>;
pub fn restore_terminal(config: &TuiRunnerConfig) -> io::Result<()>;
```

Channels

```
pub fn create_app_event_channel() -> (AppEventSender, AppEventReceiver);
pub fn create_telemetry_channel() -> (TelemetrySender, TelemetryReceiver);
```

`perspt-cli`

Clap-based CLI entry point with sixteen subcommands, plus one feature-gated.

Subcommands

Command	Description
`chat`	TUI chat (default)
`simple-chat`	Plain-text streaming chat
`agent`	PSP-9 autonomous coding agent (22 options)
`init`	Initialize project memory and policy rules
`config`	View/set/edit configuration (`--show`/`--set`/`--edit`)
`ledger`	Query and manage the Merkle ledger (`--recent`/`--rollback <session-prefix>`/`--stats`)
`status`	Show current agent status
`audit`	Delayed audit labels and conformal activation (PSP-9)

Command	Description
`providers`	Print the provider capability matrix; `--probe` runs live probes
`replay`	Deterministic, credential-free audit replay of a session (PSP-9)
`abort`	Abort a PSP-9 session by revoking its authority epoch
`resume`	Resume a paused or crashed session
`dashboard`	Launch the web monitoring dashboard
`db`	Inspect and repair the local DuckDB store (`repair`)
`prompts`	Inspect and maintain the compiled prompt section libraries (`list`/`render`/`lint`/`manifest`/`explain-session`)
`context`	Explain a session's recorded resident-context events (`explain-turn`)
`benchmark`	Optional model-backed evaluation tooling (`validate`/`run`/`aggregate`); feature-gated behind the `benchmark` feature and run manually, never in CI

See CLI Reference for the complete flag reference.

`perspt-store`

DuckDB-backed session and ledger persistence.

> **Important**
> Perspt uses **DuckDB**, not SQLite, for its session store.

Core Type

```rust
pub struct SessionStore {
    conn: Mutex<Connection>,  // duckdb::Connection
}

impl SessionStore {
    pub fn new() -> Result<Self>;           // Default path
    pub fn open(path: &Path) -> Result<Self>; // Custom path
    pub fn open_read_only(path: &Path) -> Result<Self>; // Read-only mode
    pub fn default_db_path() -> PathBuf;    // ~/.local/share/perspt/
}
```

> **Note**
> `open_read_only` uses DuckDB's `AccessMode::ReadOnly` and does **not** call `init_schema()`. This makes it safe for concurrent dashboard reads alongside the agent's write lock. The database file must already exist.

Record Types

Type	Description
`SessionRecord`	session_id, task, working_dir, merkle_root, detected_toolchain, status
`Psp9LedgerRow`	session_id, sequence, event_json, prev_hash, hash - one durable PSP-9 ledger record
`Psp9VerdictRow`	session_id, candidate_id, validator_id, stratum, missed, unsafe_label, evidence_hash
`Psp9CalibrationEpochRow`	epoch_id, stratum, target_rho, threshold, state, sample_count
`Psp9ExternalEffectRow`	idempotency_key, intent_hash, intent_json, result_json, status

The ledger accessors on `SessionStore` (append, read, and replay over the `psp9_*` tables) live in `store/psp9_ledger.rs`; `repair.rs` provides `repair_database`/`RepairReport`, the recoverable WAL quarantine behind `perspt db repair`.

DuckDB Tables

The schema is initialized by `init_schema()` through an idempotent transactional migration and comprises 11 tables: `sessions`, `schema_migrations`, `psp9_ledger_events`, `psp9_artifacts`, `psp9_authority_epochs`, `psp9_calibration_epochs`, `psp9_calibration_samples`, `psp9_context_checkpoints`, `psp9_external_effects`, `psp9_grant_policies`, `psp9_verdicts`.

`perspt-policy`

Starlark-based policy engine for agent action governance.

Core Types

```rust
pub struct PolicyEngine {
    policies: Vec<String>,
    policy_dir: PathBuf,
}

pub enum PolicyDecision {
    Allow,
    Prompt(String),
    Deny(String),
}

pub struct SanitizeResult {
    pub parts: Vec<String>,
    pub warnings: Vec<String>,
    pub rejected: bool,
    pub rejection_reason: Option<String>,
}
```

Functions

Function	Description
`PolicyEngine::new()`	Create engine instance
`PolicyEngine::load_policies()`	Load all .star files from policy directory

Function	Description
`sanitize_command(cmd)`	Parse, validate, and filter a shell command
`validate_workspace_bound(cmd, dir)`	Ensure command stays within working directory
`validate_artifact_mutation(path, root, op)`	Guard destructive artifact operations (delete/move) against protected project files

`perspt-sandbox`

Process isolation for agent-executed commands.

Core Types

```rust
pub struct CommandResult {
    pub stdout: String,
    pub stderr: String,
    pub exit_code: Option<i32>,
    pub timed_out: bool,
    pub duration: Duration,
}

pub trait SandboxedCommand: Send + Sync {
    fn execute(&self) -> Result<CommandResult>;
    fn display(&self) -> String;
    fn is_read_only(&self) -> bool;
}

pub struct BasicSandbox {
    program: String,
    args: Vec<String>,
    working_dir: Option<PathBuf>,
    timeout: Duration,
}
```

Usage

```rust
let sandbox = BasicSandbox::new("cargo", vec!["test"])
    .with_working_dir("/path/to/project")
    .with_timeout(Duration::from_secs(60));

let result = sandbox.execute()?;
assert!(result.exit_code == Some(0));
```

`BasicSandbox` can also be created from a command string:

```rust
let sandbox = BasicSandbox::from_command_string("uv run pytest -v")?;
```

`perspt-dashboard`

Real-time web dashboard for Perspt agent monitoring, built with Axum + Askama + HTMX.

Core Types

```
pub struct AppState {
    pub store: Arc<SessionStore>,        // read-only DuckDB
    pub password: Option<String>,         // optional auth password
    pub session_token: Arc<Mutex<Option<String>>>,
    pub working_dir: PathBuf,
    pub is_localhost: bool,               // controls Secure cookie flag
}

pub enum DashboardError {
    Store(anyhow::Error),     // 503 - DB unavailable
    Template(askama::Error),  // 500 - render failure
    Internal(String),         // 500 - generic
}
```

`DashboardError` implements `IntoResponse` and renders a styled HTML error page. Store errors return `503 Service Unavailable`.

Router

`build_router(state: AppState) -> Router` constructs the full Axum router:

Method	Route	Handler
GET	`/login`	`auth::login_page` - render login form
POST	`/login`	`auth::login_handler` - validate password, set session cookie
GET	`/`	`handlers::overview::overview_handler` - session list
GET	`/sessions/{session_id}`	`handlers::session_detail::session_detail_handler` - session detail
GET	`/sessions/{session_id}/topology`	`handlers::dag::topology_handler` - work-graph topology
GET	`/sessions/{session_id}/backlog`	`handlers::backlog::backlog_handler` - backlog diagnostics
GET	`/sessions/{session_id}/energy`	`handlers::energy::energy_handler` - energy convergence
GET	`/sessions/{session_id}/decisions`	`handlers::decisions::decisions_handler` - decision trace
GET	`/sessions/{session_id}/governance`	`handlers::governance::governance_handler` - governance view
GET	`/sse/{session_id}`	`sse::sse_handler` - SSE event stream

Static assets are served from `crates/perspt-dashboard/static/` under `/static` via `tower_http::services::ServeDir`.

All routes except `/login` and `/static` are behind `auth::auth_middleware`. If no password is configured, all requests pass through.

Auth Middleware

Cookie-based authentication with random session tokens:

- On successful login, generates a 32-character alphanumeric token

- Stores token in `AppState::session_token`
- Sets `perspt_session` cookie: `HttpOnly`, `SameSite=Lax`, `Path=/`, `Secure` (when not localhost)
- Middleware checks cookie value against stored token
- No password configured -> open access mode

SSE Stream

The SSE endpoint pushes named events every 2 seconds:

- `psp9-stats` - live session summary (ledger event count, measurement count, last measured energy)

Each event contains an HTML fragment suitable for HTMX `sse-swap`.

Templates

Askama templates live in `crates/perspt-dashboard/templates/`:

- `base.html` - layout with navigation, HTMX, and DaisyUI theme
- `login.html` - login form
- `session_base.html` - shared per-session layout with tab navigation
- `pages/overview.html` - session list table
- `pages/session_detail.html` - single-session summary
- `pages/dag.html` - work-graph topology view
- `pages/backlog.html` - backlog diagnostics
- `pages/energy.html` - energy component table
- `pages/governance.html` - governance view
- `pages/decisions.html` - flat Merkle-chained PSP-9 event trace

View Models

Each page has a corresponding view model in `src/views/`:

- `OverviewViewModel` - sessions, node counts, budgets
- `SessionDetailViewModel` - single-session summary
- `TopologyViewModel` - work-graph nodes and edges
- `BacklogViewModel` - backlog diagnostics
- `EnergyViewModel` - per-node energy components
- `GovernanceViewModel` - governance evidence
- `DecisionsViewModel` - the flat Merkle-chained PSP-9 event trace

`views/psp9.rs` provides the shared `LedgerProjection` these view models and the SSE stream are built from.

perspt-prompt-macros

Build-time compiler for Perspt prompt section files (PSP-10). Called from an owning crate's `build.rs`, it parses every `prompts/<stage>/NN_name.md`, runs the codegen validation list, and emits typed section structs into `OUT_DIR`. A malformed section fails `cargo build` with an error naming the offending file — never a session.

Core Types

```rust
pub struct StageDecl {
    pub dir_name: String,
    pub separator: String,
}

pub struct PromptBuildError {
    pub file: String,
    pub line: Option<usize>,
    pub message: String,
}

pub struct Generated {
    pub rust_source: String,
    pub sections: Vec<CompiledSection>,
}

pub fn compile_prompt_dir(
    root: &Path,
    stages: &[StageDecl],
) -> Result<Generated, PromptBuildError>;
```

Usage

Consumers (`perspt-core`, `perspt-coding`) call `compile_prompt_dir` from `build.rs` and include the generated source:

```rust
include!(concat!(env!("OUT_DIR"), "/prompt_sections.rs"));
```

The same validation functions (`validate_section_body`, `parse_section_file`) serve the runtime bundle scanner and `perspt prompts lint`, so external replacement bodies obey exactly the rules the build does.

`perspt-benchmark`

Optional PSP-10 evaluation tooling, independent of runtime mechanism tests. Live runs require configured routes and credentials and never run in CI. The crate is feature-gated behind `perspt-cli`'s `benchmark` feature and surfaced as `perspt benchmark` (`validate`/`run`/`aggregate`).

Core Types

```rust
pub enum BenchmarkSuite {
    Smoke,     // One production-topology arm over a small task prefix
    Adaptive,  // The paging/adaptive pair for the default-activation decision
    Full,      // The complete seven-arm diagnostic ladder
}

pub struct BenchmarkRunOptions {
    pub config_path: Option<PathBuf>,
    pub suite: BenchmarkSuite,
    pub task_limit: Option<usize>,
    pub output: Option<PathBuf>,
}
```

Functions

Function	Description
`validate_corpus_command()`	Credential-free corpus validation: structure, coverage floors, and fail-before/pass-after hidden oracles
`run_benchmark(options)`	Run a suite using the configured production topology; model names are never benchmark arguments
`aggregate_reports(paths)`	Aggregate two or more completed reports (also credential-free)

Corpus and Arms

Tasks live in `corpus/<id>/`: a `task.json` (goal, hidden check argv, tags, expectation), a `fixture/` tree the agent works in, a `hidden/` tree of withheld oracle files, and a `solution/` overlay used only by offline corpus validation. The hidden suite runs in a fresh copy of the post-run fixture with `hidden/` overlaid on top, so the oracle and solution are genuinely unseen during evaluation.

Arms are a cumulative ladder over matched tasks: `direct`, `governed`, `packets`, `paging`, `adaptive`, `multi-family`, `integration`. The report publishes paired differences with a seeded 10,000-resample percentile bootstrap; the primary outcome is hidden-test hard pass.

Developer Guide

Internals, architecture, and contribution guide for Perspt developers.

Guide	Scope
Architecture	Crate structure, data flow, and SRBN internals.
Contributing	Dev setup, coding standards, and PR workflow.
Extending Perspt	Add providers, tools, language plugins, and custom domain adapters.
Testing	Test infrastructure, patterns, and conventions.

Architecture

Perspt is a Rust workspace of fourteen crates plus a dev-only `xtask` automation crate. Eight crates make up the running program, four crates form the reusable platform layer (SDK, prompt codegen, and two domain packages), one crate is the optional feature-gated benchmark harness, and one root crate ties them together. Version 0.6.6 implements the PSP-9 governed candidate runtime and the PSP-10 typed prompt, search, and integration layers.

Workspace Layout

```
perspt/                      # Root: integration crate (perspt)
+-- crates/
|   +-- perspt-core/         # Types, config, LLM provider, events, plugins, prompts
|   +-- perspt-agent/        # Governed candidate runtime, tool loop, verifier, search
|   +-- perspt-tui/          # Ratatui TUI (chat + agent + review modal)
|   +-- perspt-cli/          # Clap CLI entry point, subcommands
|   +-- perspt-store/        # DuckDB session store
|   +-- perspt-policy/       # Starlark policy engine
|   +-- perspt-sandbox/      # Command sandboxing
|   +-- perspt-dashboard/    # Axum web dashboard
|   +-- perspt-sdk/          # Domain-neutral SRBN platform SDK
|   +-- perspt-prompt-macros/ # Build-time prompt section codegen
|   +-- perspt-coding/       # Coding domain package (first domain)
|   +-- perspt-research/     # Research domain package (second domain)
|   +-- perspt-benchmark/    # Optional credentialed evaluation harness
+-- xtask/                   # PSP code-rule & docs task runner (dev-only)
+-- tests/                   # Integration tests
+-- docs/                    # Guidedog & Typst documentation
```

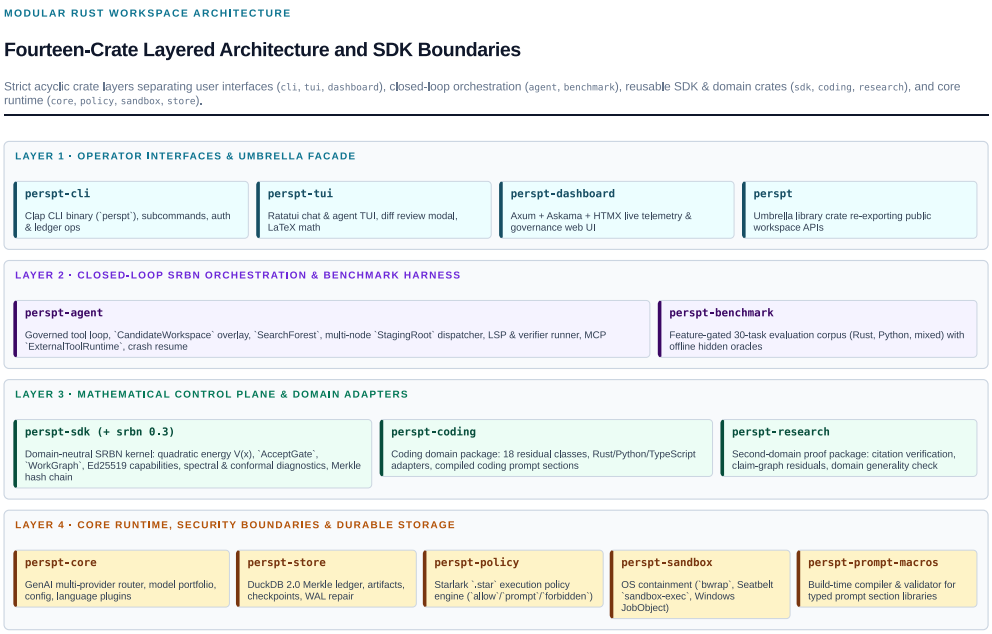
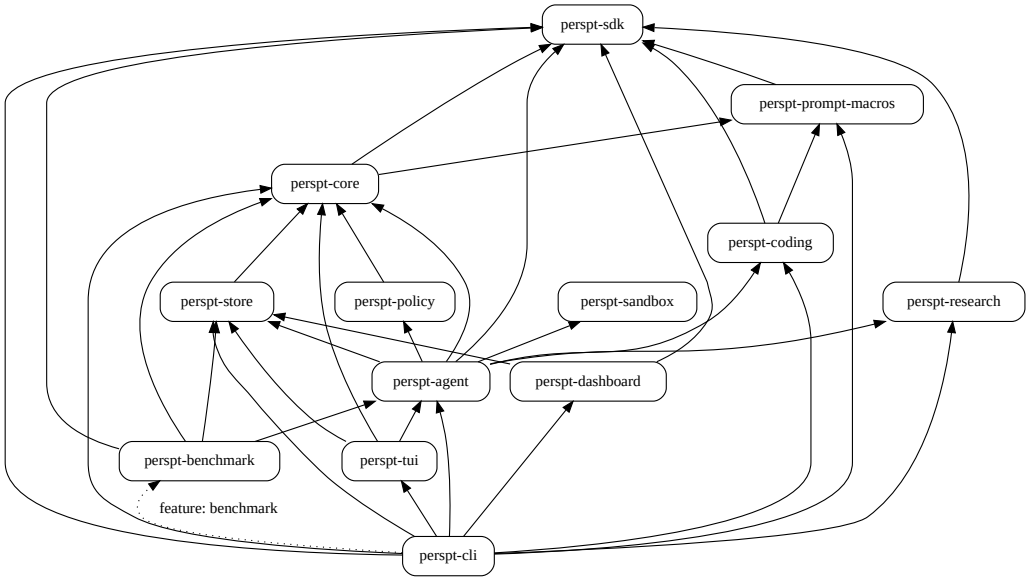


Fig. 7 Layered architecture of the fourteen-crate Perspt workspace.

Dependency Graph



PSP-9 / PSP-10 Overview

PSP-9 replaced the multi-agent orchestrator with a single governed candidate runtime: every model-issued tool call becomes a typed proposal, a deterministic admissibility kernel (in `perspt-sdk`) decides whether it may affect the reversible candidate workspace, and every gate is evaluated on the re-measured candidate, never on the model's account of it. Every event lands

in a hash-chained durable ledger (`perspt-store`), so sessions replay and resume deterministically. PSP-10 adds typed prompt section libraries with build-time codegen (`perspt-prompt-macros`), the bounded search forest with exact no-good learning, graph staging behind a global integration gate, and the optional benchmark harness (`perspt-benchmark`).

Crate: `perspt-core`

The foundation crate. Re-exports all canonical types.

Modules:

- `types` - Core shared types, split into submodules `types/{context,model,plan,policy,verification,workspace}.rs` (see Core Type Inventory below)
- `config` - `Config { provider, model, api_key, ... }` plus per-tier model overrides
- `events` - `AgentEvent` (33 variants), `AgentAction`, `NodeStatus`, `ActionType`
- `llm_provider` - `GenAIProvider` wrapping the `genai` crate; `EOT_SIGNAL`
- `portfolio` - `ModelPortfolio` with provider handles and declared caps
- `plugin` - `LanguagePlugin` trait + `PythonPlugin`, `RustPlugin`, `JsPlugin`
- `prompts` - Typed prompt section libraries (`prompts/*/`) compiled at build time by `perspt-prompt-macros`
- `memory` - `ProjectMemory` loaded from `.perspt/memory.toml`
- `normalize` - Model and provider name normalization

Key Plugin Types:

```rust
pub trait LanguagePlugin: Send + Sync {
    fn name(&self) -> &str;
    fn detect(&self, path: &Path) -> bool;
    fn get_init_action(&self, opts: &InitOptions) -> ProjectAction;
    fn test_command(&self) -> String;
    fn syntax_check_command(&self) -> Option<String>;
    fn verifier_profile(&self) -> VerifierProfile;
    fn owns_file(&self, path: &str) -> bool;
    // ... ~25 methods total
}
```

Plugins provide verifier profiles with fallback chains:

```rust
pub struct VerifierProfile {
    pub plugin_name: String,
    pub capabilities: Vec<VerifierCapability>,
    pub lsp: LspCapability,
}

pub struct VerifierCapability {
    pub stage: VerifierStage,       // SyntaxCheck | Build | Test | Lint | Format
    pub command: Option<String>,     // Primary command
    pub available: bool,
    pub fallback_command: Option<String>,
    pub fallback_available: bool,
}
```

Crate: `perspt-agent`

The governed PSP-9/PSP-10 agent runtime.

Modules:

- `runtime` - `Psp9AgentRuntime`: work-graph planning, bounded dispatch, node assembly, adjudication, staging and integration, resume, and the bounded search forest (`runtime/search/`)
- `toolloop` - The SRBN tool loop: each model-issued tool call becomes a typed proposal; the deterministic admissibility kernel decides whether it may affect the candidate
- `candidate` - `CandidateWorkspace`: reversible coding-candidate overlay
- `measure` - `CodingCandidateMeasurer`: full verifier suite at gate boundaries, cheap syntax-only pass at mutation boundaries
- `verifier` - Governed verifier sandbox: compiler/test/lint processes run under a deny-network profile with a read allow-list
- `tools` - `AgentTools` executor and sandboxing, the `CandidateHandlerRegistry` execution plane (`tools/handlers/`), and first-party tool families (`tools/families/{db,system}.rs`)
- `transport` - `GenAiTransport`: the only adapter between the SDK's provider-neutral contract and `perspt-core`'s `genai` driver
- `turn` - Universal actor turn runner (worker, explorer, architect, adjudicator, evidence summarizer, capability probe)
- `grant` - Persistent grant signing-key resolution
- `probe` - `probe_route`/`ProbeReport`: behavioral provider probes
- `promote` - Descriptor-relative workspace promotion on Unix; native write-through replacement plus best-effort reparse-point rejection on Windows
- `realize` - `SnapshotRealizer`: content-addressed workspace states
- `exploration` - Deterministic, read-only repository orientation
- `external_tools` - Official-SDK MCP 2026-07-28 client (stdio/stateless HTTP; tools, resources, prompts, roots, sampling, elicitation, subscriptions, MRTR/tasks), with local admission, replay, and separate agent/read-only-chat lifecycles
- `lsp` - `LspClient` (JSON-RPC over stdio)

Runtime Flow:

`Psp9AgentRuntime::run()` drives one governed session:

1. Exploration - deterministic read-only repository map; `--exploration-only` adds an interactive explorer tool loop and stops before any mutation
2. Planning - one forced-tool-choice architect turn, restricted to the privileged `update_graph` tool, produces the work graph; the host never fabricates multi-node graphs
3. Dispatch - bounded multi-node scheduling (`runtime/dispatch.rs`); `--max-parallel-nodes` sets the concurrency (default 1; above 1 requires `--yes`)
4. Tool loop - per node, the governed loop (`toolloop/`) turns each model tool call into a typed proposal, admits it through the deterministic kernel, and realizes it against the reversible candidate overlay (`candidate.rs`)
5. Measurement - `measure.rs` re-measures the candidate through the plugin verifier suite inside the sandboxed verifier (`verifier.rs`), or through explicitly acknowledged host-user execution in native Windows reduced- isolation mode; the gate is evaluated on the re-measured candidate, never on the model's account of it. The default evolving-test policy measures the resulting tests; backward-compatible and protected external evidence are explicit additions, never implicit assumptions about every task's contract
6. Adjudication - a tool-free validator reviews only the realized diff and records an uncalibrated verdict (`runtime/adjudicate.rs`)

7. Staging and integration - node winners stage into a graph workspace and must pass the global integration gate before descriptor-relative promotion (`runtime/integrate.rs`, `promote.rs`). Unix promotion holds ancestor directory descriptors; Windows uses write-through native replace operations and rejects observed reparse points, without claiming the same race-resistant boundary.
8. Recording - every event lands in the durable hash-chained ledger (`runtime/recorder.rs`); interrupted sessions resume from the newest durable checkpoint with exactly the remaining budgets (`runtime/resume.rs`)

Failed attempts feed the bounded search forest (`runtime/search/`): sequential, at most three branch identities, one branch attempt (quantum) at a time, with exact no-good learning that suppresses only byte-identical repeated attempts.

Crate: `perspt-store`

DuckDB-backed persistence. **Not SQLite.**

```
pub struct SessionStore {
    conn: Mutex<Connection>,  // duckdb::Connection
}
```

12.1.6.1 DuckDB Schema and Tables

The schema (`crates/perspt-store/src/schema.rs`) is applied through an idempotent transactional migration; `schema_migrations` records the applied version. Eleven tables:

Table 66 Database Tables

Table Name	Key Columns	Purpose
`sessions`	`session_id` (PK), `task`, `working_dir`, `merkle_root`, `detected_toolchain`, `status`	Life cycle of agent sessions
`schema_migrations`	`version` (PK), `applied_at`	Applied schema versions
`psp9_ledger_events`	`(session_id, sequence)` (PK), `event_json`, `prev_hash`, `hash`	The durable canonical event stream; hash-chained
`psp9_artifacts`	`content_hash` (PK), `content`, `byte_len`, `media_type`	Content-addressed artifact bytes
`psp9_authority_epochs`	`session_id` (PK), `epoch`	Single-writer authority fencing per session
`psp9_context_checkpoints`	`(session_id, covered_event_root)` (PK), `checkpoint_json`	Durable conversation checkpoints for resume
`psp9_external_effects`	`(session_id, idempotency_key)` (PK), `intent_hash`, `status`	External (MCP) effect intents and results, idempotent
`psp9_grant_policies`	`policy_id` (PK), `session_id`, `policy_json`, `revoked`	Capability grant policies
`psp9_verdicts`	`(session_id, candidate_id, validator_id)` (PK), `stratum`, `missed`, `unsafe_label`	Adjudication verdicts per candidate
`psp9_calibration_epochs`	`epoch_id` (PK), `stratum`, `target_rho`, `threshold`, `state`, `sample_count`	Conformal calibration epochs

Table Name	Key Columns	Purpose
`psp9_calibration_samples`	`(epoch_id, sample_id)` (PK), `score`, `unsafe_label`, `audit_selected`	Calibration samples; `unsafe_label` stays NULL until the delayed audit label arrives

Row types live in `store/rows.rs` (`SessionRecord`) and `store/psp9_ledger.rs` (`Psp9LedgerRow`, `Psp9VerdictRow`, `Psp9CalibrationEpochRow`, `Psp9ExternalEffectRow`). `repair.rs` backs `perspt db repair` for recovering a database with a poisoned WAL.

Crate: `perspt-tui`

Ratatui-based terminal UI with two modes:

- **ChatApp** - Interactive chat with streaming, LaTeX math transpilation, ASCII table wrapping, and markdown saving
- **AgentApp** - Agent dashboard with work-graph tree, energy display, review modal

The agent TUI is entered through `run_agent_tui_with_runtime` (`agent_app.rs`), which owns the `Psp9AgentRuntime` for the session and wires its event and action channels.

Key components:

Component	Purpose
`Dashboard`	Main agent dashboard layout
`TaskTree`	Work-graph visualization with node states
`ReviewModal`	Grouped diff viewer with approve/reject/correct
`DiffViewer`	Unified diff display
`telemetry`	TUI-side energy component display types
`FrameRateLimiter`	60fps cap, adaptive rendering

Crate: `perspt-policy`

Starlark policy evaluation:

```
pub struct PolicyEngine {
    policies: Vec<FrozenModule>,
    policy_dir: PathBuf,
}

pub enum PolicyDecision {
    Allow,
    Prompt(String),
    Deny(String),
}
```

Utility functions:

- `sanitize_command(cmd)` -> `SanitizeResult` (split, validate, filter)
- `validate_workspace_bound(cmd, working_dir)` - Ensure commands stay in scope
- `validate_artifact_mutation(path, workspace_root, operation)` - Protect root project files from delete/move

Crate: `perspt-sandbox`

Process isolation with active timeout enforcement:

```rust
pub trait SandboxedCommand: Send + Sync {
    fn execute(&self) -> Result<CommandResult>;
    fn display(&self) -> String;
    fn is_read_only(&self) -> bool;
}

pub struct BasicSandbox {
    program: String,
    args: Vec<String>,
    working_dir: Option<PathBuf>,
    timeout: Duration,  // Active: spawn + poll + kill on deadline
}
```

Core Type Inventory

All canonical shared types live in `perspt_core::types`, re-exported from `types/mod.rs`:

Workspace (``types/workspace.rs``):

Type	Description
`NodeClass`	Interface, Implementation (default), Integration
`VerifierStrictness`	Default (compile + tests), Strict (adds lint), Minimal (syntax only)

Model (``types/model.rs``):

Type	Description
`ModelTier`	Architect, Actuator, Verifier, Speculator; `default_model()` maps each tier to its recommended default route

Energy (``types/context.rs``):

Type	Fields
`EnergyComponents`	v_syn (LSP), v_str (contracts), v_log (tests), v_boot (commands), v_sheaf (cross-node); `total()` is the plain sum of the rollups

Task Planning (``types/plan.rs``):

Type	Description
`TaskPlan`	Container for `Vec<PlannedTask>` with ownership-closure, acyclicity, and implicit-dependency validation
`PlannedTask`	id, goal, context_files, output_files, dependencies, task_type, contract, command_contract, node_class, dependency_expectations
`TaskType`	Code, Command, UnitTest, IntegrationTest, Refactor, Documentation
`PlannedContract`	interface_signature, invariants, forbidden_patterns, tests
`PlannedTest`	name, criticality (informational string label)
`DependencyExpectation`	required_packages, setup_commands, min_toolchain_version

Type	Description
`CommandContract`	command, expected_exit_code, expected_files, forbidden_stderr_patterns, working_dir

Verification and Context (``types/verification.rs``):

Type	Description
`SensorStatus`	Available | Fallback { actual, reason } | Unavailable { reason }
`StageOutcome`	stage, passed, sensor_status, output
`ArtifactKind`	Signature, Schema, SymbolInventory, InterfaceSeal
`StructuralDigest`	Content hash of a compile-critical structural artifact (signatures, schemas, seals)
`SummaryDigest`	Condensed summary with hash; `SummaryKind` is IntentSummary, VerifierResults, or DesignRationale
`ContextBudget`	byte_limit (100KB default), file_count_limit (20)
`RestrictionMap`	Per-node context boundary: owned_files, sealed_interfaces, structural/summary digests, dependency commits
`ContextPackage`	The bounded, reproducible context assembled for a node
`ContextProvenance`	Audit trail of the digests and files a node's context used

Policy (``types/policy.rs``):

Type	Description
`CommandPolicyDecision`	Allow, Deny, RequireApproval
`ManifestMutationPolicy`	Allow, Deny

Events System

The event system uses unbounded tokio channels:

```
// In perspt_core::events::channel
pub type EventSender = UnboundedSender<AgentEvent>;
pub type EventReceiver = UnboundedReceiver<AgentEvent>;
pub type ActionSender = UnboundedSender<AgentAction>;
pub type ActionReceiver = UnboundedReceiver<AgentAction>;
```

`AgentEvent` has 33 variants (`crates/perspt-core/src/events.rs`):

- **Planning**: `PlanReady`, `PlanGenerated`, `PlanRevised`, `FallbackPlanner`
- **Execution**: `NodeSelected`, `BundleApplied`, `NodeCompleted`
- **Verification**: `VerificationComplete`, `DegradedVerification`, `SensorFallback`
- **Sheaf**: `SheafValidationComplete`
- **Branches**: `BranchCreated`, `InterfaceSealed`, `BranchFlushed`, `BranchMerged`, `DependentUnblocked`
- **Escalation**: `EscalationClassified`, `GraphRewriteApplied`
- **Context**: `ContextDegraded`, `ContextBlocked`, `StructuralDependencyMissing`, `ProvenanceDrift`
- **Budget**: `BudgetUpdated`

- **File Ops**: `FileDeleted`, `FileMoved`
- **UI**: `ApprovalRequest`, `TaskStatusChanged`, `EnergyUpdated`, `Log`
- **Lifecycle**: `Complete`, `Error`, `ModelFallback`, `ToolReadiness`

The sheaf and branch variants are vestigial: the TUI still renders them for older ledger streams, but the PSP-9 runtime no longer emits them.

Data Flow

```
User Input
    |
[perspt-cli]  Parse args (clap)
    |
[perspt-core]  Config + Provider init
    |
+---+---+
|       |
chat    agent
|       |
[tui]   [perspt-agent]
|       |
|       +-- Psp9AgentRuntime
|       |     +-- exploration        -> read-only repository map
|       |     +-- plan               -> architect turn (update_graph)
|       |     +-- dispatch           -> bounded multi-node scheduler
|       |     +-- tool loop          -> proposals -> admissibility kernel
|       |     +-- candidate overlay  -> reversible workspace mutations
|       |     +-- verifier/measure   -> sandboxed deterministic verification
|       |     +-- adjudicate         -> diff-only validator verdict
|       |     +-- integrate/promote  -> staging, integration gate, promotion
|       |     +-- recorder           -> hash-chained ledger events
|       |
|       +-- GenAiTransport            (provider-neutral model plane)
|       +-- CandidateHandlerRegistry  (open execution plane + tool families)
|       +-- ExternalToolRuntime       (governed MCP)
|       +-- LspClient                 (JSON-RPC stdio)
|       |
|       +-- EventSender --> [perspt-tui AgentApp / perspt-dashboard]
|       +-- ActionReceiver <-- [perspt-tui ReviewModal]
|
[perspt-store]  DuckDB persistence (ledger, checkpoints, verdicts)
[perspt-policy]  Starlark rule evaluation
[perspt-sandbox]  Process isolation
```

Streaming Contract

Both chat and agent mode use the same streaming protocol:

1. LLM requests stream chunks over `mpsc::UnboundedSender<String>`
2. End-of-response signaled by `EOT_SIGNAL` (`<|EOT|>`)
3. Provider sends EOT - UI never adds its own
4. UI batches channel messages, handles first EOT, ignores duplicates
5. Streaming buffer updates the last assistant message live
6. Pending inputs queue until EOT is received

> **Warning**
> Never block the UI select loop. Spawn LLM work on tokio tasks and send results via the channel.

Contributing

Development Setup

```
# Clone
git clone https://github.com/eonseed/perspt.git
cd perspt

# Build
cargo build

# Run tests
cargo test

# Lint
cargo clippy -- -D warnings

# Format check
cargo fmt -- --check
```

Project Structure

```
crates/
  perspt-core/           # Types, config, LLM, events, plugins, prompts
  perspt-agent/          # Candidate runtime, tool loop, verifier, dispatch
  perspt-tui/            # Chat + Agent TUI
  perspt-cli/            # CLI entry (clap)
  perspt-store/          # DuckDB persistence
  perspt-policy/         # Starlark policies
  perspt-sandbox/        # Command sandboxing
  perspt-dashboard/      # Axum web dashboard
  perspt-sdk/            # Domain-neutral SRBN platform SDK
  perspt-prompt-macros/  # Build-time prompt section codegen
  perspt-coding/         # Coding domain package
  perspt-research/       # Research domain package
  perspt-benchmark/      # Optional credentialed evaluation harness
  perspt/                # Root integration crate
xtask/                   # PSP code-rule & docs task runner (dev-only)
tests/                   # Integration tests
docs/                    # Guidedog & Typst documentation
```

Coding Standards

1. **Clippy clean** - `cargo clippy -- -D warnings` must pass
2. **Formatted** - `cargo fmt` with default settings
3. **Tests pass** - `cargo test` must pass all tests
4. **No ``println!`` in UI paths** - Use the event system or `env_logger`
5. **PSP code rules** - Files <= 1408 lines, functions <= 70 code lines, lines <= 108 columns; enforced by `./check-rules.sh check`
6. **Streaming safety** - Never block the UI select loop; spawn on tokio tasks

Commit Messages

- Describe what changed, not the sequence
- Do NOT include phase numbers or commit sequence numbers
- Keep the subject line under 72 characters

```
# Good
Add sheaf validation for cross-language boundaries

# Bad
Commit 3/7: Phase 2 - Add sheaf validation
```

PR Workflow

1. Create a feature branch from `master` (or from the preceding feature branch for an explicitly stacked pull request).
2. Make changes with passing tests
3. Run the pull-request gate:

   ```
   cargo fmt --all -- --check
   cargo clippy --locked --workspace --all-targets -- -D warnings
   cargo test --locked --workspace --all-targets -- --test-threads=1
   cargo check --locked -p perspt-cli --features benchmark
   cargo doc --locked --workspace --no-deps
   ./check-rules.sh check
   ```

   These commands use the exact DuckDB 2.0.0 shared library. CI downloads the official checksummed library matching the pinned Rust crate; local systems should install the same version. Release binaries alone use `bundled`. The final `cargo check` covers the optional benchmark command without running any credentialed evaluation.
4. Push and open a PR
5. Address review feedback

CI Tiers

The workflows separate quick review feedback from target-platform evidence:

- Every non-draft pull request runs one Ubuntu job containing format, Clippy, credential-free tests, optional-feature compilation, Rustdoc, and the PSP code rules. Stacked pull requests are supported because CI does not filter on the PR's base branch.
- A GitHub `merge_group` runs the same Ubuntu gate plus Windows and macOS tests against the temporary merge-queue commit. Enable a branch ruleset that requires the merge queue and the `PR gate (Ubuntu)`, `PSP code rules`, and `Target OS` checks to make this evidence a pre-merge requirement.
- Pushes to `master` or `develop` run the target-OS matrix as a safety net for direct merges. Maintainers can also use **Actions -> CI -> Run workflow** to request the Windows/macOS matrix before merging when a queue is not enabled. That manual target-only run does not repeat the Ubuntu quality gate.

The macOS job exercises its governed verifier backend. The native Windows job tests chat/TUI and shared functionality, proves strict coding mode fails closed, and runs a Rust verifier fixture only after explicit reduced-isolation opt-in.

Superseded PR runs are cancelled. Merge-queue and protected-branch runs are never cancelled because their result is evidence for a specific commit. PDF generation is release work rather than a pull-request gate.

Documentation

Documentation is built in pure Rust and Typst using `guidedog` and `cargo xtask`:

```
# Build Typst assets, PSP catalog, Perspt Book HTML/PDF, and unified site
cargo run -p xtask -- docs

# Build Perspt Book HTML strictly
guidedog build html docs/perspt_book -W -n

# Validate and build PSP records (HTML + Typst PDF)
guidedog dsc check --dir=docs/psps
guidedog dsc build --dir=docs/psps
guidedog build html docs/psps -W -n

# Live preview
cargo run -p xtask -- serve
```

CI treats `guidedog` warnings as errors (`-W -n`) so documentation validation is deterministic. One documentation workflow publishes the unified Guidedog site, Typst PDFs, PSP catalog, and Rustdoc API as a single GitHub Pages artifact.

Extending Perspt

Adding a Language Plugin

Language plugins implement the `LanguagePlugin` trait from `perspt-core`.

1. Create a new module in `crates/perspt-core/src/plugin/`:

```
pub struct GoPlugin;

impl LanguagePlugin for GoPlugin {
    fn name(&self) -> &str { "go" }
    fn extensions(&self) -> &[&str] { &["go"] }
    fn key_files(&self) -> &[&str] { &["go.mod", "go.sum"] }

    fn detect(&self, path: &Path) -> bool {
        path.join("go.mod").exists()
    }

    fn get_init_action(&self, opts: &InitOptions) -> ProjectAction {
        ProjectAction::ExecCommand {
            command: format!("go mod init {}", opts.name),
            description: "Initialize Go module".into(),
        }
    }

    fn test_command(&self) -> String {
        "go test ./...".into()
    }

    fn syntax_check_command(&self) -> Option<String> {
        Some("go vet ./...".into())
    }

    fn verifier_profile(&self) -> VerifierProfile {
        // Define capabilities for each verifier stage
        // with primary and fallback commands
```

```
    }
}
```

2. Register in the `PluginRegistry`
3. Add LSP config for `gopls`

Designing Domain Adapters

The platform SDK establishes a modular separation of concerns: the core SRBN control plane (in `perspt-sdk`) owns scheduling, residual scoring, capability checking, ledger tracking, and telemetry monitoring, while the domain adapter owns the domain-specific logic.

To write a custom domain extension (for example, to support research manuscript compilation, cloud deployment stability, or databases), developers must implement the `AgentDomainPackage` trait.

12.3.2.1 Implementing the AgentDomainPackage Trait

A domain package is a Rust struct that implements the `AgentDomainPackage` trait. Below is a complete implementation example of a custom research domain package:

```rust
use perspt_sdk::{
    AgentDomainPackage, DomainDetection, DomainId, DomainScope, EnergyModel,
    ResidualClass, ResidualEvent, ResidualSchema, ResidualWeight, WorkspaceSnapshot,
    CorrectionDirection, EnergyComponent, StabilityClaim
};

pub struct ResearchDomain;

impl AgentDomainPackage for ResearchDomain {
    /// Returns the unique identifier of the domain.
    fn domain_id(&self) -> DomainId {
        DomainId::new("research")
    }

    /// Detects if the current project workspace contains files corresponding to this domain.
    fn detect(&self, workspace: &WorkspaceSnapshot) -> DomainDetection {
        let mut evidence = Vec::new();
        for marker in ["paper.tex", "thesis.tex", "bibliography.bib"] {
            if workspace.has_file_named(marker) {
                evidence.push(format!("found academic manuscript file: {}", marker));
            }
        }
        let activated = !evidence.is_empty();
        DomainDetection {
            domain: self.domain_id(),
            activated,
            confidence: if activated { 0.90 } else { 0.0 },
            evidence,
        }
    }

    /// Declares the list of residual classes this domain can emit.
    fn residual_schema(&self, _scope: &DomainScope) -> ResidualSchema {
        ResidualSchema::new(vec![
            ResidualClass::Syntax,             // LaTeX syntax check
            ResidualClass::SymbolMismatch,      // Missing citations
            ResidualClass::InterfaceMismatch,   // Broken cross-references
            ResidualClass::Build,              // pdflatex build failures
        ])
    }
```

```
    /// Configures the energy model, including tolerances, bounds, and weights.
    fn energy_model(&self, scope: &DomainScope) -> EnergyModel {
        use EnergyComponent::*;
        use ResidualClass::*;

        let weights = vec![
            ResidualWeight::new(Syntax, Syn, 2.0),
            ResidualWeight::new(Build, Syn, 4.0).with_hard_threshold(0.0),
            ResidualWeight::new(SymbolMismatch, Str, 1.5),
            ResidualWeight::new(InterfaceMismatch, Str, 1.0),
        ];

        let mut model = EnergyModel::new("research", 0.10)
            .with_correction_budget(5);
        model.residual_weights = weights;
        model.energy_tolerance = 0.0;
        model.stability_claim = Some(StabilityClaim::not_claimed(format!(
            "research scope: {}",
            scope.label
        )));
        model
    }

    /// Maps residual events to directed correction prompts for the actuator.
    fn correction_directions(&self, residuals: &[ResidualEvent]) -> Vec<CorrectionDirection> {
        let mut directions = Vec::new();
        for r in residuals {
            match r.class {
                ResidualClass::Syntax => {
                    directions.push(CorrectionDirection {
                        instruction: format!(
                            "Repair LaTeX syntax error: {}. Check brackets and escape
characters.",
                            r.message
                        ),
                        ..Default::default()
                    });
                }
                ResidualClass::SymbolMismatch => {
                    directions.push(CorrectionDirection {
                        instruction: format!(
                            "Insert missing citation entry in bibliography.bib for reference:
{}.",
                            r.message
                        ),
                        ..Default::default()
                    });
                }
                _ => {}
            }
        }
        directions
    }
}
```

12.3.2.2 Registering the Domain Adapter

Once implemented, register your custom package inside the `DomainRegistry` at the composition root in `crates/perspt-cli/src/commands/agent.rs`:

```
let mut domains = perspt_sdk::DomainRegistry::new();
domains.register(std::sync::Arc::new(ResearchDomain));
```

An explicit `--domain` id wins; otherwise the registry selects the best detection, falling back to the coding domain. Every domain must implement the same interface to maintain compatibility with the ledger and dashboard.

Adding an Agent Tool

Tools live under `crates/perspt-agent/src/tools/` (catalog and executor in `mod.rs` and `executor.rs`, sandboxing in `sandbox.rs`, builtin handlers in `handlers/`, first-party families in `families/`). The extension point is the open execution plane: a *tool family* contributes catalog entries (`perspt_sdk::ToolEntry`) and registers one `CandidateToolHandler` per tool name in the `CandidateHandlerRegistry`. The governance wrapper (catalog -> validate -> budget -> certify -> execute -> re-certify) is uniform and lives outside the handlers.

```
use perspt_agent::{
    CandidateHandlerRegistry, CandidateToolHandler, CandidateWorkspace,
};
use perspt_agent::toolloop::EffectOutcome;

struct MyProbe;

#[async_trait::async_trait]
impl CandidateToolHandler for MyProbe {
    async fn apply(
        &self,
        workspace: &CandidateWorkspace,
        call: &perspt_sdk::ProviderToolCall,
        entry: &perspt_sdk::ToolEntry,
    ) -> anyhow::Result<EffectOutcome> {
        // Realize the effect against the reversible candidate.
    }
}
```

Register the family at the composition root (`crates/perspt-cli/src/commands/agent.rs`), through the same public path the shipped system and DB explorer families use:

```
let mut handlers = CandidateHandlerRegistry::with_builtins();
handlers.register("my_probe", Arc::new(MyProbe))?;
runtime = runtime
    .with_tool_family(vec![my_probe_entry()])  // catalog entries
    .with_tool_handlers(handlers);             // execution surface
```

Duplicate registration fails closed, so a family can never silently shadow a builtin. Handlers never decide admission, budgets, or certification — the admissibility kernel already did; they only realize the effect against the reversible candidate.

Adding a Prompt Section Library

Actor prompts are typed section libraries under `crates/perspt-core/prompts/` (`session_bootstrap`, `graph_plan`, `repository_explore`, `adjudicate`, `evidence_summarize`). A library is a directory of ordered Markdown sections with YAML frontmatter:

```
---
id: session_bootstrap/role
version: 1
role: system
required: true
max_bytes: 256
vars:
  domain_id: { type: "BoundedText<64>" }
---
You are a governed {{domain_id}} agent.
```

`perspt-prompt-macros` compiles the sections at build time (`frontmatter.rs` parses, `validate.rs` runs the codegen validation list, `emit.rs` generates the section constructors). After any section change:

1. `perspt prompts lint` - run the validation list (`--bundle` points it at an external bundle directory)
2. `perspt prompts manifest crates/perspt-core/prompts` - regenerate the committed `manifest.toml` with fresh content hashes

Adding a Benchmark Corpus Task

The optional evaluation corpus lives in `crates/perspt-benchmark/corpus/` (30 hidden-oracle tasks; see its `README.md`). Each task directory holds:

- `task.json` - user goal, shell-free hidden-check argv, expectation, one language tag (`rust`, `python`, `mixed`), one scale tag, and capability tags
- `fixture/` - the only tree copied into the agent's workspace
- `hidden/` - the oracle overlay copied only after the agent finishes
- `solution/` - a withheld reference overlay used only by validation

Validate with the credential-free authoring gate:

```
cargo run -p perspt-cli --features benchmark -- benchmark validate
```

The gate refuses fewer than 30 tasks, an imbalanced language/scale mix, insufficient capability coverage, a fixture that already passes its oracle, or a reference solution that does not pass.

Adding an LLM Provider

The `genai` adapter in `perspt-core/src/llm_provider.rs` handles providers. To add a new provider:

1. Add the adapter kind to `str_to_adapter_kind()`
2. Map the env var in `new_with_config()`
3. Add the provider's default model to `detect_provider_from_env()` in `perspt-core/src/llm_provider.rs`
4. Update the auto-detection priority in config

```
fn str_to_adapter_kind(provider: &str) -> AdapterKind {
    match provider {
        "openai" => AdapterKind::OpenAI,
        "anthropic" => AdapterKind::Anthropic,
        // ... existing providers ...
        "newprovider" => AdapterKind::NewProvider,
        _ => AdapterKind::OpenAI,
    }
}
```

Adding a Starlark Policy

Create a `.star` file in the policy directory (`~/.config/perspt/policies/`):

```
# custom_policy.star

def check_file_write(path, content):
    # Called before any file write.
    if path.endswith(".env"):
        return deny("Cannot write .env files")
    return allow()

def check_command(cmd):
    # Called before any command execution.
    if "curl" in cmd and "http://" in cmd:
        return prompt("Insecure HTTP request: " + cmd)
    return allow()
```

The `PolicyEngine` loads all `.star` files from the policy directory automatically.

Testing

Test Infrastructure

Perspt uses Rust's standard test framework with `#[tokio::test]` for async tests.

```
# Run all tests
cargo test

# Run tests for a specific crate
cargo test -p perspt-core
cargo test -p perspt-agent
cargo test -p perspt-store

# Run a specific test
cargo test test_name

# Run with output
cargo test -- --nocapture
```

Test Organization

Location	Type	Description
`crates/*/src/*.rs`	Unit tests	`#[cfg(test)] mod tests` blocks
`tests/`	Integration tests	Cross-crate integration
`crates/perspt-store/`	Store tests	DuckDB schema, CRUD, ledger
`crates/perspt-agent/tests/`	Mechanism checks	23 `mc_*.rs` suites plus `psp9_runtime.rs` covering the PSP-9/PSP-10 runtime mechanisms (tool loop, dispatch, search, recovery)
`crates/perspt-sdk/tests/`	Mechanism checks	`mechanism_checks.rs` and `mc_prompt_activation.rs` for the platform SDK

Test Patterns

Temporary Store:

Use `SessionStore::open()` against a temporary directory for tests that need persistence without touching the user database:

```
#[test]
fn test_session_roundtrip() {
    let dir = tempdir().unwrap();
    let store = SessionStore::open(&dir.path().join("test.db")).unwrap();
    store
        .create_session(&SessionRecord {
            session_id: "s1".into(),
            task: "demo".into(),
            working_dir: "/tmp/demo".into(),
            merkle_root: None,
            detected_toolchain: None,
            status: "active".into(),
        })
        .unwrap();
    assert!(store.get_session("s1").unwrap().is_some());
}
```

Plugin Testing:

Test plugin detection and verifier profiles:

```
#[test]
fn test_python_plugin_detection() {
    let plugin = PythonPlugin;
    let dir = tempdir().unwrap();
    std::fs::write(dir.path().join("pyproject.toml"), "").unwrap();
    assert!(plugin.detect(dir.path()));
}
```

Energy Computation:

Test energy calculation with known inputs:

```
#[test]
fn test_energy_total() {
    let energy = EnergyComponents {
        v_syn: 1.0,
        v_str: 0.0,
        v_log: 2.0,
        v_boot: 0.0,
        v_sheaf: 0.0,
    };
    assert_eq!(energy.total(), 3.0);
}
```

Event Testing:

Test event channel communication:

```
#[tokio::test]
async fn test_event_flow() {
    let (tx, mut rx) = event_channel();
    tx.send(AgentEvent::Log { message: "test".into() }).unwrap();
    let event = rx.recv().await.unwrap();
```

```
    assert!(matches!(event, AgentEvent::Log { .. }));
}
```

Quality Gates

All PRs must pass:

```
cargo fmt --all -- --check
cargo clippy --locked --workspace --all-targets -- -D warnings
cargo test --locked --workspace --all-targets -- --test-threads=1
cargo check --locked -p perspt-cli --features benchmark
cargo doc --locked --workspace --no-deps
./check-rules.sh check
```

The PR gate runs once on Ubuntu. It downloads the official DuckDB 1.5.5 shared library, verifies its published checksum, and compiles the optional `benchmark` CLI without running live model evaluation. Release builds, not CI tests, use the expensive `bundled` feature. A merge-queue commit, protected-branch push, or manual full run adds Windows and macOS tests. Platform-neutral formatting, Clippy, Rustdoc, and PSP rules are not repeated on every operating system. macOS exercises the native governed verifier. Windows covers native chat/TUI and shared code, verifies strict coding mode fails closed, and separately exercises the explicit reduced-isolation verifier.

GitHub merge queues are the pre-merge path: the `merge_group` event checks the proposed change combined with the latest target branch and any changes ahead of it in the queue. Without an enabled merge-queue ruleset, maintainers must use the CI workflow's manual full run when pre-merge cross-platform evidence is required; the protected-branch push remains a post-merge fallback.

The project currently has over 800 tests across all crates.

Benchmarks Are Not Tests

`perspt benchmark` is not part of `cargo test` or CI. The `perspt-benchmark` crate is gated behind the non-default `benchmark` feature of `perspt-cli`, requires configured model credentials, and is run manually to evaluate and compare Perspt against other coding agents. Its own unit tests are credential-free and run with the ordinary workspace suite.

Panic Safety

`main` installs a panic hook that:

1. Restores terminal (raw mode off, leave alternate screen)
2. Prints the panic message with guidance
3. Exits cleanly

Test this with `tests/panic_handling_test.rs`.

Changelog

Version 0.6.6 - “沈黛君 (Shen DaiJun)”

In memory of a mother who always gave and never asked anything in return.

This release implements PSP-9’s governed agentic platform: every stochastic action is a proposal, every effect is mediated by a deterministic kernel, every accepted state is measured on the realized candidate, and every reliability claim is conditional on recorded evidence — following the *Stability is All You Need* papers I, II, and III. On top of it, PSP-10 makes the loop’s prompts, resident context, and search governed artifacts of the same record.

Governed Tool Loop & Admissibility Kernel (PSP-9):

- **SRBN Tool Loop** - New `perspt-agent::toolloop` implements Paper II’s harness contract as an executable loop: model-issued tool calls become Paper III proposals, the five-clause admissibility kernel (Definition 3.2) decides each one, and the acceptance gate reads the re-measured candidate — never the model’s account of it. An adversarial “lying model” fixture proves a model claiming success while every effect is denied can never reach a hard pass.
- **Five-Clause Kernel** - `check_full_admissibility` replaces hardcoded contract/barrier passes with registered evaluators; an absent clause is recorded absent, never true. Promotion is one transaction debiting exactly the barrier’s certified increment `c_t` — the allowance and the debit are the same number.
- **Operational Safety Barrier** - `perspt-coding` registers six versioned policy channels (protected paths, sandbox escape, secret exposure, network reachability, dependency policy, resource limits) with exact deterministic increments; correctness evidence stays in `V`.
- **Recovery Lattice** - One shared, non-replenishing cascade budget across all five levels (Definition 8.1): exhaustion forces strict escalation, containment is unconditional, and every cascade terminates within `b + k` steps in one of Theorem 6’s four terminal classes.

Model Portfolio & Transport:

- **Multi-Provider Portfolio** - The `[providers]` table holds several credentialed routes in one process; `perspt providers` prints the capability matrix with declared-vs-probed honesty. A one-entry portfolio keeps single-provider setups unchanged.
- **ModelTransport Port** - The SDK declares the object-safe transport port; `perspt-core` provides the `genai` tool-calling driver; `perspt-agent::transport` is the single adapter — the tool loop cannot name a vendor type or credential (Gate S).
- **Portfolio Routing** - Fully qualified `provider::model` routes with capability objectives, decorrelated-family review preference, and cross-provider failover chains. Failover is recovery, not retry, and draws on the shared cascade pool.
- **Typed Tool Catalog** - Declarative footprints (`writes ⊆ reads` by construction), capability-filtered tool specs, and an open `LanguageId` adapter registry. `sed_replace`/`awk_filter` leave the model-facing catalog; `edit_file` fails closed on ambiguity. External tool servers register under the kernel and can never exceed session authority.

Measured Certificates:

- **Realizability Interface** - `srbn` upgraded 0.1.1 → 0.3.0 (`stabilize_realized`, restore-best policy, hash-chain ledger). Coding uses unmeasured realization with `projection_mismatch` telemetry; the analytic Theorem 12.4 path is admitted only when every hypothesis — including Theorem 12.1's step-size bounds — is represented by evidence.
- **Independence Statistics** - Matched-stratum pairwise misses, degenerate marginals using direct joint misses, conservative Hoeffding upper bounds, and certification gating; `ρ_eff` reports `n` beside it and the correlation-driven energy attenuation helper is removed.
- **Conformal Feasibility Floor** - A declared budget `ρ` fixes `n ≥ 1/ρ − 1` labeled samples before anything can be autonomously accepted; reject-all is reported as *insufficient calibration*, never as a satisfied budget. Calibration is keyed by full exchangeability strata with shadow/active/stale readiness epochs.
- **Audit Replay** - `perspt replay <session>` folds the durable event stream deterministically with tamper detection and no provider credentials; context compaction preserves an exact `ControlFrame` and unresolved tool calls, and a stale checkpoint is rebuilt, never patched.

Recovery, Resume & Calibration:

- **Recovery Ladder** - The runtime now holds Theorem 6's higher rungs live: level 2 revises the work graph with the exhausted attempt's evidence and re-runs at a new generation, level 3 hands the node to the configured architect route, and level 4 durably revokes the session's authority epoch — also wired to `perspt abort` and the TUI quit path.
- **Exact Mid-Loop Resume** - Every gate acceptance writes a durable candidate checkpoint (control frame plus content-addressed file artifacts). `perspt resume` rebuilds the accepted candidate from it, re-mints a fresh capability from the grant intersection at the current durable epoch, and re-enters the loop with exactly the remaining budgets; a bumped epoch refuses the resume.
- **Delayed Audit Labels & Conformal Activation** - Every promotion persists an audit-selected, unlabeled calibration sample; `perspt audit` ingests delayed labels (single-assignment) and recomputes the stratum threshold over labeled samples only. At the finite sample floor a new immutable epoch activates, after which hard-pass candidates above `theta` commit without a prompt with the certified accept ledgered; any other state backs off to approval.
- **Grant Ceilings Enforced** - `GrantPolicy::mint` intersects every live capability with the grant ceilings (effects, paths, commands, deny-by-default network, approval strictness) and fails closed; persisted grant intent is signed over canonical bytes and verified against a local trust anchor on resume, never the embedded key.
- **Behavioral Provider Probes** - `perspt providers --probe` observes each configured route on a live two-tool round trip (tool calling, multi-tool selection, parallel batching, schema validity) with evidence labelled `behavioral`; all three configured routes passed live.
- **Exploration-Only Mode** - `perspt agent --exploration-only` runs an interactive explorer tool loop under a strictly read-only capability: every call passes the kernel and mutation attempts are recorded denials.
- **Parity Gate Passed** - The governed tool loop ran against the ungoverned whole-file ablation baseline live: 100% hard-pass with zero false stability, failures preserved. No measurement justifies a typed whole-file tool yet; PSP-9 status moves to Final. The `parity_bench` harness is retired by PSP-10; evaluation moves to the optional `perspt-benchmark` crate, run manually.

- **Governance Surfaces** - A dashboard Governance page shows authority epochs, signed grant intent, calibration epochs, pending delayed audits, and validator verdicts; `perspt status`, `perspt ledger`, and `perspt abort` now read and act on the PSP-9 surfaces instead of printing placeholders.

PSP-9 Completion (all phases Done):

- **Open Execution Plane** - Tool execution moved behind an exact-name handler registry with a conformance fixture proving a new family registers at the composition root (catalog entries + handlers + derived grants) without editing the loop, the candidate, or the node assembly. The legacy 15-arm executor dispatch and its ungoverned `sed_replace`/`awk_filter`/`run_command` paths are deleted; grants derive from the assembled catalog intersected with an explicit withheld set.
- **Shipped Tool Families** - A read-only system explorer (`sys_info`, `sys_processes`, `sys_disk`, `sys_env` — variable names only, values withheld) and a local DB explorer (`db_list`, `db_schema`, `db_query` — SELECT-only over an in-memory read-only DuckDB view with statement allowlisting and row/byte caps) under two new deliberate read-only effect kinds, `SystemProbe` and `DataRead`.
- **Domain Registry Wired** - The runtime selects its domain package from the open `DomainRegistry` (`--domain` or detection); the research domain is registered and reachable; the closed `CodingLanguage` enum is retired in favor of the `LanguageId` adapter registry; `PluginRegistry` accepts registrations.
- **Governed Dependency Mutation** - `mutate_dependencies` resolves through each language plugin's dependency commands (`cargo add`, `uv add`/`uv lock``+``uv sync`, `npm install`) behind an explicit `--allow-dependency-mutation` grant, with Manifest/Lockfile footprints, package-name injection guards, pre-image journaling, and write-ahead external-effect bracketing; manifest and lockfile promote like any governed mutation.
- **Ledger-Folded Conversation (Gate O)** - Model context is a pure fold over digest-chained conversation deltas recorded before they are applied; the durable checkpoint's control frame carries the rolling fold digest, and resume refolds the ledger and refuses a conversation that is not derivable from the record.
- **Commuting Tool-Call Batches (Gate P)** - Same-turn calls are planned over declared footprints: commuting mutators run in the model's own arrival order, precise-footprint reads execute concurrently, and a colliding pair is returned as a `ToolBatchConflict` observation, never given an invented order.
- **Bounded Multi-Node Dispatch (Gate P)** - `--max-parallel-nodes` (default 1, requires `--yes`) enables a governed forced-choice `update_graph` planning turn and a dispatcher running per-node lifecycle futures on borrowed structured concurrency with cancellation tokens, one shared non-replenishing recovery pool, first live `GraphWrite` lease use, and single-flight promotion; `--max-parallel` keeps its verifier-fan-out meaning.
- **Proposal Ensembles Superseded (Gate M/T)** - The live proposal ensemble (`[ensemble]` / `--ensemble-width`) is removed by PSP-10 in favor of the bounded search forest: each branch is still one ledgered gate decision in its own reversible overlay with selection strictly by measured energy, and a lingering `[ensemble]` block is now a fatal startup error pointing to `[exploration]`.
- **Certified Pairwise Bounds (Gates R/T)** - The deterministic verifier suite records a verdict row beside the adjudicator's for the same candidate and stratum; `perspt audit` labels verdicts with the calibration sample in one pass; the independence estimator runs over labeled matched verdicts and surfaces per-pair statistics in `perspt status` and the governance dashboard — Certified with `n` or the literal `insufficient evidence`.

- **Latest MCP Client Wired Into Runtime and Chat (Gates K/L/U)** - The custom compatibility transport was replaced by the official Rust SDK and a strict MCP 2026-07-28 `server/discover` lifecycle with no legacy fallback. Stdio and stateless Streamable HTTP now cover tools, resources/templates, prompts/completions, explicit roots, opt-in sampling/elicitation, typed notifications, `subscriptions/listen`, MRTR, and bounded tasks. Remote tools remain locally admitted, bracketed, and replayed; protocol context primitives use reserved read-only operations.
- **MCP TUI and Boundary Hardening** - `/mcp` and `/help` expose status, policy rejection, and elicitation actions; chat accepts interactive form/URL requests while the non-interactive agent declines them. Sampling preserves the local model route and forwards generation controls, subscription changes re-admit catalogs, task and sampling budgets are bounded, stdio children inherit no ambient credentials beyond executable-resolution variables, and bracketed paste now inserts Unicode/multiline clipboard content into the chat input.
- **Diagnostics and Rollback (Gate R)** - The dashboard renders the work-graph revision lineage (Topology), a Backlog page with the arriving-potential gauge Φ(W) labeled as conditional diagnostics, and the governance independence pair table; `perspt status` shows Φ(W), the withheld `topology_gap`, and certification state; `perspt ledger --rollback <session>` restores promoted pre-images through the hardened descriptor-relative path, extends the hash chain with the rollback event, and labels the candidate UNSAFE — the delayed-label source for calibration and independence.
- **Promotion TOCTOU Hardening** - Promotion, resume recovery, and rollback share one engine. Unix uses `openat`/`renameat`/`unlinkat` with `O_NOFOLLOW` component walks and held directory descriptors. Native Windows uses write-through replacement and rejects observed reparse points in its explicitly weaker reduced-isolation mode. Rollback failures are reported, never swallowed. The scheduler now tracks refined generations exactly.
- **Recovery Continuity** - Transient provider errors (429, 5xx, timeouts) are retried with bounded exponential backoff inside the transport, per Gate P's rule that rate-limit waits are throughput delays; recovery-ladder rungs reseed from the previous attempt's best accepted candidate state instead of re-paying the whole cost (`ladder_reseeded`); and containment caused by exhausted transport preserves the session's authority epoch so `perspt resume` remains viable after an outage, while governance containment still revokes.
- **Legacy Retirement** - ~19,000 lines of orphaned PSP-5 orchestrator code, dead TUI entry points, the legacy dashboard LLM/Sandbox pages, and the empty legacy status branch are removed; remaining pages read the PSP-9 ledger.

Typed Prompts, Paged Context & Bounded Search (PSP-10):

- **Typed Prompt Sections & Codegen** - The new `perspt-prompt-macros` crate compiles typed prompt sections into prompt programs; every compiled program and every model call is ledgered with ordered section provenance, the tool-surface hash, and its digest. `perspt prompts` (`list`, `render`, `lint`, `manifest`, `explain-session`) inspects and maintains the section libraries, and `perspt agent --allow-experimental-prompts` substitutes validated `[prompts]` bundle sections live — experimental until a change record passes paired evaluation (Gate AE).
- **Resident-Context Paging** - Model context is a paged resident set with reserves declared in `[context]`: evicted pages stay addressable through the `context_recall` tool, every compaction

and refusal is a recorded context event, and `perspt context explain-turn` explains a session's recorded events.

- **Bounded Search Forests** - After a gate failure the runtime can open a bounded search forest (`[exploration]`; at most three branch identities, sequential eager branches): exact no-goods are learned from rejected candidates, and every branch's workspace reservation is durable and pre-action with cost-free refusals.
- **Multi-Node Dispatch with Integration Gate** - `--max-parallel-nodes` dispatches work-graph nodes into per-node staging overlays behind one global integration gate, so concurrent nodes promote through a single measured decision point.
- **Evolving Test Evidence** - Production coding gates now default to `[verification] test_policy = "evolving"`: the resulting implementation, tests, and project configuration are measured together, so an intentional contract change is not forced to satisfy a superseded test. `backward-compatible` explicitly adds the restored historical regression view, while `external-oracle` overlays separately protected acceptance material on a private candidate copy and adds its command as a blocking test-stage verdict. The selected policy is shared by node, search-preview, and integration measurements and recorded at session start; benchmark hidden oracles remain withheld and fixed.
- **Crash Resume** - `perspt resume` rebuilds a crashed session from the durable record, judging the work graph by its latest recorded state and refolding the conversation by digest chain.
- **Optional Benchmark Crate** - The new `perspt-benchmark` crate (perspt-cli Cargo feature `benchmark`, non-default) evaluates and compares Perspt with other coding agents over a 30-task hidden-oracle corpus: suites `smoke` (1 arm, 8 tasks), `adaptive` (2 arms), and `full` (7 arms). It is credentialed and run manually only — never part of `cargo test` or CI.

Engineering Discipline:

- **Reproducible Toolchain** - Added `rust-toolchain.toml` pinned to Rust 1.97.1 while retaining edition 2021; workspace packages inherit the same MSRV.
- **DuckDB 1.5.5 Boundary** - Exact-pinned `duckdb` and `libduckdb-sys` at `1.10505.0`, added dynamic ABI rejection, transactional versioned migrations, catalog-checked nullability changes, and post-DDL checkpoints.
- **Recoverable WAL Repair** - `perspt db repair --db-path PATH --discard-wal` records WAL size and SHA-256, makes durable timestamped database/WAL backups, quarantines the WAL, verifies recovery read-only, and restores the original WAL on failure.
- **Shared MCP Runtime Foundation** - Introduced the governed external-runtime boundary, mode filtering, local effect/footprint admission, namespacing, bounded observations, environment-referenced secrets, replay-only observations, and explicit reconciliation. The current release supersedes its initial tool-only transport with the official latest SDK client described above.
- **PSP-5 API Retirement** - Removed the legacy orchestrator and Merkle-ledger facade from the public agent crate and refused resume of PSP-5 sessions; their existing rows remain inert forensic data.
- **PSP Code Check** - New `xtask` checker enforces file ≤ 1408 lines, function ≤ 70 code lines (measured with `syn`, not brace counting), and line ≤ 108 columns, with a shrink-only baseline ratchet wired into CI (`./check-rules.sh`).

- **Workspace Decomposition** - The 5,889-line orchestrator, `types.rs`, `plugin.rs`, `store.rs`, `tools.rs`, `ledger.rs`, and `verification.rs` split into focused submodules; workspace version fields now inherit from `[workspace.package]`.
- **Native Windows Capability Contract** - TUI chat, simple-chat, MCP, and the coding workflow compile natively. Coding process execution stays fail-closed by default and requires `--allow-unisolated` or `[verification] allow_unisolated = true` until a restricted-token/ACL sandbox backend exists; CI tests strict refusal and the acknowledged native verifier path separately.
- **Audit-Driven Hardening** - A full-code audit closed a candidate-restore gap (files created after a checkpoint now roll back exactly via a pre-image journal), separated read paths from the promotable mutated set, routed promotion itself through the five-clause kernel, made the policy engine consume canonical command lines, inverted network-scope delegation to deny-by-default, and made ledger appends O(1) via stage-then-commit.

Version 0.6.2 - “Hózhó”

Hózhó (Navajo) — A state of perfect balance, harmony, and continuous self-improvement.

This iteration focuses on bringing greater stability, balance, and refinement to the core architecture. It introduces the full platform SDK implementation alongside the mathematical constructs outlined in the *Stability is All You Need* paper series (Papers I, II, and III) and formalizes PSP-8.

Stability Is All You Need & Platform SDK (PSP-8):

- **SDK Platform Crates** - Introduced `perspt-sdk`, `perspt-coding`, and `perspt-research` to separate domain-neutral orchestration logic from target execution domains.
- **Measured Acceptance Gate** - Implemented the PSP-8 spectral acceptance gate using symmetric eigensolvers (via `nalgebra`) to compute energy-slope convergence guarantees.
- **Goal Verification Loop** - Integrated target validation kernels with directed LLM corrections and structured `GoalVerdict`/`TaskType` contracts.
- **Symbol Extraction & Verification** - Added runtime parsing, dependency tracing, and semantic verification to enforce correct output states.

Core Capabilities & Providers:

- **Vertex AI Support** - Added seamless Google Cloud Vertex AI provider support with automatic OAuth2 Bearer token resolution via `gcp_auth` ADC.
- **29 LLM Adapters** - Expanded provider configurations to handle 29 unique genai adapters, modernizing model defaults (e.g., `gemini-3.5-flash`, `gpt-5.5`, `claude-fable`).
- **Pipenv Integration** - Added support for `pipenv` environment isolation to the Python execution plugin.

Documentation Overhaul:

- **Stability Foundations** - Reworked core concept and developer documentation to align with the theoretical foundations of the Stability paper series.
- **Sphinx Book Polish** - Redesigned the Sphinx book layout featuring a retro scientific textbook LaTeX cover, custom plots, and enhanced tables.
- **Refined Clinical Tone** - Rewrote the user manual, tutorials, and configuration guides in a formal, clinical documentation style.

Workspace Maintenance & Upgrades:

- **Ratatui Upgrade** - Bumped `ratatui` dependency to `0.30.2` in `perspt-tui` to support modern terminal layouts and components.
- **TUI Textarea Fork Migration** - Migrated from the unmaintained `tui-textarea` to `tui-textarea-2` version `0.11.0` to resolve API compatibility conflicts.
- **Dependency Upgrades** - Bumped `rustyline` to `18.0`, `toml` to `1.1`, and `tower-http` to `0.7` across the workspace.
- **GitHub Actions Security** - Upgraded GitHub Action workflows to use `actions/checkout@v7` for improved stability and security.

Version 0.6.1 - "AKU"

AKU - sharp fixes from sharp ears.

Config Coherency & Schema-Driven Settings:

- **TOML Config Schema & Resolution** - Added a robust TOML configuration schema with smart config-driven provider resolution.
- **Unified Provider Binding** - Bound the configured provider cohesively across all modes including `chat` (TUI), `simple-chat` (CLI), and the autonomous `agent` mode.
- **Refined Config Commands** - Redesigned *perspt config* command to be fully structured, interactive, and improved key initialization/init prompts.

TUI & CLI Slash Commands:

- **CLI Simple-Chat Enhancements** - Integrated `rustyline` file-based history and introduced interactive slash commands inside simple-chat.
- **TUI Input Navigation & Persistence** - Added persistent history paths and implemented fully UTF-8 safe input navigation in TUI inputs.
- **Conversation Persistence** - Added new conversation save commands to easily export dialogues to local files.

Advanced Terminal UI Rendering:

- **LaTeX Math Transpilation** - Integrated real-time LaTeX mathematical transpilation into the markdown rendering pipeline.
- **Intelligent ASCII Table Wrapping** - Implemented self-wrapping logic for ASCII tables in the chat UI to ensure readable presentation on narrow views.

Documentation:

- **Comprehensive Feature Guides** - Fully documented TUI/CLI slash commands, config schemas, and keyboard shortcut matrices in the user guides.
- **PSP-8 Stability Documentation** - Added a dedicated Perspt Book chapter explaining how the three *Stability is All You Need* papers shape agent mode, the SDK-first roadmap, and future domain plugins.

Workspace Maintenance:

- **Crate Version Alignment** - Bumped version of all workspace crates to `0.6.1`.

Version 0.6.0 - “kukuza”

Nurturing the foundation, empowering the core.

Workspace-Wide Dependency Upgrades:

- **duckdb Upgrade** - Bumped `duckdb` requirement to `=1.10503.1` in the workspace root.
- **askama Upgrade** - Bumped `askama` requirement to `0.16` in `perspt-dashboard`.
- **diffy Upgrade** - Bumped `diffy` requirement to `0.5` in `perspt-agent`.
- **genai Upgrade** - Bumped `genai` requirement to `0.6.1` in `perspt-core`.
- **starlark Upgrade** - Bumped `starlark` requirement to `0.14` in `perspt-policy`.

API Alignments & Adaptations:

- **genai API Alignment** - Updated `all_model_names` inside `llm_provider.rs` to pass `()` for the new `ProviderConfig` parameter.
- **starlark API Alignment** - Adapted policy loading in `engine.rs` to use `Module::with_temp_heap` since `Module::new` was deprecated and made private in `0.14`.

Specification & Ecosystem Maintenance:

- **PSP-7 Finalization** - Formally transitioned the PSP-7 specification to `Final` status under the PSP-000001 process.

Version 0.5.9 - “xinli guanghua”

Perfecting the essence until the work needs no words to shine.

PSP-7: Robust Correction Loop Contracts:

- **Structured Artifact Bundle Format** - Switched correction prompt from free-form `File: ...` output to a strict JSON `{ artifacts: [], commands: [] }` schema. Includes exact target paths from evidence so the LLM targets the correct files, reducing parse failures.
- **AgentTools Integration** - Routed correction commands through `execute_correction_command()` to integrate with plugin policy, user approval gates, and tool failure tracking.
- **Typed Parse Pipeline** - Replaced Option-based bundle extraction with a 5-layer fail-closed typed parse pipeline. Added `RetryClassification` (Retarget, MalformedRetry, SupportFileViolation, Replan) population in `CorrectionAttemptRecord`.
- **Manifest Policy Enforcement** - Added semantic validation to prevent implicit mutation of root manifests (Cargo.toml, package.json) unless explicitly listed as output targets, while preserving legal support files.
- **Strict Budget Exhaustion** - Widened budget exhaustion checks from cost-only to `any_exhausted()` to properly respect step and revision caps before attempting LLM calls.

LLM Provider Maintenance:

- **genai Upgrade** - Bumped `genai` dependency from 0.5.1 to 0.5.3 (stable patch release with bug fixes).
- **Dead Code Cleanup** - Removed `generate_response_with_history()` and `generate_response_with_options()` which had zero callers across the workspace and were the only methods leaking `genai::ChatOptions` into the public API surface.

- **Clippy Fixes** - Fixed *clippy::unnecessary-sort-by* and applied *clippy::collapsible-match* auto-fixes for Rust 1.95 compatibility.

Version 0.5.8 - “Qualitaetsveredelung”

Orchestration State Overhaul Release

> “Qualitaetsveredelung - the craft of refining what exists until its quality speaks for itself. Not new features, but the quiet discipline of making every state transition truthful, every metric honest, and every dead path removed.”

Orchestration Correctness (Refs: #112, #113, #114, #116):

- **SessionOutcome enum** - New `SessionOutcome` type (Success, PartialSuccess, Failed) derived from actual completed/escalated node counts. The `Complete` event now carries truthful outcomes instead of unconditional `success: true`.
- **NodeOutcome enum** - `execute_node()` returns `Result<NodeOutcome>` where `NodeOutcome` is `Completed` or `Escalated`, replacing the previous `Result<()>` that could not distinguish outcomes.
- **Correct session outcome derivation** - `run_orchestration()` and `run_resumed_inner()` track completed/escalated counts per node and derive the final `SessionOutcome` accordingly.
- **Always-on LLM telemetry** - `call_llm_with_logging()` now records token usage (in/out), latency, and estimated cost via `record_llm_usage()` after every LLM call, regardless of `--log-llm`. The flag now only controls verbose prompt/response text persistence.
- **Budget envelope persistence** - `upsert_budget_envelope()` called after each `BudgetUpdated` event to persist cost/step tracking to the database.
- **Sandbox-aware context retrieval** - `ContextRetriever` in `step_speculate()` uses `effective_working_dir(idx)` (the node’s sandbox directory) instead of the workspace root. Sandbox file tree listings included in actuator and correction prompts for better generation grounding.

Type-Safe State Management (Refs: #114):

- **NodeState::from_display_str()** - Case-insensitive canonical parser with legacy aliases (“running” -> Coding, “stable” -> Completed, “retrying” -> Retry). Replaces all ad-hoc string parsing across the codebase.
- **NodeState helpers** - `is_success()` (true only for Completed), `is_active()` (true for Coding, Verifying, Planning, Retry, SheafCheck, Committing), and `Display` impl producing lowercase labels.
- **CLI state cleanup** - All string-based state comparisons in `status.rs`, `agent.rs`, and `resume.rs` replaced with `NodeState::from_display_str()` and type-safe helper methods.

Dead Code Elimination:

- Removed 16 unused functions across `perspt-core`, `perspt-store`, `perspt-agent`, `perspt-policy`, `perspt-tui` (~234 lines)
- Downgraded `canonicalize` to `pub(crate)` in `perspt-policy`
- Removed orphaned `sha2` dependency from `perspt-store`

Bug Fixes (Refs: #107, #111):

- **Session status stuck at RUNNING** - Status now persisted in `end_session()` with `COALESCE`-based finalization guarantee (#111)
- **LLM token counts always zero** - Real provider token usage (prompt + completion tokens) extracted from `genai ChatResponse::usage` and persisted per request (#107, #110)

Tests:

- 7 new tests covering `NodeState` parsing round-trips, `SessionOutcome` equality, `NodeOutcome` discriminants, and session outcome derivation from completed/escalated counts
- Total test count: 359 (up from 352)

Documentation:

- Updated README: fixed crate count (8 -> 9), deduplicated dashboard command, added missing agent flags (`--single-file`, `--verifier-strictness`, `--output-plan`, all `--*-fallback-model`), aligned contributing commands with CI gates
- Updated Perspt Book: SRBN architecture (Phase 7 -> Commit & Outcome), CLI reference (`logs` always shows token metrics), developer architecture guide (orchestrator lifecycle, NodeOutcome, SessionOutcome in type inventory and data flow), workspace crates (removed dead `is_safe_for_auto_exec`)
- Updated PSP-5: execution flow steps 8-11 with Completed/Escalated paths and SessionOutcome derivation, headless output with `OUTCOME` line, added Orchestration State Overhaul implementation appendix

Version 0.5.7 - “navikaran”

Dashboard UX Polish Release

> “Bridging the purpose of Ikigai with the momentum of Kaizen - renewal through continuous, intentional refinement.”

Dashboard UX Improvements (PSP-6 continued):

- **Custom DaisyUI 5 themes** - `perspt-light` and `perspt-dark` themes with orange/pink oklch palette (WCAG AA compliant), powered by `@plugin "daisyui/theme"` blocks
- **Theme toggle** - Navbar button with sun/moon icons, localStorage persistence, and migration from legacy theme names
- **Friendly session names** - Deterministic human-readable names (e.g. “bold-hawk”) derived from session UUIDs via hash-indexed adjective+noun arrays
- **Breadcrumb friendly names** - All six session sub-pages show friendly name with UUID-on-hover tooltip
- **Session card layout** - Stacked vertical cards with `btn-outline` sub-page buttons replacing ghost buttons
- **Task text formatting** - `whitespace-pre-line` rendering for readable multi-line task descriptions
- **Collapse arrow fix** - `pe-10` padding on DAG and LLM collapse summaries to prevent arrow overlap with text
- **Decisions page resilience** - All six store queries use `unwrap_or_default()` instead of `?` early-return, preventing 503 errors on partial data

- **Paginated overview** - 20 sessions per page with DaisyUI `join` pagination controls, backed by `list_sessions_paginated()` and `count_sessions()` store methods
- **Login page theme** - Updated to `perspt-light` default

CI & Build:

- **Node.js in CI** - Added `actions/setup-node@v4` (Node 22) to CI test matrix and release workflows so `npx @tailwindcss/cli` runs on all runners

Store:

- `list_sessions_paginated(limit, offset)` - LIMIT/OFFSET SQL for paginated session listing
- `count_sessions()` - Total session count for pagination controls

Version 0.5.6 - “ikigai”

SRBN Sandbox Revision Flow Release

> “A reason for being - the happiness of always being busy with what you love.”

Real-Time Web Dashboard (PSP-6):

- **perspt-dashboard crate** - Axum 0.8 + Askama 0.15 + HTMX 2 + Tailwind v4/DaisyUI 5 web interface for monitoring agent execution
- **Read-only store access** - `SessionStore::open_read_only()` with DuckDB `AccessMode::ReadOnly` for safe concurrent reads alongside the agent
- **Six monitoring pages** - Overview (sessions), DAG (task graph), Energy (Lyapunov components), LLM (request telemetry), Sandbox (provisional branches), Decisions (escalations, sheaf validations, rewrites, plan revisions, repairs, verifications)
- **SSE live updates** - Server-Sent Events stream node statistics every 2 seconds
- **Password authentication** - Random token, HttpOnly/SameSite cookie, Secure flag on non-localhost deployments
- **``perspt dashboard`` CLI command** - Launches the dashboard server on a configurable port
- **12 integration tests** - Route smoke tests, SSE content-type, auth flow
- **Store extensions** - `get_session_energy_history()`, `get_all_sheaf_validations()`, `get_all_repair_footprints()`

SRBN Sandbox Revision Flow (PSP-5 Phases 3-12):

- **PlanningPolicy** - Adaptive agent gating with 5 policies (LocalEdit, FeatureIncrement, LargeFeature, GreenfieldBuild, ArchitecturalRevision). `needs_architect()` and `needs_speculator()` gate agent tier activation
- **FeatureCharter auto-creation** - Policy-derived file/module/revision limits created before architect planning so the plan gate has bounds to enforce
- **Speculator lookahead gating** - Speculator tier only activates for LargeFeature, GreenfieldBuild, and ArchitecturalRevision policies
- **BudgetEnvelope session restore** - Step/cost/revision caps restored from DB during `resume` so interrupted sessions honour the original limits
- **Bundle path normalization** - `filter_bundle_to_declared_paths` uses `normalize_artifact_path` for correct comparison of path variants (e.g. `./src/main.rs` vs `src/main.rs`)

- **NodeState::Superseded** - New terminal state for plan amendment (Phase 14 preparation). Updated `is_terminal()`, `parse_node_state()`, and `NodeStatus` conversion
- **Orchestrator module extraction** - `orchestrator.rs` split into 9 submodules: `mod.rs`, `bundle.rs`, `commit.rs`, `convergence.rs`, `init.rs`, `planning.rs`, `repair.rs`, `solo.rs`, `verification.rs`
- **Centralized prompts** - All 15 agent prompts consolidated in `prompts.rs` with constants and `render_*` helpers; duplicates removed from `agent.rs`
- **RepairFootprint-backed correction** - `build_correction_prompt` uses `RepairFootprint` for precise, grounded repair context
- **Greenfield bootstrap ordering** - Plugin-driven project initialization with correct pre-sheafify plugin re-detection
- **Provisional branch lifecycle** - Sandbox-first execution with branch creation, merge, and flush cascade
- **Escalation classification** - 5 categories with 9 rewrite actions and graph surgery support

Documentation:

- Updated architecture docs with PlanningPolicy, FeatureCharter, and NodeState::Superseded documentation
- Added Planning Policy and Feature Charter sections to SRBN architecture guide
- Updated workspace crates docs with orchestrator submodule structure
- Added speculator lookahead and budget restore documentation to agent mode guide
- Fixed energy weight default (gamma=2.0) in advanced features guide
- Refreshed all version references to 0.5.6

Version 0.5.5

PSP-5 Cross-Platform and CI Stabilization Release

Cross-Platform Fixes:

- **Windows sandbox path normalization** - `list_sandbox_files` now returns forward-slash-separated relative paths on all platforms
- **Windows workspace-bound validation** - `validate_workspace_bound` correctly detects absolute paths with Windows drive prefixes (`C:\...`) and normalizes backslash path separators before POSIX shell tokenization
- **Clippy and fmt CI compliance** - Resolved `items_after_test_module` and `useless_vec` warnings that were failing CI on all platforms

Build and CI Improvements:

- **Removed accidental eval workspace member** - `.perspt-eval/rust_cli` removed from workspace members; `.perspt-eval/` added to `.gitignore`
- **Stabilized cargo doc** - Added `doc = false` to CLI bin target to prevent output collision with the `perspt` library crate
- **Removed deprecated atty dependency** - Replaced with `std::io::IsTerminal` for TTY detection
- **Lockfile refresh** - Cleared hard `cargo audit` vulnerability failures via dependency updates

Documentation:

- Updated workspace coding instructions to match current multi-crate architecture
- Refreshed all version references across the Perspt Book and Sphinx configuration

Version 0.5.4

PSP-5 Compliance Release

SRBN Agent Enhancements:

- **Per-node error recovery** - Retry logic respects `ErrorType` classification (`Compilation`, `ToolFailure`, `ReviewRejection`) with separate counters
- **Multi-file extraction** - Actuator reliably extracts artifact bundles with multiple files from LLM responses
- **Multi-artifact bundles** - Bundle protocol correctly handles write, diff, and command artifacts in a single node
- **Plugin-driven project initialization** - All plugins use `uv init --lib` (Python), `cargo init` (Rust), `npm init` (JS) for proper project scaffolding
- **Degraded verification mode** - When tool binaries are missing, falls back to heuristic verification with clear warnings
- **Sheaf validation** - 7 validator classes for cross-node contract verification

Bug Fixes:

- Fixed `uv init` to `uv init --lib` in `plugin.rs` (2 locations) and `test_runner.rs` (1 location) for correct `src/` layout with `[build-system]`
- Removed dead `test_check_workspace_requirement` test from `orchestrator.rs`

Documentation:

- Complete rewrite of the Perspt Book for PSP-5 accuracy
- Updated developer guide with full type inventory and architecture diagrams
- Added tutorials for headless mode, Python ETL, Rust CLI, and scientific computing
- Updated all model names and provider defaults to current versions

Version 0.5.3

- SRBN energy convergence improvements
- Provisional branch lifecycle management
- Interface seal and flush cascade support
- DuckDB migration from SQLite

Version 0.5.2

- Initial SRBN orchestrator implementation
- Per-tier model selection (`--architect-model`, etc.)
- Merkle ledger with DuckDB backend
- Plugin system for Python, Rust, and JavaScript
- Agent TUI with dashboard, task tree, and review modal
- Starlark policy engine
- Basic sandbox command execution

- 10 CLI subcommands (chat, agent, simple-chat, init, config, ledger, status, abort, resume, logs)

Version 0.5.1

- Multi-provider chat support
- TUI markdown rendering with code blocks
- Simple CLI mode for scripting
- Streaming response protocol with EOT signal

Version 0.5.0

- Initial release
- Basic chat interface with OpenAI
- Configuration auto-detection from environment

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Core Dependencies

Crate	License	Description
tokio	MIT	Async runtime for Rust
ratatui	MIT	Terminal user interface library
serde	MIT/Apache-2.0	Serialization framework
clap	MIT/Apache-2.0	Command line argument parser
anyhow	MIT/Apache-2.0	Error handling library
thiserror	MIT/Apache-2.0	Error derive macros

LLM Integration

Crate	License	Description
genai	MIT/Apache-2.0	Unified LLM provider interface
reqwest	MIT/Apache-2.0	HTTP client library

Terminal and UI

Crate	License	Description
crossterm	MIT	Cross-platform terminal library
unicode-width	MIT/Apache-2.0	Unicode character width calculation
textwrap	MIT	Text wrapping and formatting

Development Dependencies

Crate	License	Description
criterion	MIT/Apache-2.0	Benchmarking library
mockall	MIT/Apache-2.0	Mock object library
tempfile	MIT/Apache-2.0	Temporary file management

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- **Enterprise solutions** - Build enterprise tools that use Perspt

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- **GitHub Issues**: License Questions
- **Include** specific details about the concern or question

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See also

- Acknowledgments - Credits and thanks to contributors
- Contributing - How to contribute to the project
- GNU Project - Official LGPL v3 License text

Acknowledgments

Perspt is built on the shoulders of giants. We extend our gratitude to the many open-source projects, libraries, and communities that made this project possible.

Theoretical Foundation

The SRBN engine at the heart of Perspt's experimental agent mode is based on the *Stability is All You Need* paper series:

Paper I: *Lyapunov-Guided Hierarchies for Long-Horizon LLM Reliability*
Paper II: *SRBN-Control Beyond Harness Engineering*
Paper III: *The SRBN Platform Contract*

The series introduces a topological and control-theoretic framework that reformulates LLM agency as measured proposal, correction, and acceptance. Key theoretical contributions adopted in Perspt include:

- **Input-to-State Stability (ISS)** - bounded reasoning errors yield bounded system deviation (paper proof)
- **Flow Matching Barriers** - project diverging trajectories back onto the safe manifold (paper result)
- **Adaptive Flow Speculation** - latency reduction via predictive branch execution
- **Logarithmic Reliability Scaling** - the papers predict $O(\log N)$ retry cost instead of exponential decay

Perspt serves as an experimental testbed for exploring these theoretical ideas in real-world multi-file coding workloads. The mathematical framework is mature; empirical benchmarks on Perspt's implementation have not yet been published.

Core Dependencies

AI and LLM Integration

genai

The foundation of Perspt's multi-provider support. This exceptional crate provides unified interfaces to multiple AI providers and automatically stays up-to-date with new models and capabilities.

- **Project**: genai
- **License**: MIT/Apache 2.0
- **Impact**: Enables seamless integration with OpenAI, Anthropic, Google, Groq, Cohere, XAI, DeepSeek, and Ollama providers

serde & serde_json

Rust's premier serialization framework, powering Perspt's configuration management and API communication.

- **Project**: serde
- **License**: MIT/Apache 2.0
- **Impact**: JSON configuration parsing, API request/response handling

User Interface and Terminal

ratatui

The modern, feature-rich TUI framework that powers Perspt's interactive terminal interface.

- **Project**: ratatui
- **License**: MIT
- **Impact**: Rich terminal UI, markdown rendering, scrollable chat interface

crossterm

Cross-platform terminal manipulation library enabling consistent behavior across operating systems.

- **Project**: crossterm
- **License**: MIT
- **Impact**: Keyboard input handling, terminal control, cross-platform compatibility

Async Runtime and Concurrency

tokio

The asynchronous runtime that enables Perspt's responsive, non-blocking architecture.

- **Project**: tokio
- **License**: MIT
- **Impact**: Async/await support, concurrent LLM requests, responsive UI

Error Handling and Utilities

anyhow

Elegant error handling that makes Perspt's error messages helpful and actionable.

- **Project**: anyhow
- **License**: MIT/Apache 2.0
- **Impact**: Comprehensive error context, user-friendly error messages

clap

Command-line argument parsing that makes Perspt easy to use and configure.

- **Project**: clap
- **License**: MIT/Apache 2.0
- **Impact**: CLI interface, help generation, argument validation

Documentation Tools

Guidedog

The Rust-native documentation, RFC (*guidedog dsc*), and static site engine (*guidedog ssg*) powering the Perspt Book, PSP Catalog, and unified site.

- **Project**: Guidedog

- **Impact**: Native reStructuredText & MyST compilation, RFC governance, multi-mount static site generation

Typst
: The modern typesetting system used to compile the Perspt Book PDF, PSP technical papers, and vector SVG architecture diagrams.

- **Project**: Typst
- **License**: Apache 2.0
- **Impact**: High-contrast editorial PDF typesetting, vector diagrams, mathematical notation

Development Tools

Rust Language
: The systems programming language that makes Perspt fast, safe, and reliable.

- **Project**: Rust
- **License**: MIT/Apache 2.0
- **Impact**: Memory safety, performance, excellent tooling ecosystem

cargo
: Rust's package manager and build system that makes development smooth and dependency management effortless.

- **Project**: Part of Rust toolchain
- **License**: MIT/Apache 2.0
- **Impact**: Dependency management, build automation, testing framework

Community and Inspiration

AI Provider Communities

OpenAI
: For creating GPT models and establishing many of the patterns that define modern AI interaction.

Anthropic
: For Claude models and their pioneering work in AI safety and helpful, harmless, and honest AI systems.

Google
: For Gemini models and their contributions to accessible AI technology.

Groq
: For ultra-fast inference infrastructure and democratizing AI speed.

Cohere
: For enterprise-grade language models and excellent developer tools.

XAI
: For Grok models and advancing conversational AI capabilities.

DeepSeek
: For their contributions to the open-source AI ecosystem.

Ollama
: For making local AI model hosting accessible and user-friendly.

Open Source Ecosystem

GitHub
: For providing the platform that enables collaborative development and open-source sharing.

crates.io
: Rust's package registry that makes sharing and discovering Rust libraries effortless.

docs.rs
: For automatically generating and hosting documentation for Rust crates.

Terminal and CLI Inspiration

The terminal and CLI interface draws inspiration from many excellent tools:

- **htop** - For showing how terminal UIs can be both beautiful and functional
- **tmux** - For terminal multiplexing concepts and keyboard navigation patterns
- **vim/neovim** - For modal editing concepts and efficient keyboard shortcuts
- **fzf** - For demonstrating responsive, interactive terminal interfaces

Rust Community Projects

Many patterns and approaches in Perspt were learned from studying excellent Rust projects:

- **ripgrep** - For performance optimization and user experience design
- **bat** - For beautiful terminal output and syntax highlighting
- **exa/eza** - For modern CLI design and colored output
- **gitui** - For TUI application architecture and event handling

Testing and Quality Assurance

Users and Beta Testers
: The early adopters and users who provided feedback, reported bugs, and suggested improvements.

Security Researchers
: For responsible disclosure of security issues and helping make Perspt more secure.

Documentation Reviewers
: For helping improve the clarity and completeness of this documentation.

Special Thanks

AI Safety Research Community
: For ongoing work to make AI systems more reliable, interpretable, and aligned with human values.

Open Source Contributors
: To everyone who contributes to open-source projects, from major features to documentation fixes.

Rust Community
: For creating and maintaining an inclusive, helpful community that makes Rust development a joy.

Terminal Enthusiasts
: For keeping the art of terminal-based applications alive and pushing the boundaries of what's possible in text-based interfaces.

Contributing Back

Perspt aims to be a good citizen of the open-source ecosystem. We contribute back by:

Open Source Release
: Perspt itself is released under the LGPL v3 license, allowing anyone to use, modify, and distribute it.

Documentation Standards
: This comprehensive documentation serves as an example of thorough project documentation.

Best Practices Sharing
: Through blog posts, talks, and code examples, we share what we've learned building Perspt.

Upstream Contributions
: When we find bugs or missing features in dependencies, we contribute fixes and improvements back to those projects.

License Information

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All dependencies are used in accordance with their respective licenses. We are grateful to all the authors and maintainers who choose to share their work under permissive open-source licenses.

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: Help improve Perspt by reporting bugs, suggesting features, or improving documentation.

Contribute Code
: See our Contributing guide for how to contribute code improvements.

Share Knowledge
: Write blog posts, create tutorials, or give talks about your experience with Perspt.

Support Dependencies
: Consider contributing to the open-source projects that Perspt depends on.

Spread the Word
: Help others discover Perspt and the amazing ecosystem of Rust and AI tools.

—

Thank you to everyone who makes open-source software development possible. Your contributions, large and small, make projects like Perspt possible.